# BIOPHYSICAL MODELS OF TRANSCRIPTIONAL REGULATION FROM SEQUENCE DATA

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# Abstract

In the post-genomics era, DNA sequence itself is becoming a medium by which to probe biological phenomena. With the advent of microarray technology, and ultra-high-throughput sequencing more recently, large sequence data sets are becoming standard products of day-to-day research. Yet as software for analyzing such data proliferates, a fundamental understanding of how DNA sequence should be used to gain biological insight is missing from the literature.

The focus of this thesis is on developing tools for characterizing the biophysical interactions underlying transcriptional regulation – the ability of cells to control which genes they transcribe to mRNA, and thus express as protein. We begin by presenting basic principles for the analysis of DNA sequence data – specifically, data for which each sequence $\sigma$ is accompanied by a (perhaps very noisy) measurement $z$ of biophysical functionality. A salient feature of experiments which produce such $\sigma z$ data is the difficulty of characterizing experimental noise *a priori*. We overcome this obstacle by introducing error-model-averaged (EMA) likelihood, which allows biophysical models of arbitrary functional form to be rigorously fit to $\sigma z$ data. EMA likelihood is closely related to mutual information, but its probabilistic interpretation provides some advantages. We demonstrate EMA likelihood's utility on previously published microarray data, using Metropolis Monte Carlo sampling to infer models for the DNA-binding energy of transcription factor proteins.

The ability to properly analyze $\sigma z$ data leads us to propose a new experimental assay, called Sort-Seq. This technique uses ultra-high-throughput sequencing to probe the protein-DNA and protein-protein interactions underlying transcriptional regulation at specific genomic loci. We present data from a proof-of-principle Sort-Seq experiment probing the *lacZ* promoter of *E. coli*, data we use to characterize the sequence-dependent binding energy of the transcription factor CRP. We then discuss what one can, in principle, infer from large Sort-Seq data sets. We show that, with enough $\sigma z$ data probing the binding of multiple proteins per sequence, one should be able to infer both protein-DNA and protein-protein interaction energies in absolute thermal units. We conclude that, with the advent of ultra-high-throughput sequencing, DNA sequence itself might provide a very sensitive means by which to probe *in vivo* biophysics.

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I also thank Michael Lässig and Ville Mustonen, with whom I enjoyed a productive and enlightening collaboration studying the evolution of transcription factor binding sites. The gist of our collaborative work deviates from the line of argument pursued in this thesis, and has (with a few misgivings) been left out as a result. Instead, I encourage any interested reader to refer to our paper, which was recently published [2].

I thank Bill Bialek for conversations which, though intermittent, had a great effect on me an my work. Although Bill was never my official advisor, there is little in this thesis that did not see his influence.

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*To Antoinette, with love.*

# Chapter 1

# Introduction

Every cell in the human body (to a good approximation) contains the same DNA encoding the same genes with the same sequence of nucleotides. Why, then, are neurons, liver cells, leukocytes,[1] fibroblasts,[2] and embryonic stem cells so different? Since the pioneering work of Jacob and Monod (see [3]), it has been understood that cells can choose, based on environmental signals or lineage history, which genes they express as proteins and at what rate these proteins are produced. Understanding the mechanisms by which cells effect gene regulation has been a central problem in biology ever since [4].

According to molecular biology's central dogma, the expression of a gene requires that that gene's coding sequence first be *transcribed* from DNA to mRNA (Fig. 1.1a). This mRNA is then *translated* into a sequence of amino acids comprising the protein (Fig. 1.1b). Regulation of gene expression can occur at many points along this path [5, 6, 7]. Cells have the ability to target specific proteins for degradation. Regulatory RNAs transcribed from elsewhere in the genome can interfere with mRNA translation, and in many organisms can also direct the degradation of specific mRNA transcripts. In eukaryotes, mRNA must be *spliced* before being exported from the nucleus. During splicing, a substantial fraction of the mRNA transcript is removed while the remaining bits – called *exons* – are pasted together, and the choice of which exons to include can be regulated by the cell. Eukaryotes also have the ability to sequester coding sequences in a protein-DNA structure called heterochromatin, which severely limits the ability of RNA polymerase to initiate mRNA transcription.

Despite this myriad of regulatory mechanisms, the control of mRNA transcript initiation by proteins called *transcription factors* – a regulatory mechanism first discovered by Jacob and Monod roughly 50 years ago – remains the canonical mechanism by which cells can regulate gene expression [4]. This is also the one form of gene regulation for which simple biophysical models have not just been proposed [8, 9, 10], but have successfully accounted for experimental observations in quantitative detail [11, 12, 13].

The last decade has seen an intense effort put towards the problem of characterizing transcriptional regulation. In particular, the development of microarray tech-

[1] White blood cells.

[2] Found in connective tissues.

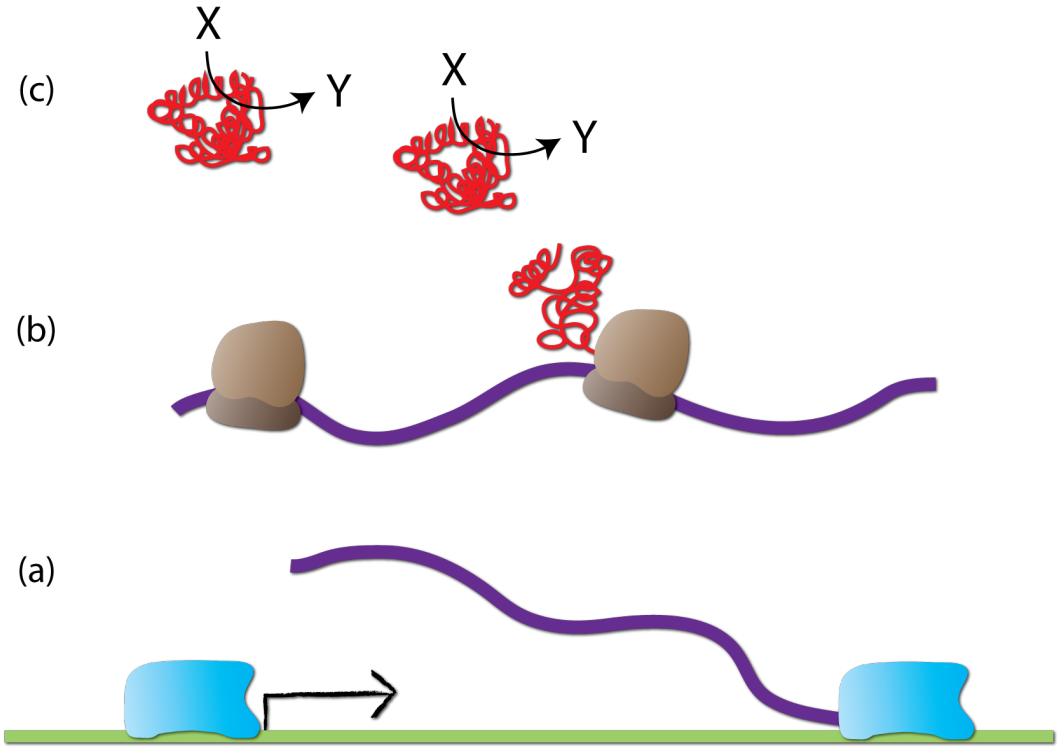


Figure 1.1: The central dogma of molecular biology. (a) DNA (green) is transcribed into mRNA (purple) by RNA polymerase (cyan). (b) mRNA is translated into a sequence of amino acids (red) by ribosomes (brown) (c) The folded amino acid sequence comprises the functional protein which, for instance, can catalyze reactions.

nology [7], and more recently of ultra-high-throughput sequencing (e.g. [14]), has enabled assays which simultaneously probe different aspects of transcriptional regulation genome-wide [15]. But obstacles to our understanding of transcriptional regulation on a detailed mechanistic level remain. First, the methods available for analyzing DNA sequence data produced by these microarray and sequencing technologies have yet to mature; specifically, a simple, transparent, and principled approach for inferring biophysical models of transcriptional regulation from such data is lacking. Furthermore, these genome-wide assays provide only rough information about how specific transcriptional regulatory sequences function. History has shown that scientific understanding is often best advanced by the detailed characterization of specific systems, and less so by the rough characterization of many diverse systems. And at this point in time, classic molecular biology techniques, often requiring much labor and serial experimentation, are still required to provide the detailed biophysical understanding of transcriptional regulation that we seek.

In this dissertation, we present theoretical, computational, and experimental methods by which DNA sequence data can be used to build biophysical models of transcriptional regulation. This work is motivated largely by the recent advent of ultra-high-throughput sequencing. Our hope is that simple, stereotyped assays producing vast amounts of sequence data can, in combination with principled computational analyses, provide detailed pictures of regulatory biophysics.

We begin by reviewing transcriptional regulation at the *lacZ* promoter of the gut bacterium *Escherichia coli*. We then illustrate how simple biophysical models can be used to model regulation at this locus. Central to these models is an understanding of transcription factor affinity for specific DNA sequences, and we review experimental methods used to characterize this. This is followed by a review of the basic ideas behind existing computational tools for DNA sequence analysis.[3] We end chapter 1 by discussing the modeling of transcription factor specificity, explicitly addressing the differences between the common statistical approach and more physically based models; this difference has often been obscured in the literature, and we hope this discussion will help clear the air.

In chapter 2 we propose mutual information maximization as a principled approach to the analysis of DNA sequence data consisting of a set of sequences $\{\sigma_i\}_{i=1}^N$ where each sequence $\sigma_i$ is accompanied by a measurement of functionality $z_i$. Such data appear often in the study of transcriptional regulation, and we henceforth refer to these as $\sigma z$ data. We then draw a connection between maximizing mutual information and maximizing likelihood: for large $N$, maximizing mutual information is equivalent to maximizing likelihood when the experimental error model is uncertain. This leads us to propose error-model-averaged (EMA) likelihood as an objective function with which biophysical models of transcriptional regulation can be fit. While EMA likelihood reduces to mutual information in the large $N$ limit, it has a rigorous interpretation in terms of likelihood and can be evaluated analytically for any $N$.

[3] The reliance of these methods on either generative models of DNA sequence or simplistic assumptions about experimental noise leads us, in chapter 2, to reexamine fundamental aspects of statistical inference from DNA sequence data.

Requiring essentially no assumptions about experimental noise processes, EMA likelihood is an ideal objective function for Metropolis Monte Carlo sampling of model parameters.

We then use EMA likelihood to infer energy matrix models of the *Saccharomyces cerevisiae* transcription factor Abf1. For this we analyze two data sets, one from an *in vitro* (protein binding microarray, or PBM) assay [16] and one from an *in vivo* (ChIP-chip) assay [17]. We find that Metropolis Monte Carlo sampling using EMA likelihood works quite well. In fact, independent analyses of these two data sets produce very precise models of Abf1's sequence-dependent binding energy which are, nevertheless, almost completely consistent between these two experiments.

In chapter 3 we propose a new assay – called Sort-Seq – which uses ultra-high-throughput sequencing to produce $\sigma z$ data probing the detailed biophysical interactions governing transcriptional regulation at specific genomic loci. We then present data from a proof-of-principle Sort-Seq experiment on the *E. coli lacZ* promoter. By fitting models to the resulting $\sigma z$ data, we demonstrate the ability to characterize the *in vivo* sequence-dependent binding energy of the *E. coli* transcription factor CRP. We also demonstrate how the completeness of such inferred models – i.e. their ability to account for all sequence-dependent information in the experiment – can be directly measured.

Returning to theoretical issues, we discuss what, in principle, mutual information maximization can and cannot reveal. We show that, when analyzing $\sigma z$ data probing single protein binding sites per sequence, one cannot infer binding energies in natural thermal units; this is due to the presence of *diffeomorphic modes* in parameter space. However, when analyzing data probing multiple binding events per sequence, natural nonlinearities in the formulas of the statistical mechanics break these diffeomorphic models, allowing one to characterize protein-DNA and protein-protein interaction energies in absolute thermal units through mutual information maximization.

We conclude by discussing the outlook for sequence-based assays in the age of ultra-high-throughput sequencing. While this revolutionary technology is bringing unprecedented sensitivity to genome-wide assays, Sort-Seq suggests that the availability of so much sequence data has the potential to permit qualitatively new ways of probing cellular function.

## 1.1 Review of transcriptional regulation

### 1.1.1 The *lac* operon

The *lac* operon [3, 4] of *E. coli* contains three genes involved in the metabolism of the sugar lactose, all of which are transcribed one-after-the-other on the same mRNA transcript (Fig. 1.2a). The first gene in the line of transcription is *lacZ*, which encodes $\beta$-galactosidase. $\beta$-galactosidase is an enzyme which cleaves the disaccharide lactose into glucose and galactose (both monosaccharides) as the first step in lactose metabolism. Second is *lacY*, which encodes $\beta$-galactoside permease, a protein residing in the membrane which transports lactose into the cell. The final gene in the operon

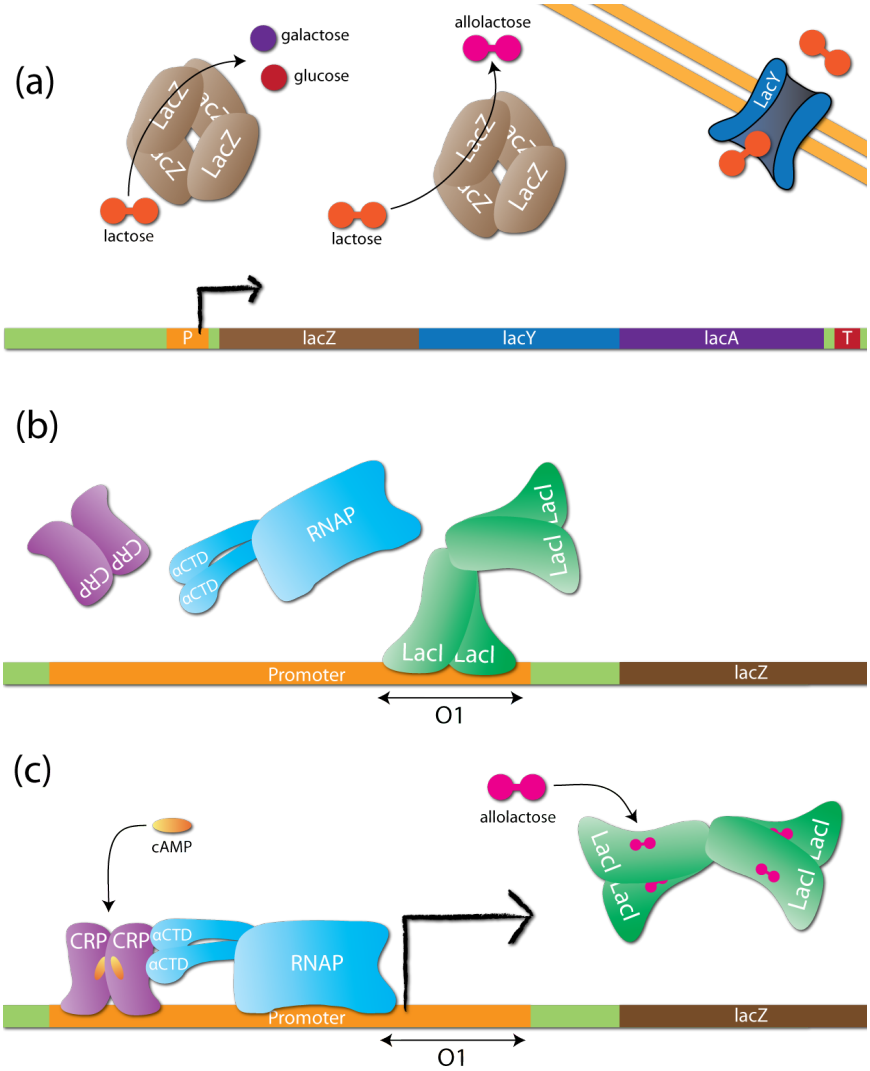


Figure 1.2: The *lac* operon. (a) This operon consists of a promoter (P), the coding sequences of the genes *lacZ*, *lacY*, and *lacA*, and a terminator sequence (T). Transcription is regulated by the promoter, and proceeds through the three coding sequences to the terminator. After translation, LacZ protein (i.e. $\beta$-galactosidase) forms a homotetramer and catalyzes the cleavage of lactose to glucose and galactose. LacZ also converts lactose to allolactose in a side reaction. LacY ($\beta$-galactoside permease) is a membrane protein which allows the cell to intake lactose from the environment. The function of LacA is less well understood. (b) In the absence of allolactose, the LacI tetramer bids site O1 on the *lac* promoter, blocking the binding of RNAP. In the absence of cAMP, CRP is also unable to bind to its recognition site. (c) In the presence of allolactose, LacI is induced and cannot specifically bind O1. This allows RNAP to bind its target site and initiate transcription. cAMP-bound CRP can also bind upstream of RNAP and interact with the $\alpha$CTD subunits of the holoenzyme, thereby increasing RNAP occupancy and up-regulating operon transcripiton.

is *lacA*, which encodes $\beta$-galactoside transacetylase. This enzyme transfers an acetyl group to $\beta$-galactoside sugars, but is not necessary for lactose metabolism.

The *lac* operon is transcribed by the RNA polymerase (RNAP) protein complex. The core enzyme of RNAP, which is capable of transcription, contains five subunits: one $\beta$, one $\beta'$, one $\omega$, and two $\alpha$ subunits. The RNAP holoenzyme consists of this core enzyme, together with a $\sigma$ subunit which gives RNAP specificity for certain DNA sequences. At the *lac* operon, RNAP containing the $\sigma^{70}$ subunit is responsible for initiating transcription from a specific site which resides upstream of the *lacZ* coding region.[4]

RNAP bound to DNA at this site initiates transcription spontaneously, then typically runs through the entire operon until it hits a terminator sequence, at which point transcription is terminated. We note that some promoters require a conformational change in either the RNAP holoenzyme (e.g. activation of the *glnA* promoter by the factor NtrC) or in the DNA itself (e.g. regulation of the *merT* promoter by MerR) for transcription to initiate [4]. The fact that RNAP's presence at the *lacZ* promoter suffices for transcription [4] motivates us to model transcriptional activity as being proportional to RNAP's equilibrium occupancy at its binding site; we will pursue this line of reasoning in the next section.

Transcription from the *lac* promoter is regulated by proteins called *transcription factors*. One of these transcription factors is LacI, also known as the *lac* repressor. LacI is present in the cell as a dimer of homodimers. Each dimer is capable of binding DNA tightly at specific sequences. In its *uninduced* confirmation, the LacI tetramer will bind strongly to a site, called O1, just downstream of (and partially overlapping with) RNAP's binding site (Fig. 1.2b). The presence of this protein occludes RNAP, thus preventing transcription. However, when allolactose[5] is present in the cell, it binds LacI and effects a conformational change which makes the repressor unable to bind DNA specifically, thereby relieving transcriptional repression (Fig. 1.2c). In this state, LacI is said to be *induced*.

At the same time, the binding of the transcription factor CRP (cAMP receptor protein) just upstream of the RNAP binding site increases the rate of transcription. CRP is a homodimer, which binds the small molecule cAMP (cyclic adenosine monophosphate). The presence of cAMP induces an allosteric change in the CRP protein, enabling it to bind to specific DNA sequences (as a dimer). When bound to its primary binding site on the *lac* promoter, the CRP dimer interacts energetically with the carboxy-terminal domains of the $\alpha$ subunits (the $\alpha$CTDs) of the RNAP holoenzyme. This interaction increases the equilibrium occupancy of RNAP on the promoter, thereby increasing the rate of transcription (Fig. 1.2c).

These transcription factors, CRP and LacI, can thus alter the rate at which lactose metabolism proteins are synthesized: transcription is activated by the presence of cAMP, and repressed by the absence of allolactose. Why regulate lactose metabolism

[4]The RNAP binding site itself was first termed the *promoter*, but we will use this term to refer to the entire region upstream of *lacZ* containing sequences which affect *lac* operon transcription; this usage is more in line with what is generally meant by the "promoter" of other genes.

[5]Allolactose is a derivative of lactose produced in a side-reaction catalyzed by $\beta$-galactosidase (Fig. 1.2a).

genes in response to these signals? As mentioned, allolactose is a derivative of lactose. If there is no lactose in the environment there will be no allolactose, and LacI will not be induced; instead, LacI will specifically bind its target site, block transcription of the *lac* operon, and the cell will not waste resources producing lactose metabolism proteins.[6] The sensitivity of *lac* operon transcription to cAMP is somewhat more puzzling. Low cAMP levels have been associated with the presence of glucose in the environment. Glucose is more directly digested than lactose, and *E. coli* will exhaust the glucose in a mixed medium before resorting to lactose metabolism.[7] It therefore makes sense that the cell would wait until glucose was low (supposedly indicated by high levels of cAMP) before fully expressing the *lac* operon. In reality, the relationship between intracellular cAMP and environmental glucose is more complicated and less direct, but this issue will not be pursued here.

The *lac* operon has other important components which substantially complicate our understanding [18]. There are two other LacI binding sites: O2 (located within the *lacZ* coding sequence) and O3 (located just upstream of the CRP binding site). LacI tetramers will typically bind to both O1 and one of these other sites, thereby looping the DNA and increasing the efficiency of repression. CRP also bends DNA by about 90° [19], effectively increasing the Hill coefficient of the promoter's response to allolactose [13]. The upstream region of *lacZ* also contains at least two alternative RNAP binding sites from which transcription is initiated *in vivo* [18].

### 1.1.2 Thermodynamic models

Despite these complications, simple biophysical models of transcriptional regulation at the *lacZ* promoter can be gainfully employed. The simplest such models are so-called *thermodynamic models* [11, 12, 8, 9, 10, 13].[8] The essential feature of such models is that all relevant interactions – including protein-DNA interactions, protein-protein interactions, and protein-ligand interactions – are purely energetic with fast on- and off-rates. We further assume that transcription initiation happens randomly at some fixed rate when RNAP is bound; this assumption renders transcription proportional to RNAP occupancy. With these assumptions, one can determine the rate of transcription (up to a proportionality constant) by computing the fractional contribution to the system's partition function by states in which RNAP occupies its DNA binding site.

Activation of *lacZ* transcription by CRP (when LacI is fully induced) permits the simplest model of transcriptional regulation by a transcription factor. We illustrate this model in Fig. 1.3. Let $\alpha$ be the DNA-binding energy of RNAP at its binding site (in units of $kT$ and after subtracting away the RNAP chemical potential), $\beta$ be the DNA-binding energy of CRP (ditto), and $\gamma$ be the interaction energy between RNAP

[6] Of course, allolactose will not be present if there isn't any $\beta$-galactosidase around either. The *lac* operon presents more than one of these "chicken or egg" type problems, none of which will be addressed here.

[7] This phenomenon is called *diauxie* [3].

[8] These models actually involve statistical mechanics, not thermodynamics *per se*.

Figure 1.3: A thermodynamic model for CRP activation of transcription by RNAP at the *lacZ* promoter. Transcription is assumed to be proportional to RNAP occupancy at its binding site. Interaction energies are assumed to be in units of $kT$ and measured relative to the associated chemical potential. $\alpha$ denotes RNAP-DNA interaction energy, $\beta$ denotes CRP-DNA interaction energy, and $\gamma$ denotes the interaction energy between DNA-bound CRP and DNA-bound RNAP. Binding sites for RNAP and CRP are shown in black.

and CRP. In the simplest scenario, this system permits four molecular configurations:

| RNAP | CRP | energy |
|---|---|---|
| off | off | 0 |
| off | on | $\beta$ |
| on | off | $\alpha$ |
| on | on | $\alpha + \beta + \gamma$ |

RNAP occupancy is given by the fractional contribution to the partition function of the two states in which RNAP is bound:

$$\mathcal{O}_{RNAP} = \frac{e^{-\alpha} + e^{-\alpha-\beta-\gamma}}{1 + e^{-\beta} + e^{-\alpha} + e^{-\alpha-\beta-\gamma}} = \frac{1}{1 + F^{-1}}, \quad F = e^{-\alpha}\frac{1 + e^{-\beta-\gamma}}{1 + e^{-\beta}}. \tag{1.1}$$

Following Bintu et al. [10], we refer to $F$ as the system's *regulation factor*.[9] We use this quantity because it has a more transparent mathematical form than the raw RNAP occupancy $\mathcal{O}_{RNAP}$. For instance, it is clear from this formula that interactions with CRP effectively renormalize the interaction of RNAP to DNA:

$$\alpha \to \alpha - \log \frac{1 + e^{-\beta-\gamma}}{1 + e^{-\beta}} \tag{1.2}$$

When $\gamma = 0$ (i.e. CRP and RNAP do not interact), CRP binding has no effect on regulation. When $\gamma > 0$, $\alpha$ is effectively increased, causing lower RNAP occupancy and resulting in less transcription. Conversely, $\gamma < 0$ – which obtains at the real *lacZ* promoter – effectively decreases $\alpha$, thereby increasing transcription.

Simple thermodynamic models such as this were first proposed by Shea and Ackers for modeling the promoter controlling the lysogenic/lytic switch of the bacteriophage $\lambda$ [11, 12]. Recently, Kuhlman et al. [13] validated a thermodynamic model for the *lacZ* promoter, following a first attempt by Setty et al. [20]. This model included the

[9] Our use of the term *regulation factor* actually differs from Bintu et al.'s definition, but not substantially.

above interaction between RNAP and CRP, as well as the CRP-cAMP interaction, the LacI-IPTG[10] interaction, the occlusion of RNAP by LacI, and LacI-mediated DNA looping, with looping between O1 and O3 facilitated by CRP's bending of DNA. To do so, Kuhlman et al. created a triple mutant ($\Delta lacY$ $\Delta cpdA$ $\Delta cyaA$) in which the $lacY$-mediated feedback loop was disabled, and cAMP regulation was debilitated enough that the intracellular concentration of cAMP was reliably proportional to the cAMP concentration in the growth medium. By measuring $\beta$-galactosidase activity in this mutant strain grown under various concentrations of cAMP and IPTG, Kuhlman et al. showed that a model defined by the above interactions (with appropriate parameters) impressively fit the data. Thanks to this work, it now appears that we understand – on a quantitative level – the basic physical mechanisms of transcriptional regulation at the *lac* promoter.

Ideally, we would like to have such a quantitative understanding of how every transcriptional regulatory sequence works. However, the $\lambda$ switch and the *lac* promoter – the two best-studied transcriptional regulatory systems in molecular biology – appear to be the only loci for which thermodynamic models have been tested in quantitative detail. The primary reason for this is the difficulty of making the necessary measurements. In Kuhlman et al.'s work, not only was the knowledge that the relevant transcription factors could be controlled by cAMP and IPTG necessary (knowledge that took many years to obtain [3]), but the authors had to create a triple mutant in order to control internal levels of cAMP.

But for many transcriptional regulatory sequences, the lack of quantitative thermodynamic models is not due to the difficulty of making quantitative measurements so much as it is due to the sheer lack of knowledge of how these sequences *qualitatively* function. There are a relatively small number of transcriptional regulatory sequences for which all of the basic regulating proteins and their corresponding DNA binding sites are known. And even in many well-studied systems the precise ways in which regulatory proteins interact with each other and RNAP is poorly understood.

The traditional approach to studying a transcriptional regulatory sequence of interest is *promoter bashing*. Typically, the sequence of interest is used to control the expression of a reporter – either a fluorescent protein, or an mRNA molecule which can be detected through an assay known as *in situ* hybridization. The transcriptional regulatory sequence is then fiddled with; sequence elements are removed, mutated, etc., and the resulting effect on expression is observed. While promoter bashing is very powerful – a power exemplified by what it has taught us about the *eve* stripe 2 enhancer of *Drosophila melanogaster* [21, 22, 23] – it is also slow and requires experiments to be done iteratively. It is of particular note that, while the genomics revolution in molecular biology has greatly aided the identification of likely transcriptional regulatory sequences [24, 25], it has done relatively little to increase the ease with which promoter bashing experiments can be carried out.

An alternative strategy for understanding gene regulation is to focus on individual transcription factors instead of specific regulatory sequences. Like all protein

[10]IPTG, or isopropyl $\beta$-D-1-thiogalactopyranoside, is an alternative inducer of LacI which, unlike allolactose, cannot be metabolized by the cell.

molecules, there is a lot to learn about transcription factors: the ligands they bind, the post-translational modifications (such as phosphorylation) which influence their behavior, their structure, and so on. But given that each transcription factor typically regulates multiple genes, we have a particular interest in characterizing the DNA-binding properties which cause different factors to mediate transcriptional regulation at different regulatory loci. We now review some experimental techniques used to probe this sequence-specificity.

## 1.2 Experimental techniques

Transcription factors typically bind DNA with a footprint of about 10 to 20 bp. The DNA sequence within this footprint, however, is not uniquely specified; a given transcription factor can bind to many different sequences, with nucleotides at different positions within these sequences constrained to different extents. This degeneracy is largely responsible for the difficulty investigators have had in identifying where transcription factors bind.

There are two differing approaches to characterizing a transcription factor's DNA-binding behavior. One is to directly identify the *locations* of transcription factor binding sites in the genome. The other is to characterize the *sequence specificity* of the transcription factor – usually *in vitro*. We emphasize that using *in vitro* binding data to predict *in vivo* binding sites is nontrivial. While the *in vitro* sequence specificity of some transcription factors appears sufficient to dictate *in vivo* binding, many factors – such as the *D. melanogaster* transcription factor *bicoid* – show far too little specificity *in vitro* to pinpoint their *in vivo* binding sites. Indeed, explaining binding site locations from knowledge of a factor's specificity is one important task for biophysical modeling (see, for example, [24, 26, 27, 28]).

### 1.2.1 Characterizing factor specificity

Many methods have been developed for experimentally characterizing the specificity of transcription factors, but only a small number of techniques have gained widespread use, or been applied in a wide range of systems. Here we detail a few of the primary methods, to which much of our knowledge of transcription-factor-DNA specificity is due.

#### DNase I footprinting

DNase I footprinting [29] is the classic *in vitro* method for determining where, on a given sequence of DNA, a factor will bind. One starts with a solution of double-stranded DNA oligos, where one of the strands is labeled at its 5' end with $P^{32}$. After incubating these oligos with the factor of interest, one performs a brief digestion with the endonuclease DNase I. This endonuclease creates nicks in the DNA, disrupting the phosphate bond between adjacent nucleotides. The key to this assay is that DNase I is blocked by DNA-bound proteins.

DNase I treatment is brief, so that each oligo receives about one nick on average. The nicked oligos are then run on a denaturing polyacrylamide gel and, much like Sanger sequencing, the different length DNA fragments reveal, though autoradiography, the locations of the DNase I nicks. Gaps in the DNase I banding pattern thereby indicate the location of protein binding sites. Running these fragments on lanes adjacent to those produced by dideoxy sequencing thereby reveals the sequences of transcription-factor-bound sites. Since its introduction in 1978, the directness of the DNase I footprinting method has made it the standard method for verifying predicted DNA binding sites *in vitro* (e.g. see [30]).

### SELEX

Another method for determining the *in vitro* specificity of a transcription factor is SELEX.[11] First proposed in 1990, and applied to the identification of protein-DNA binding sites in 1991 [32], this technique uses multiple rounds of PCR to exponentially amplify DNA (or RNA) sequences which bind molecules of interest [33, 34]. Here we focus on the application of SELEX to protein-DNA sequence recognition.

In SELEX, one begins with a synthesized library of single-stranded DNA (ssDNA) oligos containing specified sequences at the 5' and 3' ends. These sequences are separated by a few dozen bp of random sequence, in which each position is synthesized using a mixture of all four bases (A,C,G, and T). PCR amplification using primers targeting the 3' and 5' ends of these oligos produces double stranded DNA (dsDNA). The factor of interest is incubated with this oligo library, after which protein-bound oligos are enriched (e.g. by nitrocellulose filter binding, immunoprecipitation, or gel extraction). These enriched oligos are amplified with PCR, after which they are incubated with the transcription factor again, and so on. Repeated rounds of PCR exponentially amplify oligos containing factor binding sites. Some number of these oligos are then sequenced, and from these sequences one can infer models of protein-DNA interaction.

Recently, Roulet et al. [33] made two substantial improvements to SELEX. One problem with SELEX is that, in practice, one often ends up sequencing many copies of a very small number of distinct sequences. This is due to over-enrichment. Roulet et al. overcame this problem by continually adding initial library oligos to the enriched oligos, thereby generating a diverse, quasi-stationary distribution of sequences. Roulet et al. also combined SELEX with a SAGE-like sequencing procedure [35], in which many different enriched oligos were concatenated before sequencing. This allowed the authors to sequence over 10,000 oligos produced by this modified version of SELEX. Of course, the need for this latter step will likely be obviated by Illumina's ultra-high-throughput DNA sequencer which can, as we will soon describe, sequence tens of millions of DNA oligos in a single run.

[11] Systematic evolution of ligands by exponential enrichment; [31].

### Protein binding microarrays

The development of microarray technology – in which tens or hundreds of thousands of different DNA sequences can be arrayed on the surface of a glass slide – enabled the study of protein binding to many defined DNA sequences at once. Martha Bulyk's group has pioneered this capability, developing the protein binding microarray (PBM) [36, 37, 16, 38, 39].[12] PBMs are microarrays containing dsDNA features,[13] to which the factor of interest is bound *in vitro*. A fluorescent antibody is then targeted[14] against the factor. The microarray is scanned, and features containing factor binding sites fluoresce.

This technology was first reported by Mukherjee et al. [16] in 2004, who used microarrays spotted with all intergenic regions of *S. cerevisiae* to probe the specificity of the *S. cerevisiae* transcription factors Abf1, Rap1, and Mig1. Since this work, Berger et al. [38] have moved to double-stranded Agilent arrays containing all 10-mers. This technology was initially tested on five factors from yeast, mouse, worm and human, and has recently been used to characterize the binding specificities of most (i.e. 168) homeodomain transcription factors in mouse [39].

### *E. coli* one-hybrid

An alternative method for characterizing transcription-factor-DNA specificity was recently developed by Meng, Brodsky, and Wolfe [41]. As opposed to *in vitro* methods, this assay tests the specificity of eukaryotic factors inside of *E. coli*, bypassing the need to purify proteins. The DNA binding domain of the transcription factor of interest is fused to the $\alpha$ (or $\omega$) subunit of the core RNAP enzyme. The coding sequence for this fusion protein is then transformed into *E. coli* on a *bait* plasmid. A library of *prey* plasmids, with each plasmid containing a region of randomized DNA upstream of a weak RNAP binding site, is also transformed. This weak RNAP binding site is positioned to initiate transcription of a positive marker (the yeast *HIS3* gene, whose expression is necessary to overcome inhibition by 3-amino-triazole (3-AT) and allow growth on minimal media) and a negative marker (the yeast *URA3* gene, whose protein causes 5-fluoro-orotic acid (5-FOA) to be converted into a toxin). By adding 3-AT or 5-FOA to growth media, one can select for cells which transcribe from the prey promoter only when the factor fusion protein is expressed.[15] Prey plasmids selected from the initial library in this way should thus contain binding sites for the factor of interest (in the randomized region).

This method was first used to characterize the specificity of several factors, including a few factors involved *Drosophila* morphogenesis [41]. Recently this method has been used to characterize the specificities of 78 *Drosophila* homeodomains [42], as well as most of the transcription factors involved in *Drosophila* segmentation [43].

[12] Though see also the work of Warren et al. [40].

[13] Such microarrays are to be contrasted with ssDNA arrays, which are often used to study gene expression.

[14] Typically through the use of an epitope tag engineered into the transcription factor.

[15] The negative marker, *URA3*, is needed to avoid selecting constitutive promoters.

### 1.2.2 Locating genomic binding sites

There is only one widely used technique for identifying the genome-wide binding sites of a given transcription factor *in vivo*: ChIP-chip, and its sequencing-based incarnation ChIP-Seq.

#### ChIP-chip

ChIP-chip was first reported by Ren et al. in 2000 [44]. In this assay, cells are infused with formaldehyde, which cross-links interacting molecules. In particular, it cross-links DNA to DNA-bound proteins. Genomic DNA is then isolated, sheared into segments $\sim$ 500 bp long, and enriched using an antibody specific for the factor of interest (a process called chromatin immunoprecipitation, or ChIP). This enriched DNA is then amplified, fluorescently labeled, and hybridized to a microarray in competition with alternatively-labeled unenriched DNA.

Two years after Ren et al.'s groundbreaking article, Lee et al. [17] reported ChIP-chip results for most transcription factors in yeast. Harbison et al. [45] later expanded this data set to include ChIP-chip experiments done in a variety of conditions. ChIP-chip has since become a standard tool for studying DNA-binding proteins, and has been used to locate the binding sites of many transcription factors in many organisms.

The data from a ChIP-chip experiment are a set of microarray intensities, each corresponding to some location on the genome. The greater the observed enrichment of ChIP-ed DNA at a given microarray probe, the greater the supposed nearby occupancy of the transcription factor. Lee et al. used spotted microarrays, where each microarray spot contained DNA from a different intergenic region of *S. cerevisiae* (these were the same arrays used by Mukherjee et al. [16]). These regions typically range from 500 bp to 1000 bp – the resulting resolution of Lee et al.'s experiment. Spotted microarrays have since fallen out of favor; arrays containing orders of magnitude more probes which *tile* the genome are now available for many organisms. The sheared DNA precipitated in ChIP-chip assays is still about 200 bp - 500 bp in length, but the stereotyped shape of ChIP-chip intensities, when laid out along the genome, often allows one to pinpoint binding site locations which substantially higher resolution.

#### ChIP-Seq

Recently, a new technique called ChIP-Seq has been developed [46, 47]. In this assay, protein-bound fragments of DNA are ChIP-ed, but are then directly sequenced instead of being hybridized to an array [48]. This technology is made possible by recently developed ultra-high-throughput sequencing platforms. ChIP-Seq is more accurate than ChIP-chip, and is likely to supplant ChIP-chip as the price of ultra-high-throughput sequencing drops.

### 1.2.3 Ultra-high-throughput sequencing

Recent years have witnessed a revolution in DNA sequencing technology. While the primary aim of these technologies has been to enable the sequencing of individual human genomes on a commercial scale, the data these technologies are capable of generating have many other applications. In particular, various *sequence census* assays have been developed [15]. These assays produce large libraries of DNA sequence which, using ultra-high-throughput sequencing, are then mapped back to a reference genome. This effectively produces a large list of genomic locations which can be used to identify the locations of DNA-bound proteins (ChIP-Seq), quantify mRNA transcript levels (RNA-Seq), or identify methylated DNA sequences (Methyl-Seq, BIS-Seq [49]). We now review two of these technologies: 454 pyrosequencing (from Roche) and Solexa sequencing (from Illumina).

#### 454 pyrosequencing

454's pyrosequencing technology, first reported in 2005 by Margulies et al. [14], was the first ultra-high-throughput sequencing platform to market. 454 Life Sciences has since been bought by Roche, and their most recent sequencer, the Genome Sequencer FLX, can produce $10^6$ reads, each $\sim$ 400 bp long, in one $\sim$ 12 hr run. Later in this thesis we will discuss how this sequencing technology might be used to interrogate the detailed architecture of specific transcriptional regulatory sequences.

454's technology is facilitated by a technique known as emulsion PCR (emPCR). In emPCR, single DNA molecules are captured on separate plastic beads. A water-in-oil emulsion is then prepared, which isolates each bead in its own microscopic PCR reaction well. During PCR thermocycling, those beads with an initial attached DNA molecule become coated with a lawn of $\sim 10^7$ clonal DNA molecules. Only one strand of each oligo is attached to the bead; the second strand is melted off, and a sequencing primer is annealed in its place.

DNA-coated beads are then deposited onto an array (called a *picotiter plate*) containing hundreds of thousands of picoliter wells.[16] Smaller beads carrying surface-bound ATP sulphurylase and luciferase enzymes are then added to these wells. Each well is etched from the end of one optic fiber, and is large enough to hold just one DNA bead. The picotiter plate is made from many such fibers fused together. One surface of this plate is covered with beads; the opposing surface is placed against a high-resolution CCD camera, which monitors light emanating from each well during the pyrosequencing reaction.

Pyrosequencing, which was first developed by Nyrén and colleagues [51, 52, 53] (see [54] for a review, and [55] for autobiographical account), allows the real-time monitoring of nucleotide incorporation into a growing DNA sequence. When DNA polymerase covalently attaches a dNTP molecule to the 3' end of a growing DNA strand, one molecule of $PP_i$ (inorganic phosphate; $P_2O_7^{4-}$) is released. ATP sulphurylase reacts $PP_i$ and APS (adenosine 5' phosphosulfate) to produce ATP. This

[16] Interestingly, it appears that these picoliter wells were initially intended – at least in part – to serve as PCR reaction chambers [50].

ATP is then hydrolyzed by the enzyme luciferase during the conversion of luciferin to oxyluciferin, releasing one photon of visible light in the process. These photons are corralled by the optic fiber from which the picoliter well is etched, and detected by the CCD camera.

Sequencing is initiated from an annealed sequencing primer. Different nucleotides are flowed over the picotiter plate in succession, and emitted light is observed from wells in which nucleotides are incorporated. The integrated intensity of this emission determines the number of homotypic nucleotides incorporated during a given nucleotide flow. The series of light emissions from each picoliter well thus provides the sequence of the DNA oligos attached to each resident bead.

#### Solexa sequencing

Another ultra-high-throughput sequencing platform is that invented by Solexa (now owned by Illumina). Instead of using beads, Illumina uses a flat surface covered in a lawn containing two different PCR primer sequences. Adapters complementary to these primers are ligated onto the DNA to be sequenced. These adapted oligos are then sparsely hybridized (via their ends) to the complementary primers on the flat substrate, and an ensuing *bridge amplification* process creates, from each hybridized oligo, a cluster of millions of identical single-stranded oligos. These clusters are spaced far enough apart to be resolved by a microscope.

After a sequencing primer is annealed to the clustered oligos, fluorescent nucleosides (labeled by four differently-colored fluorophores) are flowed over the solid substrate, and incorporate at the 3' end of the primer. These nucleosides are protected, so no more than one nucleotide will incorporate at each oligo in a given flow. A microscope is then used to photograph the differently colored clusters. Afterward, the newly incorporated nucleotides have their fluorophores cleaved and are de-protected, allowing for extension in subsequent nucleotide flows. Image stacks of the fluorescent clusters are then used to determine the sequence of each cluster's oligo.

Each run of Illumina's sequencer currently provides $\sim 4 \times 10^7$ reads of $\sim$ 35-50 bp, but these read lengths will likely double by the end of 2008. One run of the machine takes about three days.[17] Illumina's read lengths are much shorter than 454's, but the vastly greater number of sequences the platform produces make it preferable for sequence census assays (e.g. [46, 47, 49]).

## 1.3 Sequence analysis

We now discuss approaches to analyzing data produced by these experimental methods – PBM and ChIP-chip in particular. In its most basic form, the data from a PBM experiment consists of a set of $N$ sequences $\{\sigma_i\}_{i=1}^N$, with each sequence $\sigma_i$ assigned a measurement $z_i$ of (normalized) microarray fluorescence; we will refer to such data henceforth as $\sigma z$ data. In the case of $\sigma z$ data from PBM experiments, higher fluorescence values $z$ generally indicate higher levels of transcription-factor binding to

[17] Run time increases with increasing read length.

the corresponding sequence $\sigma$. The precise mapping between these two quantities, however, is usually unclear. ChIP-chip data has a similar form.

The typical approach for inferring models of transcription factor specificity from such data is called *motif finding*. Motif finding is a very general term, encompassing a number of statistical methods, with each method implemented by a variety of different software packages, and each software package replete with various features, tricks, and approximations. The one feature unifying most if not all motif finders is the pursuit of a *generative* model of DNA – in the simplest form, a statistical description of DNA sequence as a mixture of binding sites and *background* or *null* DNA. Here we review the basic principles behind two of the primary motif finding approaches: expectation maximization and Gibbs sampling. We illustrate these two approaches by showing how they can be used to infer the same generative model, depicted in Fig. 1.4.

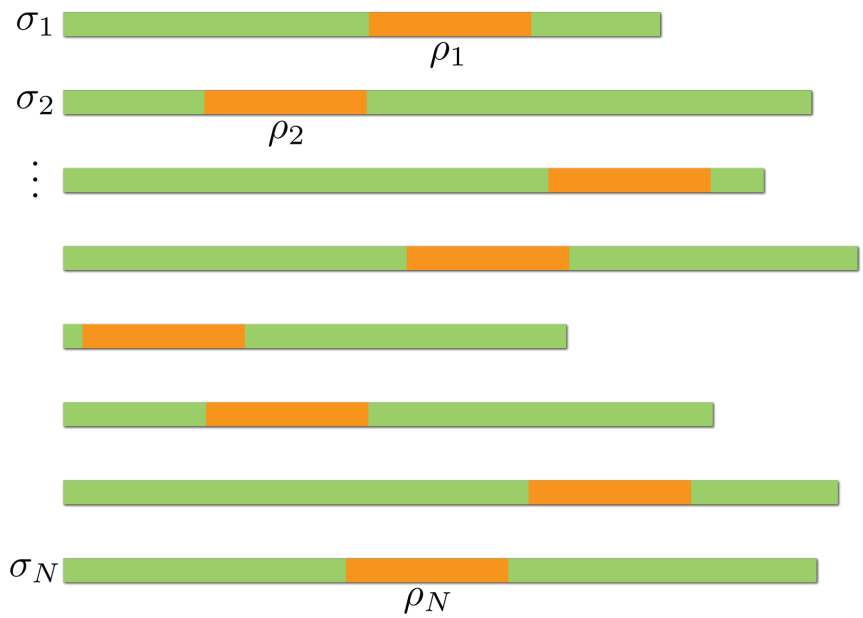


Figure 1.4: A generative model of DNA sequence. Each sequence $\sigma_i$ consists of one binding site $\rho_i$ (drawn from the motif $\{q_{bl}\}$ and indicated in orange) embedded in background DNA (generated from the single-nucleotide probabilities $\{q_{b,null}\}$ and shown in green).

Suppose we have a set of sequences $s = \{\sigma_i\}$, each of which, we believe, contains precisely one transcription factor binding site. We propose the following generative model: that each sequence $\sigma_i$ consists of one binding site $\rho_i$ of length $L$, with the rest of $\sigma_i$ made up of background DNA. $\rho$ is drawn from a distribution of possible binding sites in which base $b \in \{A, C, G, T\}$ occurs at position $l \in \{1, 2, \ldots, L\}$ with probability $q_{bl}$, and each base occurs independently of its neighbors. Nucleotides of the null DNA, on the other hand, are drawn according to the single-base probabilities $\{q_{b,null}\}$. Given $s = \{\sigma_i\}$, we aim to learn the parameters $\theta = \{q_{bl}, q_{b,null}\}$ of this generative model, as well as the identities $r = \{\rho_i\}$ of the binding sites within these sequences.

In what follows we will use a matrix representation for DNA sequence: a sequence $\rho$ of length $L$ is represented by a $4 \times L$ matrix with entries $\{\rho_{bl}\}$, where $\rho_{bl} = 1$ if base $b$ occurs at position $l$, and $\rho_{bl} = 0$ otherwise. In particular, this means that the log likelihood ratio of a given binding site $\rho$ being generated according to $\{q_{bl}\}$ as opposed $\{q_{b,null}\}$ is

$$\log \frac{p(\rho|\text{site})}{p(\rho|\text{background})} = \sum_{bl} \rho_{bl} \log \frac{q_{bl}}{q_{b,null}} \tag{1.3}$$

Such log likelihood ratios are typically used in bioinformatics to *score* possible binding sites.

We emphasize that this approach describes a transcription factor's DNA-binding specificity using a *distribution on the space of binding site sequences*. Such a distribution is called a *motif*.

### 1.3.1 Expectation maximization

The expectation maximization (or EM) algorithm was first described in full generality by Dempster, Laird, and Rubin in 1977 [56], and first applied to motif finding by Lawrence and Reilly in 1990 [57]. A variation of this technique was later implemented in the MEME algorithm [58], which has seen widespread use in the biological community.

The stated goal of EM is to maximize

$$p(s|\theta) = \sum_r p(s, r|\theta), \tag{1.4}$$

the likelihood of the observed sequences $s$ given the generative model defined by $\theta$. One does this by treating the identities of the binding sites $r$ as missing data, hypothesizing an initial set of parameters $\theta_0$, then following an iterative algorithm which increases $p(s|\theta_n)$ at each step $\theta_n \to \theta_{n+1}$.

We first describe the basic principle behind EM. Given $\theta_n$, one chooses

$$\theta_{n+1} = \text{argmax}_\theta \left\langle \log p(s, r|\theta) \right\rangle_{r|s,\theta_n} \tag{1.5}$$

where the subscript on the bracket indicates averaging over $r$ according to $p(r|s, \theta_n)$. Factoring $p(s, r|\theta) = p(r|s, \theta)p(s|\theta)$, we can write

$$\log p(s|\theta) = \left\langle \log \frac{p(s, r|\theta)}{p(r|s, \theta)} \right\rangle_{r|s,\theta'} \tag{1.6}$$

which holds for all choices of parameters $\theta'$. We can now show that each step $\theta_n \to \theta_{n+1}$ increases the likelihood of $s$:

$$\begin{aligned} \log \frac{p(s|\theta_{n+1})}{p(s|\theta_n)} &= \left[ \left\langle \log p(s, r|\theta_{n+1}) \right\rangle_{r|s,\theta_n} - \left\langle \log p(s, r|\theta_n) \right\rangle_{r|s,\theta_n} \right] \\ &\quad + \left\langle \log \frac{p(r|s, \theta_n)}{p(r|s, \theta_{n+1})} \right\rangle_{r|s,\theta_n} . \end{aligned} \tag{1.7}$$

The first term on the right is nonnegative by definition of $\theta_{n+1}$ in Eq. 1.5. The second term is a Kullback Leibler divergence,[18] making it nonnegative as well. In fact, this second term is equal to zero only when $p(r|s,\theta_n) = p(r|s,\theta_{n+1})$ for all $r$. We therefore find that

$$p(s|\theta_{n+1}) \geq p(s|\theta_n) \tag{1.8}$$

where equality obtains if and only if the models described by $\theta_{n+1}$ and $\theta_n$ are, for all intents and purposes, identical. This monotonic increase of $p(s|\theta_n)$ with $n$ will typically lead one to a *local* maximum in parameter space. There is, of course, no guarantee that EM will locate the global maximum. In practice, EM is often initiated many times with different starting parameters.

To use EM for motif finding, we must have an explicit formula for $p(s,r|\theta)$. From our generative model,

$$\log p(s,r|\theta) = \sum_{bl} c_{bl} \log \frac{q_{bl}}{q_{b,null}} + \sum_b c_b^{tot} \log q_{b,null} - C \tag{1.9}$$

where $c_{bl} = \sum_i \rho_{bl}^i$ is the total number of binding sites $\rho_i \in r$ for which base $b$ occurs at position $l$, $c_b^{tot}$ is the total number of occurrences of base $b$ in *all sequences* $\sigma_i \in s$, and $C$ is the log number of ways binding sites $r$ can be positioned within the sequences $s$. For the E step of the EM algorithm, we need to compute the expected log likelihood,

$$\langle \log p(s,r|\theta) \rangle_{r|s,\theta_n} = \sum_{bl} \langle c_{bl} \rangle_{r|s,\theta_n} \log \frac{q_{bl}}{q_{b,null}} + \sum_b c_b^{tot} \log q_{b,null} - C \tag{1.10}$$

For this we need $\langle c_{bl} \rangle_{r|s,\theta_n}$, which can be computed rather easily; one simply examines each possible site $\rho_i$ in $\sigma_i$, records $\rho_{bl}^i$, and weights this contribution to $\langle c_{bl} \rangle_{r|s,\theta_n}$ by $p(\rho_i|\sigma_i,\theta_n)$:

$$\langle c_{bl} \rangle_{r|s,\theta_n} = \sum_r p(r|s,\theta_n) \sum_i \rho_{bl}^i = \sum_i \sum_{\rho_i \in \sigma_i} p(\rho_i|\sigma_i,\theta_n) \rho_{bl}^i \tag{1.11}$$

where $\theta_n = \left\{ q_{bl}^n, q_{b,null}^n \right\}$ and

$$p(\rho_i|\sigma_i,\theta_n) = \frac{1}{Z_i} \prod_{bl} \left( \frac{q_{bl}^n}{q_{b,null}^n} \right)^{\rho_{bl}^i}, \qquad Z_i = \sum_{\rho \in \sigma_i} \prod_{bl} \left( \frac{q_{bl}^n}{q_{b,null}^n} \right)^{\rho_{bl}}. \tag{1.12}$$

For the $M$ step we maximize Eq. 1.10 with respect to $\theta$, which can be done analytically. The resulting EM algorithm is $\theta_n \to \theta_{n+1} = \left\{ q_{bl}^{n+1}, q_{b,null}^{n+1} \right\}$ where

$$q_{bl}^{n+1} = \frac{1}{N} \langle c_{bl} \rangle_{r|s,\theta_n}, \quad q_{b,null}^{n+1} = \frac{1}{M} \left[ c_b^{tot} - \langle c_{bl} \rangle_{r|s,\theta_n} \right], \tag{1.13}$$

$M = \sum_b c_b^{tot}$ being the sum length of all sequences $\sigma_i$.

[18] See Cover and Thomas [59] chapter 2 for a review of Kullback Leibler divergence and mutual information; we use to these two quantities extensively throughout this dissertation.

### 1.3.2 Gibbs sampling

Gibbs sampling was first proposed for motif finding by Lawrence et al. in 1993 [60]. Many different implementations of Gibbs sampling have since become available, most notably AlignACE [61] and BioProspector [62]. Taking a Bayesian approach, Gibbs sampling aims to sample from the distribution of model parameters $\theta$ and binding sites $r$ given the sequences $s$, i.e. from

$$p(\theta, r|s). \tag{1.14}$$

This is done by iteratively sampling from the marginal distributions of $p(\theta, r|s)$. First one posits an initial set of sites $r_0 = \{\rho_i^0\}$, and samples parameters $\theta_0$ from the marginal distribution $p(\theta_0|r_0, s)$. One then chooses $j \in \{1, 2, \ldots, N\}$, and replaces $\rho_j^0 \in r_0$ with a site $\rho_j^1$ drawn from $p(\rho_j^1|\theta_0, s, \{\rho_i^0\}_{i\neq j})$, thus giving $r_1$. One then samples $\theta_1$ from $p(\theta_1|r_1, s)$, etc. The resulting series

$$\cdots \to (r_{n-1}, \theta_{n-1}) \to (r_{n-1}, \theta_n) \to (r_n, \theta_n) \to \cdots \tag{1.15}$$

forms a Markov chain whose stationary distribution is $p(\theta, r|s)$.

The distribution $p(\rho_j|\theta, s, \{\rho_i\}_{i\neq j}) = p(\rho_j|\sigma_i, \theta)$ was computed in Eq. 1.12 and is easy to sample. To sample from $p(\theta|r, s)$ we assume a Dirichlet prior on parameter space, i.e.

$$\begin{aligned} p(\theta) &= p(\{q_{bl}\})p(\{q_{b,null}\}), && (1.16)\\ &= \left[\frac{\Gamma(4\beta)^L}{\Gamma(\beta)^{4L}} \prod_l \delta\left(1 - \sum_b q_{bl}\right) \prod_b q_{bl}^{\beta-1}\right] && (1.17)\\ &\quad \times \left[\frac{\Gamma(4\gamma)}{\Gamma(\gamma)^4} \delta\left(1 - \sum_b q_{b,null}\right) \prod_b q_{b,null}^{\gamma-1}\right] \end{aligned}$$

for some $\beta$ and $\gamma$ of our choosing. Bayes's rule can then be used to compute the posterior distribution

$$\begin{aligned} p(\theta|r, s) &= \left[\frac{\Gamma(4\beta + N)^L}{\prod_{bl} \Gamma(\beta + c_{bl})} \prod_l \delta\left(1 - \sum_b q_{bl}\right) \prod_b q_{bl}^{\beta + c_{bl} - 1}\right] && (1.18)\\ &\quad \times \left[\frac{\Gamma(4\gamma + M - NL)}{\prod_b \Gamma(\gamma + c_b^{tot} - \sum_l c_{bl})} \delta\left(1 - \sum_b q_{b,null}\right) \prod_b q_{b,null}^{\gamma + c_b^{tot} - \sum_l c_{bl} - 1}\right] \end{aligned}$$

from which one can directly sample $\theta$.[19,20]

[19] One can sample $\{q_{bl}\}$ from this distribution by first drawing $\{r_{bl}\}$, where each $r_{bl}$ is drawn from the gamma distribution

$$p(r_{bl}) = r_{bl}^{c_{bl}+\beta-1} e^{-r_{bl}} / \Gamma(c_{bl} + \beta), \tag{1.19}$$

then setting $q_{bl} = r_{bl} / \sum_{b'} r_{b'l}$. $\{q_{b,null}\}$ can be sampled similarly.

[20] We note, however, that Lawrence et al. [60] did not do this; they kept $\{q_{b,null}\}$ fixed, and given $\{\rho_i\}$ they simply assigned $q_{bl}$ to its maximum likelihood estimate,

$$q_{bl} = \frac{c_{bl} + \beta}{N + 4\beta}. \tag{1.20}$$

### 1.3.3 Correlative models

The above approaches to motif finding take as input only a subset of all the sequences probed – those sequences $\{\sigma_i\}$ which, by dint of the measurements $\{z_i\}$, appear likely to contain factor binding sites. But ChIP-chip and PBM assays assign a measurement $z_i$ to *every* sequence $\sigma_i$, and it would be preferable to use all of this data instead of just a user-selected subset of it. Indeed, $\sigma z$ data enables a completely different way of inferring transcription-factor-DNA specificity: one can build a biophysical model which, given parameters $\theta$, assigns to each sequence $\sigma$ a prediction $x(\sigma;\theta)$ of transcription factor occupancy which best accounts for *all* of the sequence-specific measurements $\{z_i\}$.

Such an approach was first taken by Stormo, Schneider, and Gold in 1986 [63]. The austhors took previously-published data probing three different phenomena (context-dependent suppression of the amber stop codon, context-dependent mutagenisis using 2-aminopurine, and context-dependence of translation initiation), and fit models whose predictions best correlated with measured sequence activity. A similar approach was later used by Barrick et al. [64] to characterize ribosome binding sites. Bussemaker, Li, and Siggia [65] were the first to fit correlative models to microarray data, and their work was later extended in scope by Foat et al. [66].[21] Here we review Foat et al.'s MatrixREDUCE algorithm.

MatrixREDUCE directly models the sequence-dependent binding energy of the transcription factor of interest. To do so, it uses the *energy matrix* approximation: it assumes that, within a biding site $\rho$ of length $L$, each position contributes independently and additively to the protein-DNA binding energy. These single-nucleotide energy contributions constitute the elements of a $4 \times L$ energy matrix $\{\epsilon_{bl}\}$, which assigns to $\rho$ a binding energy

$$\epsilon(\rho) = \sum_{bl} \rho_{bl}\epsilon_{bl}. \tag{1.21}$$

Given the ChIP-chip or PBM intensity ratios intensity ratios $\{z_i\}$ for a set of sequences $\{\sigma_i\}$, MatrixREDUCE seeks to infer parameters $\theta$ consisting of an energy matrix $\{\epsilon_{bl}\}$, as well as two constants $A$ and $B$, which minimize

$$\chi^2 = \sum_i [z_i - x(\sigma_i;\theta)]^2, \quad x(\sigma;\theta) = A \sum_{\rho\in\sigma} \exp\left(-\sum_{bl} \rho_{bl}\epsilon_{bl}\right) + B. \tag{1.22}$$

Here, $x(\sigma;\theta)$ is the (affinely transformed) equilibrium occupancy of transcription factors on the sequence $\sigma$ in thermal equilibrium.[22] Note that this requires no generative model of DNA. Instead, model parameters are fit directly to the probed sequences $\{\sigma_i\}$ and measurements $\{z_i\}$.

[21] These authors analyzed mRNA expression data, not ChIP-chip or PBM data. Correspondingly, the models that they fit were models of transcription-factor-activated gene expression, not models of raw transcription factor occupancy. However, the basic concept behind their algorithm REDUCE is essentially the same as the concept behind MatrixREDUCE, which has been applied directly to ChIP-chip and PBM data.

[22] This assumes a low transcription factor concentration.

This method of inference is equivalent to maximum likelihood if the experimental errors are gaussian, that is,

$$p(z|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\chi^2/2\sigma^2\right] \tag{1.23}$$

for some variance $\sigma^2$. If experimental errors are not gaussian, however, minimizing Eq. 1.22 can lead to a misestimation of parameters. We discuss this issue in more depth in the next chapter.

While it is clear that, for both ChIP-chip and PBM experiments, greater microarray probe fluorescence should correlate with increased transcription factor occupancy on the probed sequences, the quantitative stochastic mapping between these two quantities is difficult to model. In PBM experiments, factor binding is not performed and visualized in equilibrium; on the contrary, in Mukherjee et al.'s experiments [16], PBMs were washed after incubation with the transcription factor, and washed again after antibody staining before being scanned. Microarray intensities were then averaged over replicates, and normalized for the mass of DNA in each feature.[23] A further Lowess-normalization step was used before each sequence was assigned a final *log intensity ratio*. While these steps might improve the microarray signal, they make it difficult, if not impossible, to model the noise processes in this experiment and develop reliable error models *a priori*.

In the ChIP-chip experiments of Lee et al. [17], the possible sources of error are a more fundamental problem. Aside from the many experimental steps needed to generate a library of ChIP-ed DNA fragments, *in vivo* effects such as cooperative binding and occlusion of binding sites by chromatin make it virtually impossible to create quantitatively reliable noise models describing the probabilistic map between the DNA occupancies predicted by a biophysical model of binding and the actual observed microarray intensities.

## 1.4 Modeling factor specificity

We thus have two alternative ways of modeling transcription factor specificity for DNA. One can either describe the sequence-dependent free energy of binding to DNA, or one can give a statistical description of binding site sequences. Certainly, the sequences of protein binding sites will depend strongly on the protein's sequence-dependent binding energy; in some situations this connection is nearly isomorphic. But these two types of models are fundamentally different, and this difference will become manifest when we discuss the principles of inferring biophysical models of transcriptional regulation from arbitrary $\sigma z$ data.

### 1.4.1 Berg and von Hippel

The connection between the sequence-dependent binding energy of a transcription factor and the statistics of its binding site sequences and was first made by Berg and

[23] Not the number of oligos.

von Hippel [67, 68]. Here we present an overview of their basic argument; an explicit derivation is presented in the appendix to this chapter.

Imagine that a DNA site of length $L$ is free to evolve via point mutations, and does so in such a way that all four bases occur with equal likelihood at every position. We can think of each position as a particle with four possible states corresponding to the four bases it can host.[24] Mutation acts to thermalize these particles.

Now picture a DNA site of length L constrained by natural selection to bind a given factor. Biological function will require that the binding site's energy fall within a certain range. Further assume that each position within the binding site contributes independently and additively to the overall binding energy, so that the contribution of base $b$ at position $l$ is given by an energy matrix element $\epsilon_{bl}$. How does this constraint on binding energy affect the probability of each base occurring in each position?

Berg and von Hippel elegantly suggested [67] viewing the energy constraint as placing the entire sequence in a *microcanonical ensemble*. If there are enough positions within the site which play a substantial role in influencing the binding energy, one can then focus on a single position and view the rest of the site as a heat bath. Doing so puts the position of interest in the *canonical* ensemble, giving

$$q_{bl} \approx \frac{\exp[-\beta\epsilon_{bl}]}{\sum_{b'} \exp[-\beta\epsilon_{b'l}]} \tag{1.24}$$

for some effective temperature $1/\beta$. This temperature should be approximately the same across all positions. Although Berg and von Hippel did not mention this, their result can be easily modified to include the case where unconstrained positions exhibit different bases with different frequencies; this is done by assigning a prior probability $\{q_{b,null}\}$ to each position:

$$q_{bl} \approx \frac{q_{b,null} \exp[-\beta\epsilon_{bl}]}{\sum_{b'} q_{b',null} \exp[-\beta\epsilon_{b'l}]}. \tag{1.25}$$

Such prior probabilities would result, in the physical picture, from different base states having different entropies.

Having in hand a collection of binding sites allowing one to estimate $q_{bl}$ directly, one can solve for the elements of the energy matrix:

$$\epsilon_{bl} = -\frac{1}{\beta} \log \frac{q_{bl}}{q_{b,null}} \tag{1.26}$$

where $\beta$ is an unknown constant.[25] Eq. 1.26 reveals a simple connection between a factor's energy matrix $\{\epsilon_{bl}\}$ and the motif $\{q_{bl}\}$ describing the statistics of the factor's binding sties. This relationship appears to be a very good approximation in

[24] For instance, we can think of each position of this DNA sequence as a spin 3/2 particle, with the base identities corresponding to different values of the spin's z-component.

[25] We note that, in defining an energy matrix, it is common practice to shift the values in each column $l$ so that $\min_b(\epsilon_{bl}) = 0$. The fact that binding energy only has physical meaning up to an additive constant gives one the freedom to do this – unless one defines binding energy with respect to a chemical potential.

practice; indeed, in chapter 3 we will show this explicitly for the *E. coli* transcription factor CRP. However, it also appears to be the source of much confusion between the statistical view of transcription factor specificity and the biophysical view. While Berg and von Hippel provide an elegant and useful result, this result is not a fundamental connection between sequence statistics and binding energy, but is rather a product of the simplifying assumptions they make.

### 1.4.2 Binding site evolution

We illustrate this point by considering an explicit model of binding site evolution, first put forth by Berg, Willmann, and Lässig [69]. Consider a population of $N$ haploid individuals, each of which has a transcription factor binding site of sequence $\rho$ at a specific locus. Also assume that these individuals have a Malthusian fitness $f(\epsilon)$[26] which depends only on the binding energy $\epsilon(\rho)$ of this site. As these individuals reproduce and die (in such a way that the total population stays constant at $N$), the set of binding sites among these individuals will tend to become less and less polymorphic, until eventually one sequence fixates in the population. Now let us include mutation: we assume a sequence $\rho$ will mutate to a sequence $\rho'$ at a rate of $\mu(\rho \to \rho')$ per generation per individual.

We restrict our focus to the limit in which binding site mutations either go extinct or fixate much faster than they arise in the population. Given a mutation $\rho \to \rho'$, the Kimura fixation probability [70] is

$$p_{fix}(\Delta f) = \frac{1 - e^{-2\Delta f}}{1 - e^{-2N\Delta f}} \tag{1.27}$$

where $\Delta f = f(\epsilon(\rho')) - f(\epsilon(\rho))$ is the change in fitness due to the change in binding site energy.[27]

Now consider the ensemble of binding sites realized over a large number of mutations and fixations. Assuming detailed balance, the observed frequencies $p_{site}(\rho)$ and $p_{site}(\rho')$ are related by

$$p_{site}(\rho)\mu(\rho \to \rho')p_{fix}(\Delta f) = p_{site}(\rho')\mu(\rho' \to \rho)p_{fix}(-\Delta f). \tag{1.28}$$

This gives a time-averaged sequence-space distribution $p_{site}(\rho)$ for binding sites $\rho$ under the fitness constraint $f(\epsilon(\rho))$,

$$p_{site}(\rho) = p_{null}(\rho)\exp[2(N-1)f(\epsilon(\rho)) + C] \tag{1.29}$$

[26] By *fitness* we mean that if two individuals harbor sites with respective binding energies $\epsilon_1$ and $\epsilon_2$, then the ratio of organism one's reproductive rate to that of organism two is $\exp[f(\epsilon_1) - f(\epsilon_2)]$. We use *Malthusian* to emphasize that, regardless of which binding sites occur in the population, the number of individuals in the population remains fixed.

[27] It is instructive to verify the limits of this equation. For $1 \gg \Delta f \gg 1/N$, $p_{fix} \approx 2\Delta f$, so the fixation probability is proportional to $\Delta f$. For $\Delta f \sim 1$, the probability of the mutation *not* fixating is $1 - p_{fix} \approx e^{-2\Delta f}$. On the other hand, a deleterious mutation for which $-1/N \gg \Delta f$ gives $p_{fix} \approx 2|\Delta f|e^{-2N|\Delta f|}$ which decays exponentially with population size; it is hard for deleterious mutations to fixate in large populations, whereas fixation in smaller populations is facilitated by the statistical vagaries of survival and reproduction.

where $p_{null}(\rho)$ is the distribution of sites under no functional constraint, and $C$ is some constant chosen to normalize the distribution $p_{site}$. Note that the value of the fitness $f(\epsilon)$ matters here only up to an overall additive constant.

Berg and von Hippel's formula, Eq. 1.26, is exactly retrieved when $p_{null}(\rho)$ factors across the positions of $\rho$, $\epsilon(\rho)$ is properly represented by an energy matrix, and fitness $f$ is a linear function of energy $\epsilon$. While all these assumptions are suspect, the last is particularly so. Whatever the transcription factor does for the cell, its effect is going to depend on its binding site *occupancy* which, quite generally, will be very nonlinear in binding energy; in the simplest case, factor occupancy will be given by a fermi function, and more complicated scenarios involving multi-protein interactions are liable to produce even greater nonlinearities. In fact, we recently showed that binding sites for the *S. cerevisiae* transcription factor Abf1 appear to be under highly nonlinear fitness constraints [2].

Berg and von Hippel did not argue that fitness was linear in binding energy. Nor did they assume that bases at different positions within a binding site occur independently of one another. All they said was that the frequency with which each base occurs at a given position can be approximated by a thermal distribution in the energy contribution of that base, and that the different positions would experience roughly equal effective temperatures. This argument is perfectly valid, but it needs to be understood as an approximation. In particular, while this argument provides a way of inferring approximate models of a factor's sequence-dependent binding energy from a set of known sites, it does not allow one to infer $p_{site}(\rho)$ from knowledge of $\epsilon(\rho)$, any more than knowing the single-particle distributions of a coupled system allows one to infer the statistics of the entire system.

### 1.4.3 Stormo and Fields

In 1998, Stormo and Fields [71, 72] (building off of previous work by Heumann, Lapedes, and Stormo [73]) offered an alternative argument connecting sequence statistics to binding energy. Their argument can be summarized as follows: suppose that $N$ factor molecules, well described by an energy matrix, are put in solution with a random DNA polymer generated by the null model $\{q_{b,null}\}$. Given a set of known binding site sequences $\{\rho_i\}_{i=1}^N$, the probability that the $N$ transcription factor molecules, if all bound to the random DNA polymer, will be found at all $N$ of *these* sequences is

$$p(\{\rho_i\} \mid \{\epsilon_{bl}\}) = \frac{1}{C} \frac{\prod_i p_{null}(\rho_i) \exp\left[-\sum_{bl} \rho^i_{bl} \epsilon_{bl}\right]}{\left[\sum_\rho p_{null}(\rho) \exp[-\sum_{bl} \rho_{bl} \epsilon_{bl}]\right]^N} \tag{1.30}$$

where the sum in the denominator is taken over all sequences of length $L$ and the energy matrix elements $\{\epsilon_{bl}\}$ are given in units of $kT$.[28] Here, $C$ is a combinatorial factor which depends on how many of the sequences are identical or reverse-compliments of one another (e.g. $C = N!$ if all $N$ sequences are distinct).[29]

[28] This objective function does not appear is in Stormo and Fields's 1998 paper [71], but can be inferred from the paper of Heumann, Lapedes, and Stormo [73].

[29] This constant $C$ does not appear in [73].

Given the sequences $\{\rho_i\}$ of known binding sites, Stormo and Fields claimed that the best-fitting energy matrix $\{\epsilon_{bl}\}$ will maximize Eq. 1.30. If we further assume that $p_{null}(\rho)$ factors to the single-nucleotide distribution $\{q_{b,null}\}$, the best-fit energy matrix $\{\epsilon_{bl}\}$ can be found analytically:

$$\epsilon_{bl} = -\log \frac{q_{bl}}{q_{b,null}} \tag{1.31}$$

where $q_{bl} = c_{bl}/N$ is the *empirical* occurrence frequency of base $b$ in position $l$ within the set of sites $\{\rho_i\}$. Note that, unlike in Berg and von Hippel's result, there is no unknown proportionality constant; Stormo and Fields claim that this log ratio gives binding energies in units of $kT$.

Stormo and Field's paper [71] has since become a standard reference for the relationship between transcription-factor-DNA binding energy and the sequence statistics of binding sites. This result, in fact, is often interpreted as showing the equivalence of sequence statistics and binding energy. We argue, however, that Stormo and Field's formula, Eq. 1.31 is probably less valid than Berg and von Hippel's (Eq. 1.26) in most situations.

First, the objective function (Eq. 1.30) proposed by Stormo and Fields only applies in an artificial *in vitro* scenario. Certainly, the selection procedure they outline is not how a cell chooses which binding sites to include in its genome. This makes it hard to justify the Bayesian step

$$p(\{\epsilon_{bl}\} \mid \{\rho_i\}) \propto p(\{\rho_i\} \mid \{\epsilon_{bl}\}) p(\{\epsilon_{bl}\}) \tag{1.32}$$

implicit in their argument. Berg and von Hippel, on the other hand, do not posit such an objective function; rather, they assume binding sites will have maximum sequence entropy consistent with some functional constraint on binding energy. They then supplement with appropriate (and not unreasonable) approximations.

Furthermore, Berg and von Hippel claim that the sequences of known binding sites can be used to determine binding energy *only up to an unknown multiplicative constant* $\beta$, whereas Stormo and Fields claim that sequence statistics directly give binding energy in units of $kT$ – i.e. that $\beta = 1$. To be fair, it is hard to believe that maximizing the objective function proposed by Stormo and Fields would be wildly incorrect. In practice, a constant $\beta$ of order 1 is likely to work for most transcription factors, especially considering that differences between elements $\epsilon_{bl}$ at important positions $l$ within binding sites are typically $\sim 1 - 3kT$ [74]. Nevertheless, the value of $\beta$ presents a substantive difference between Eq. 1.26 and Eq. 1.31.

Figure 1.5 illustrates the relationship between Berg von Hippel binding energy predictions (with $\beta = 1$) and measured binding energies for two transcription factors: the Mnt repressor of the *Salmonella* phage P22, and the CRP protein of *E. coli*. In 1997, Fields, He, Al-Uzri, and Stormo used a quantitative version of Sanger sequencing to measure the binding energy of Mnt to all single-point mutants of the wild-type Mnt operator site. At the same time, the authors also performed a SELEX experiment with this protein, and constructed a Berg von Hippel energy matrix from the resulting binding sites. Fig. 1.5a plots these measured sequence-dependent binding energies

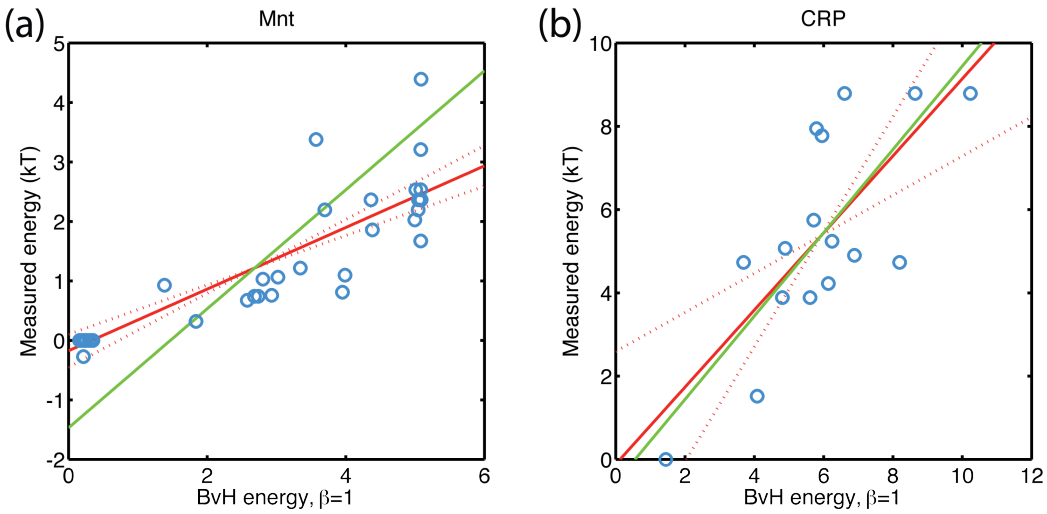


Figure 1.5: Measured binding energies vs. Berg-von-Hippel-predicted energies with $\beta = 1$. (a) DNA binding energies of the Mnt repressor were measured by Fields et al. [75]. The corresponding Berg von Hippel energy matrix was constructed from SELEX-selected binding sites (also from [75]) using a pseudocount of 1. Least squares regression gives a best estimate of $\beta = 1.9$ (red solid line) and a 95% confidence interval $1.6 < \beta < 2.4$ (red dotted lines). Clearly, a value of $\beta = 1$ (green line) is incompatible with these measurements. (b) DNA binding energies for CRP were measured by Gunasekera, Ebright, and Ebright [76]. The Berg von Hippel CRP energy matrix was constructed from the 49 sites listed in DPInteract [30] (extended to 26 bp, with the reverse-complement of each sequence added to the list, and using a pseudocount of 1). Least squares regression gives an optimal value of $\beta = 1.1$ (red solid line) and a 95% confidence interval $0.7 < \beta < 2.1$ (red dotted lines). This data appears to be impressively concordant with $\beta = 1$.

against their corresponding Berg-von-Hippel-predicted binding energies.[30] There is a significant correlation ($\rho = .86,\ P \ll 10^{-5}$) between these predicted and measured energies, but as shown in in the figure, these data are clearly inconsistent with $\beta = 1$.[31]

The situation is different for CRP, as shown in Fig. 1.5b. The Berg von Hippel energy matrix constructed from 49 experimentally verified binding sites in the *E. coli* genome [30] makes energy predictions which are plotted against the nirocellulose filter binding measurements of Gunasekera, Ebright and Ebright [76]. The correlation coefficient is $\rho = 0.74$ ($P = 9.3 \times 10^{-4}$), and the optimal value $\beta = 1.1$ matches well with the claim of Stormo and Fields that $\beta = 1$. Thus there do appear to be situations in which $\beta = 1$ is (surprisingly) accurate.

### 1.4.4 Moving forward

We conclude this section with the following sentiment: while the relation offered by Berg and von Hippel between the sequences of functional binding sites and the corresponding factor's sequence-dependent binding energy is simple, useful, and probably accurate in a number of cases, it is not a fundamental relationship between sequence statistics and binding energy. Indeed, a precise relationship between sequence statistics and binding energy requires assumptions about mutational processes, population dynamics, and the quantitative dependence of organism fitness on binding energy (e.g. [77, 69, 78, 2]). Having it in mind to understand how transcriptional regulation works in the cell – where factors bind and what they do – we will focus our attention on describing the energetics of protein-DNA interactions (as well as protein-protein interactions) relevant for transcriptional regulation. In particular we will model the sequence-dependent binding energy of transcription factors, typically employing energy matrix models as the natural starting point, and will not dwell any further on the concept of motifs.

This still leaves us with the problem presented by MatrixREUDCE: how should one fit models to $\sigma z$ data when experimental error processes are uncertain. We tackle this problem in the next chapter. In doing so we introduce *error-model-averaged (EMA) likelihood*, which can be used in conjunction with Metropolis Monte Carlo sampling to infer biophysical models of arbitrary form from $\sigma z$ data [1]. EMA likelihood is closely related to mutual information, and we show that, when the number $N$ of sequences probed is large, the most likely model will maximize the mutual information between its sequence-dependent predictions $x(\sigma;\theta)$ and the sequence-specific measurements $z$. But while mutual information estimation can be cumbersome, a simple analytic formula exists for EMA likelihood.

We then use Metropolis Monte Carlo sampling with EMA likelihood to infer energy matrix models for the *S. cerevisiae* transcription factor Abf1 from PBM data and ChIP-chip data. This method of inference allows us to derive error bars for all energy matrix elements. We find that our ChIP-chip and PBM analyses produce surprisingly precise models which largely agree between these two very different assays.

[30] A small random number was added to each predicted binding energy in order to separate overlapping data points.

[31] This fact was not noted by Fields et al. [75].

In the third chapter we use our newfound ability to properly analyze $\sigma z$ data to motivate a new assay, which we call Sort-Seq. Sort-Seq uses ultra-high-throughput sequencing to probe the detailed biophysical interactions at specific transcriptional regulatory sequences. We demonstrate Sort-Seq with a proof-of-principle experiment probing CRP binding to its site in the *lacZ* promoter. We also address further theoretical issues, and show that when $\sigma z$ data probes protein binding to one site per sequence, one cannot, by maximizing mutual information (or EMA likelihood), infer interaction energies in absolute thermal units. However, natural nonlinearities in the formulas of statistical mechanics *can* allow one to access absolute energy scales when $\sigma z$ data probes the binding of multiple proteins per sequence. This could potentially allow one to measure *in vivo* protein-DNA and protein-protein interaction energies, as well as changes in transcription factor concentration, simply by sorting cells and sequencing DNA.

## 1.5 Appendix: derivation of Berg and von Hippel's formula

Berg and von Hippel's derivation, as presented here, requires four assumptions.

1. The probability $p_{null}(\rho)$ of an *unconstrained* site factors across its positions:
$$p_{null}(\rho) = \prod_l \sum_b \rho_{bl} q_{b,null}. \tag{1.33}$$

2. A site's binding energy can be represented by an energy matrix,
$$\epsilon(\rho) = \sum_{bl} \rho_{bl} \epsilon_{bl}. \tag{1.34}$$

3. The distribution $p_{site}(\rho)$ of sites under functional constraint is maximally entropic for a some (unknown) distribution of binding energies, i.e. there is some function $F(\epsilon)$ such that
$$p_{site}(\rho) = p_{null}(\rho) \exp\left[-F(\epsilon(\rho))\right]. \tag{1.35}$$
However, we require no specific assumptions about how these sites evolve; i.e. how $F(\epsilon)$ comes about.

4. This function $F$ is roughly linear in the range of most sites' energies, i.e. for some constant $\beta$,
$$F'(\epsilon(\rho)) \approx \beta \quad \text{for most} \quad \rho \sim p_{site}(\rho). \tag{1.36}$$

With these assumptions, consider the probability $q_{bl}$ of a functional site $\rho \sim p_{site}(\rho)$ having base $b$ at position $l$,

$$q_{bl} = \sum_\rho \rho_{bl} p_{null}(\rho) \exp[-F(\epsilon(\rho))] \tag{1.37}$$
$$= \frac{1}{4} \sum_\rho p_{null}(\hat{\rho}) \exp[-F(\epsilon(\hat{\rho}))] \tag{1.38}$$

where we've used $\hat{\rho}$ to denote the sequence $\rho$ but with base $b$ at position $l$, i.e.

$$\hat{\rho}_{b'l'} = \begin{cases} \rho_{b'l'} & \text{if } \; l' \neq l \\ \delta_{bb'} & \text{if } \; l' = l \end{cases} . \tag{1.39}$$

From this we can rederive Eq. 1.25:

$$\begin{aligned} q_{bl} &= \frac{1}{4}\sum_{\rho} p_{null}(\rho)\frac{q_{b,null}}{\sum_c \rho_{cl} q_{c,null}} \exp\left\{-F\left(\epsilon(\rho) + \epsilon_{bl} - \sum_c \epsilon_{cl}\rho_{cl}\right)\right\} && (1.40) \\ &\approx \frac{1}{4}\sum_{\rho} \frac{p_{null}(\rho) q_{b,null}}{\sum_c \rho_{cl} q_{c,null}} \exp\left\{-F(\epsilon(\rho)) - F'(\epsilon(\rho))\left[\epsilon_{bl} - \sum_c \epsilon_{cl}\rho_{cl}\right]\right\} && (1.41) \\ &\approx q_{b,null} \exp\left[-\beta\epsilon_{bl}\right]\left[\sum_{\rho} \frac{p_{null}(\rho)}{4\sum_c \rho_{cl} q_{c,null}} \exp\left\{-F(\epsilon(\rho)) + \beta\sum_c \epsilon_{cl}\rho_{cl}\right\}\right] && (1.42) \\ &\approx \frac{q_{b,null}\exp\left[-\beta\epsilon_{bl}\right]}{\sum_c q_{c,null}\exp\left[-\beta\epsilon_{cl}\right]}. && (1.43) \end{aligned}$$

Assumption 4 was used to justify the Taylor expansion in Eq. 1.41 and the removal of $\exp[-\beta\epsilon_{bl}]$ from the sum in Eq. 1.42. As a result, the term in brackets in Eq. 1.42 is independent of $b$, and is thus fixed by the normalization requirement $\sum_b q_{bl} = 1$, giving Eq. 1.43. This allows us to solve for $\epsilon_{bl}$ (up to an unphysical $l$-dependent constant), giving Eq. 1.26.

It is interesting to note that, according to Eq. 1.29, the evolutionary model of Berg, Willmann, and Lässig [69] gives

$$\beta \approx 2Nf'(\langle\epsilon\rangle), \tag{1.44}$$

i.e. Berg and von Hippel's inverse temperature $\beta$ is roughly twice the population size $N$ times the derivative of the fitness $f$ at the mean energy $\langle\epsilon\rangle$ of functional sites. This gives an explicit situation in which $\beta$ can differ substantially from 1.

# Chapter 2

# Analysis of $\sigma z$ data

## 2.1 Theoretical considerations

We now address a basic problem in sequence analysis: given a $\sigma z$ data set – one consisting of sequences $\{\sigma_i\}_{i=1}^N$, with each sequence $\sigma_i$ assigned a measurement of functionality $z_i$ – how should we infer models of the biophysics underlying sequence function?

$$\sigma \xrightarrow{\text{experiment}} z \tag{2.1}$$

For example, PBM data consists of a set of microarray probe sequences, with each sequence assigned a corresponding fluorescence. How do we infer from this data a model of the assayed protein's sequence-dependent binding energy?

### 2.1.1 Mutual information

To tackle this problem, we make the assumption that each measurement $z$ results from a noisy measurement of some underlying biophysical quantity, $x^*$, which is completely specified by the sequence $\sigma$. We wish to model the sequence-dependent function $x^*(\sigma)$, so we hypothesize a specific function $x(\sigma;\theta)$ specified by parameters $\theta$. This sets up the following Markov chain,

$$x \xleftarrow{x(\sigma;\theta)} \sigma \xrightarrow{x^*(\sigma)} x^* \xrightarrow{p(z|x^*)} z. \tag{2.2}$$

The arrow mapping $\sigma$ to $x$ follows from the fact that, whichever function and parameters $x(\sigma;\theta)$ we choose, the value $x$ assigned to each sequence will depend solely on the sequence and not independently on $x^*$ or $z$. The arrows connecting $\sigma$ to $x^*$ to $z$ result from our assumption of an intermediate variable $x^*(\sigma)$ of which $z$ is a noisy measurement. Note that, while $x^*(\sigma)$ is assumed to be a deterministic function, $p(z|x^*)$ – the experimental *error model* – can be any probabilistic mapping. The mere fact that Eq. 2.2 is a Markov chain implies, by the data processing inequality (see [59]), that

$$I(z;x) \leq I(z;\sigma) = I(z;x^*). \tag{2.3}$$

Here, $I(a;b)$ denotes the mutual information between any two variables $a$ and $b$, and is given by

$$I(a;b) = \int da\ db\ p(a,b) \log \frac{p(a,b)}{p(a)p(b)} \tag{2.4}$$

where $p(a,b)$ is the joint distribution of $a$ and $b$, and $p(a)$ and $p(b)$ are the corresponding marginal distributions. If $N$ is large enough, the variables $z$ and $x$ can be quantized (if they aren't naturally discrete), and $I(z;x)$ can be estimated by

$$I(z;x) \quad = \quad \sum_{z,x} f(z,x) \log \frac{f(z,x)}{f(z)f(x)} \tag{2.5}$$

where $f(z;x)$ is the fraction of sequences in the $\sigma z$ data set assigned a measurement of value $z$ and a prediction of value $x$, and $f(z)$ and $f(x)$ are the corresponding marginal frequencies. $I(z;x^*)$ can be computed similarly, if one knows the function $x^*(\cdot)$. $I(z;\sigma)$, while similarly defined, can be harder to estimate from $\sigma z$ data, unless one has multiple measurements for each sequence $\sigma$.[1]

The inequality in Eq. 2.3 suggests that the *best* model of the biophysics underlying the measurements $z$ should maximize the mutual information $I(z;x)$ between model predictions and measurements. In particular, no model will exhibit greater mutual information than the correct model $x^*(\cdot)$. So given a hypothesized functional form $x(\cdot,\theta)$, one should, at the very least, choose parameters $\theta$ which maximize $I(z;x)$ (to within error bars). Note that doing so requires no assumptions about the experimental error model $p(z|x^*)$; all one needs is enough data to accurately estimate $I(z;x)$.

### 2.1.2 Likelihood

Maximizing mutual information, though, is not typically how one fits models to data; instead, one generally seeks to find model parameters which maximize the likelihood of the data, either directly or in the context of Bayesian inference.

The method of maximum likelihood invokes the following Markov chain,

$$\sigma \xrightarrow{x(\sigma;\theta)} x \xrightarrow{p(z|x)} z. \tag{2.6}$$

One assumes that $x(\sigma;\theta)$ correctly describes the underlying biophysics, then chooses the parameters $\theta$ which maximize the probability of the data, $p(\{z_i\} \,|\, \{\sigma_i\}, \theta)$. For Bayesian inference, one further assumes a prior distribution over possible parameters $p(\theta)$, and uses likelihood to compute a posterior distribution on $\theta$ given the data,

$$p(\theta|\, \{z_i\}, \{\sigma_i\}) \propto p(\{z_i\} \,|\, \{\sigma_i\}, \theta) p(\theta). \tag{2.7}$$

In either case, being able to accurately compute the likelihood $p(\{z_i\} \,|\, \{\sigma_i\}, \theta)$ is essential.

[1] Estimating $I(z;x)$ with appropriate error bars is not generally a trivial task. We will discuss this issue in more depth in chapter 3.

Assuming the variables $z$ and $x$ only take on discrete values, this likelihood can be rewritten as

$$p(\{z_i\} \mid \{\sigma_i\}, \theta) = \prod_{z,x} p(z|x)^{c_{zx}} = e^{N[I(z;x)-H(z)-D(f||p)]}, \tag{2.8}$$

where $x_i = x(\sigma_i, \theta)$, $p(z|x)$ is the experimental error model, and $c_{zx} = Nf(z,x)$ is the number of sequences $\sigma_i$ for which $z_i = z$ and $x_i = x$. In the second line, we've rewritten this likelihood in terms of $I(z;x)$, given in Eq. 2.5, as well as

$$H(z) = -\sum_{z,x} f(z) \log f(z), \tag{2.9}$$

$$D(f||p) = \sum_{z,x} f(z,x) \log \frac{f(z|x)}{p(z|x)}. \tag{2.10}$$

Here, $H(z)$ is the entropy of the measurements $\{z_i\}$, and does not depend on $\theta$. $D(f||p)$ is the Kullback Leibler divergence between the conditional frequencies $f(z|x) = f(z,x)/f(x)$ observed for parameters $\theta$, and the error model $p(z|x)$. Note that we cannot compute likelihood without assuming a specific error model $p(z|x)$ *a priori*.

Eq. 2.8 reveals an important relationship between mutual information and likelihood in the large $N$ limit; maximizing likelihood is equivalent to maximizing $I(z;x) - D(f||p)$.[2] By maximizing likelihood, one effectively tries to accomplish two goals simultaneously: finding the model which is most *informative* about the data, but whose predictions also best conform to one's prior expectations about experimental noise.

In the previous section, we showed that, in the large $N$ limit, the correct model will always always be maximally informative – it will maximize $I(z;x)$. According to Eq. 2.8, the correct model will also be the most likely if the conditional frequencies $f(z|x)$ it generates match the assumed error model $p(z|x)$; in this case $D(f||p) = 0$, so the correct model will minimize $D(f||p)$ and maximize $I(z;x)$ simultaneously. However, if one assumes an incorrect error model, then in searching for the most likely model one allows $I(z;x)$ to be decreased for a compensating decrease in $D(f||p)$. This is dangerous because the correct model will maximize $I(z;x)$ regardless of $p(z|x)$. Postulating error models is particularly problematic when analyzing data from assays such as PBM and ChIP-chip; such experiments often exhibit noise that cannot be accurately characterized *a priori*, since it might vary from protein to protein, from lab to lab, or even from day to day.

On the other hand, using likelihood as an objective function has certain advantages over using mutual information. Mutual information is not easy to estimate when data is limiting (an issue we will discuss in depth later on; see also [79, 80, 81]), whereas likelihood has a simple analytic form. Furthermore, Eq. 2.7 allows us to estimate a posterior distribution on model parameters given data, a distribution which can be sampled using computational methods like Metropolis Monte Carlo. Error bars on model parameters can then be computed from the sampled ensemble of parameters. Finally, Bayesian posteriors from multiple experiments can be used to refine model parameters simply by multiplying the resulting distributions together.

[2] We restrict this statement to the large $N$ limit because it is only in this limit that Eq. 2.5 accurately gives the mutual information between $z$ and $x$; Eq. 2.8, however, is valid for *all* $N$.

### 2.1.3 Error-model-averaged likelihood

Fortunately, we can circumvent the problem of having to assume a specific error model *a priori* by explicitly including the error model itself in the inference problem. By postulating an error model prior $P[p(z|x)]$, the likelihood of a given model $x(\cdot,\theta)$ is equal to the likelihood of the model *and* a particular error model $p(z|x)$, averaged over all possible error models. We thus define *error-model-averaged (EMA) likelihood* [1],

$$\begin{aligned} p_{ema}(\{z_i\} \mid \{\sigma_i\}, \theta) &= \int \mathcal{D}[p(z|x)] p(\{z_i\}, p(z|x) \mid \{\sigma_i\}, \theta) && (2.11)\\ &= \int \mathcal{D}[p(z|x)] P[p(z|x)] p(\{z_i\} \mid p(z|x), \{\sigma_i\}, \theta) && (2.12)\\ &= \int \mathcal{D}[p(z|x)] P[p(z|x)] \prod_{z,x} p(z|x)^{c_{zx}} && (2.13)\\ &= e^{N[I(z;x)-H(z)]} \int \mathcal{D}[p(z|x)] P[p(z|x)] e^{-ND(f||p)} && (2.14) \end{aligned}$$

Note that $\exp[NI(z;x)]$, which does not depend on $p(z|x)$, factors out of the integral while $\exp[-ND(f||p)]$ stays inside; when $N$ is large, only error models very close to $f(z|x)$ will contribute substantially to Eq. 2.14.

We are at liberty to choose a reasonable error model prior, $P[p(z|x)]$. Choosing a "flat" prior

$$P[p(z|x)] = \Gamma(m)^n \prod_x \delta\left(1 - \sum_z p(z|x)\right), \tag{2.15}$$

where $m$ is the number of possible $z$-values and $n$ is the number of possible $x$-values, we can evaluate EMA likelihood analytically:

$$p_{ema}(\{z_i\} \mid \{\sigma_i\}, \Theta) = \frac{\Gamma(m)^n \prod_{z,x} \Gamma(c_{zx}+1)}{\prod_x \Gamma(c_x+m)} = e^{N[I(z;x)-H(z)-\Delta]} \tag{2.16}$$

where $c_{zx} = Nf(z,x)$ and $c_x = Nf(x)$. In Eq. 2.16 the integral over all error models effectively replaces $D(f||p)$ by a correction term, $\Delta$, which is given by

$$\Delta = \frac{1}{N} \log\left[\frac{\prod_x \Gamma(m+c_x) c_x^{-c_x}}{\Gamma(m)^n \prod_{z,x} \Gamma(c_{zx}+1) c_{zx}^{-c_{zx}}}\right]. \tag{2.17}$$

We now prove that $\Delta$ becomes negligible (compared to $I(z;x)$) as $N \to \infty$.

$$\Delta = \frac{1}{N} \log\left[\frac{\prod_x (m-1+c_x)! c_x^{-c_x}}{(m-1)!^n \prod_{z,x} c_{zx}! c_{zx}^{-c_{zx}}}\right] = \frac{1}{N} \log\left[\prod_x \prod_{i=1}^{m-1} (i+c_x)\right] + \frac{\log AB}{N} \tag{2.18}$$

where $A = (m-1)!^{-n} \leq 1$ and

$$B = \left[\frac{\prod_x c_x!}{\prod_{z,x} c_{zx}!}\right] \prod_{z,x} \left(\frac{c_{zx}}{c_x}\right)^{c_{zx}} \tag{2.19}$$

is a term in the binomial expansion of

$$1 = \prod_x \left( \sum_z \frac{c_{zx}}{c_x} \right)^{c_x} , \tag{2.20}$$

and so is $\leq 1$ as well. The last term in Eq. 2.18 cannot, as a result, be positive, and so

$$\Delta \leq \frac{1}{N} \sum_{i=1}^{m-1} \sum_x \log(i + c_x) \leq \frac{(m-1)n}{N} \log[m + N]. \tag{2.21}$$

This is a loose bound, but it shows explicitly that $\Delta \to 0$ as $N \to \infty$, thereby proving that, in the large $N$ limit, maximizing EMA likelihood is equivalent to maximizing mutual information.

In the special case where all $c_{zx} \gg 1$, Stirling's approximation gives

$$N\Delta = \left[ \frac{(m-1)n}{2} \log \frac{N}{2\pi} - n \log \Gamma(m) \right] + \tag{2.22}$$

$$\left[ \frac{m-1}{2} \sum_x \log f(x) - \frac{1}{2} \sum_{zx} \log f(z|x) \right] + o\left(\frac{1}{N}\right). \tag{2.23}$$

The first term on the right grows as $\log N$ but is independent of $\theta$. The second term is $\theta$-dependent, but stays constant as $N$ grows. So in this case, $\exp(-N\Delta)$ essentially acts as a prior on $f(z, x)$. Note: using the fact that the logarithm is a concave function, we see that $\exp(-N\Delta)$ is weighted toward models for which some $f(x)$ are very small, and for which the conditional probabilities $f(z|x)$ are uniform over $z$. This effective prior therefore favors joint distributions $f(z, x)$ exhibiting little mutual information.

One can also compute EMA likelihood using a Dirichlet error model prior biased toward a specific error model $p^*(z|x)$. Using a set of nonnegative weights $\{W_x\}$, we define

$$P[p(z|x)] \equiv \mathcal{N} \prod_x \delta \left( 1 - \sum_z p(z|x) \right) \prod_z p(z|x)^{W_x p^*(z|x)}, \tag{2.24}$$

which requires the normalization constant

$$\mathcal{N} = \frac{\prod_x \Gamma(W_x + m)}{\prod_{z,x} \Gamma(W_x p^*(z|x) + 1)} . \tag{2.25}$$

Given the counts $\{c_{zx}\}$ we can again compute EMA likelihood explicitly, this time using using the Dirichlet prior in Eq. 2.24,

$$p_{ema}(\{z_i\} \mid \{\sigma_i\}, \theta) = \mathcal{N} \frac{\prod_{z,x} \Gamma(c_{zx} + W_x p^*(z|x) + 1)}{\prod_x \Gamma(c_x + W_x + m)} . \tag{2.26}$$

Up to a multiplicative constant, this is exactly what one gets by computing EMA likelihood with the uniform error model prior after "spiking" each count $c_{zx}$ with $W_x p^*(z|x)$ additional data points. Indeed, when $W_x = 0$ for all $x$, one recovers Eq. 2.16. On the other hand, fixing $\{c_{zx}\}$ while sending all $W_x \to \infty$ gives

$$p_{ema}(\{z_i\} \mid \{\sigma_i\}, \theta) \longrightarrow \prod_{z,x} p^*(z|x)^{c_{zx}} , \tag{2.27}$$

which is the likelihood one gets by assuming the error model $p(z|x) = p^*(z|x)$. Dirichlet error model priors thus provide a convenient way of interpolating between an analysis using a single, well defined error model and an analysis that is completely agnostic about error models. Such an interpolation, however, may not be necessary in practice; later in this chapter we will show that the agnostic approach can lead to a very meaningful analysis when enough data are available.

Again, $\Delta \to 0$ as $N \to \infty$ for the Dirichlet prior in Eq. 2.24. In fact, it seems pretty clear that $\Delta \to 0$ as $N \to \infty$ if $P[p(z|x)]$ is continuous and nonzero almost everywhere on the space of properly normalized error models (though we do not provide a proof here).

Before moving on to applications of EMA likelihood, we note that the bias of $\exp[-N\Delta]$ toward uninformative models might be counteracted by using an error model prior consisting of a mixture of Dirichlet priors. A similar strategy was used by [81] (a work we will detail later) for estimating entropy, and seems to work quite well. Doing so would require the evaluation of a numeric integral when computing EMA likelihood, but might make EMA likelihood useful on smaller amounts of data. It might also help to overcome some of the problems with EMA likelihood we will encounter in chapter 3.

### 2.1.4 The quantization problem

One salient issue to address before moving on is how to properly deal with continuous model predictions $x(\sigma;\theta)$. In computing EMA likelihood, we are free to quantize $z$ however we like; indeed, the quantization of $z$ can be thought of as part of the experiment. We are not generally free, however, to quantize an $x$ which is naturally continuous. The reason is that any quantization $x \to x_q$ will split the Markov chain (Eq. 2.6) used in the likelihood calculation, giving

$$\begin{tikzcd} \sigma \arrow[r] \arrow[dr] & x_q \arrow[r] & z \\ & x - x_q \arrow[ur] & \end{tikzcd} \quad . \tag{2.28}$$

Computing EMA likelihood using $x_q$ requires us to ignore the $x - x_q$ branch of this graph, and assume that

$$p(z|\sigma,\theta) = p(z|x) = p(z|x_q, x - x_q) = p(z|x_q). \tag{2.29}$$

The last equality is not generally true, even when $x$ correctly describes the underlying biophysics.

The quantization of $x$ should also cause us to reflect on the applicability of our error model prior $P[p(z|x)]$. In Eq. 2.15 and 2.24, we assume no correlation between the distributions $p(z|x)$ for neighboring values of $x$. This will effectively reduce our sensitivity to the data; more data will be needed to constrain the error model than

if we had assumed an error model prior that incorporated a notion of smoothness in the $x$ coordinate.

Unfortunately, this dissertation does not develop the machinery needed to deal properly with this problem. We briefly note, however, that a field-theoretic strategy, similar to that of Bialek, Callan, and Strong [82] might prove fruitful. Writing the error model as $p(z|x) = \exp[-\phi_z(x)]$, i.e. in terms of a set of continuous fields $\{\phi_z\}$, a seemingly reasonable error model prior is

$$P[\{\phi_z\}] \propto \int \mathcal{D}[\lambda(x)] \exp\left\{-\frac{l}{2}\int dx \sum_z (\partial\phi_z)^2 + i\int dx\ \lambda(x)\left[1-\sum_z e^{-\phi_z(x)}\right]\right\} \tag{2.30}$$

where $l$ is a length scale below which fluctuations are severely damped, and $\lambda(x)$ is a Lagrange multiplier field, enforcing the normalization $\sum_z p(z|x) = 1$ at every $x$. This error-model prior defines a field theory, in which EMA likelihood is an $N$-point function of the exponentiated fields $\{\phi_z\}$:

$$\begin{aligned} p_{ema}(\{z_i\} \,|\, \{\sigma_i\}, \theta) &= \langle \prod_i \exp[-\phi_{z_i}(x_i)]\rangle && (2.31)\\ &= \int \mathcal{D}[\phi_z] P[\{\phi_z\}] \prod_i \exp\{-\phi_{z_i}(x_i)\} && (2.32)\\ &= \frac{1}{Z}\int \mathcal{D}[\phi_z, \lambda] \exp\{-S[\{\phi_z\}, \lambda]\} && (2.33)\end{aligned}$$

where, scaling $x$ so that $l = 1$,

$$S[\{\phi_z\}, \lambda] = \frac{1}{2}\int dx \sum_z (\partial\phi_z)^2 + \sum_i \phi_{z_i}(x_i) + i\int dx\ \lambda(x)\left[1-\sum_z e^{-\phi_z(x)}\right]. \tag{2.34}$$

The equations of motion for the classical path, after solving for $\lambda(x)$, are

$$\partial^2\phi_z = \sum_{z'} \frac{e^{-\phi_z-\phi_{z'}}}{\sum_{z''} e^{-2\phi_{z''}}}(\partial\phi_{z'})^2 + \sum_i \delta(x-x_i)\left[\delta_{zz_i} - \frac{e^{-\phi_z-\phi_{z_i}}}{\sum_{z''} e^{-2\phi_{z''}}}\right]. \tag{2.35}$$

For a semiclassical computation of EMA likelihood, these equations for $\{\phi_z\}$ must be integrated numerically (as Nemenman and Bialek did, see [83]), after which one still must compute a functional determinant. While this field theory formulation of EMA likelihood is similar to the distribution inference method of Bialek, Callan, and Strong, the required functional determinant appears to be substantially more difficult to compute.

### 2.1.5 Sampling model parameters

Given a model $x(\cdot;\theta)$ and a $\sigma z$ data set, EMA likelihood (together with an assumed prior $p(\theta)$ on parameter space) allows us to sample model parameters $\theta$ from the posterior distribution,

$$p(\theta|\,\{\sigma_i\}, \{z_i\}) \propto p_{ema}(\{z_i\} \,|\, \{\sigma_i\}, \theta)p(\theta). \tag{2.36}$$

A useful computational method for doing this is Metropolis Monte Carlo (MMC).[3] Starting from an initial choice of model parameters, $\theta_0$, MMC stochastically wanders from parameter set $\theta_{t-1}$ to parameter set $\theta_t$ in such a way that the resulting ensemble $\Theta = \{\theta_1, \theta_2, \ldots, \theta_T\}$ of parameter sets is distributed according to a desired distribution (in our case, Eq. 2.36). The expected value of any $\theta$-dependent quantity $F(\theta)$, given the observations $\{z_i\}$ and sequences $\{\sigma_i\}$, can then be estimated by the ensemble average

$$\langle F \rangle \equiv \int d\theta \; F(\theta) p(\theta | \{z_i\}, \{\sigma_i\}) \approx \frac{1}{T} \sum_{t=1}^{T} F(\theta_t), \tag{2.37}$$

More specifically, MMC sampling of $\Theta$ is accomplished as follows. In step $t$ of the algorithm, MMC randomly perturbs the parameter set $\theta_{t-1}$ to generate a new parameter set $\tilde{\theta}_t$. MMC then sets $\theta_t = \tilde{\theta}_t$ with probability

$$\min\left(1, \frac{p(\tilde{\theta}_t | \{z_i\}, \{\sigma_i\})}{p(\theta_{t-1} | \{z_i\}, \{\sigma_i\})}\right) \tag{2.38}$$

Otherwise the algorithm chooses $\theta_t = \theta_{t-1}$. Note that we do not need to know the constant of proportionality in Eq. 2.36 to perform this sampling. We also have a great deal of freedom in choosing the perturbations $\theta \to \tilde{\theta}$, so long as the probability of this perturbation equals that of $\tilde{\theta} \to \theta$.

### 2.1.6 $\chi^2$ consistency test

Suppose we have multiple $\sigma z$ data sets probing the same biophysical system. While we do note expect $p(\theta | \{\sigma_i\}, \{z_i\})$ to be the same for these different data sets, we do expect that there should be some choice of parameters consistent with both posterior distributions. The existence of such a $\theta$ provides an assessment of the consistency of two such data sets.

We perform this assessment of consistency between $K$ data sets as follows. Suppose we have well sampled model ensembles $\Theta_1, \ldots, \Theta_K$, generated by MMC runs on $K$ different data sets. Let $\mu_{\alpha k}$ and $\sigma^2_{\alpha k}$ denote the mean and variance of the model parameter $\theta_\alpha \in \theta$ appearing in the ensemble $\Theta_k$. Let us also assume, for these purposes, that the distributions of all such parameters are gaussian. The value $\theta^*_\alpha$ for $\theta_\alpha$ that maximizes the joint likelihood over all $K$ ensembles is then the value that minimizes the $\chi^2$ statistic

$$\chi^2_\alpha = \sum_{k=1}^{K} \frac{(\theta^*_\alpha - \mu_{\alpha k})^2}{\sigma^2_{\alpha k}}, \tag{2.39}$$

which is given by the weighted average of means,

$$\theta^*_\alpha = \frac{\sum_k \mu_{\alpha k}/\sigma^2_{\alpha k}}{\sum_k 1/\sigma^2_{\alpha k}}. \tag{2.40}$$

[3] This is commonly referred to as Markov chain Monte Carlo (MCMC; e.g. by [1]), though this latter term actually has a more general connotation.

This is the one solution most consistent with all of the ensembles $\Theta_k$, but it may not be a good solution: if $\chi^2_\alpha$ is too large, then $\theta^*_\alpha$ is improbable according to at least some of the $\Theta_k$. For all ensembles to be consistent with respect to parameter $\theta_\alpha$, the minimized $\chi^2_\alpha$ should not be much larger than its expected value in the $\chi^2$ distribution with $K$ degrees of freedom. The p-value, or the probability of finding values $> \chi^2_\alpha$, therefore provides a convenient parameter-by-parameter assessment of the consistency of multiple MMC runs. Since the gaussian assumption is generally not accurate, these p-values should not be interpreted too literally. They do, however, provide a useful and easy-to-compute diagnostic.

## 2.2 Analysis of PBM data

We first apply EMA likelihood to the PBM data of Mukherjee et al. [16] for the yeast transcription factor Abf1. In this assay, epitope-tagged factors were bound directly to dsDNA spotted on a glass microarray and then visualized with fluorescent antibodies. The fluorescent intensity observed for each of the $\sim 6000$ microarray spots (representing virtually all the intergenic regions of *S. cerevisiae*) was then normalized by the amount of DNA in each spot. After averaging the data over replicates and further processing, Mukherjee et al. reported log intensity ratios (LIRs) for $N = 5812$ of these intergenic sequences. To analyze these data, we partitioned the 5812 fluorescence intensities into $m = 292$ bins, resulting in $\sim 20$ values per bin.[4] The resulting set of intergenic sequences $\{\sigma_i\}_{i=1}^N$ and corresponding bin numbers $\{z_i\}$ constituted the $\sigma z$ data input to our analysis.

### 2.2.1 Modeling factor binding to PBMs

From this $\sigma z$ data we sought to infer energy matrix models for Abf1's sequence-dependent binding energy. We note that, while simplistic, the assumption that each nucleotide contributes independently to overall binding energy has proven surprisingly accurate for a number of factors [84, 85, 86], though there are cases where this is known to be a bad approximation [87, 88].

Because such a model $x(\sigma;\theta)$ assigns energies to each of the $\sim 10^3$ possible overlapping sites in each intergenic region $\sigma_i$, we need some way of collapsing all these energies into a discrete set of values $\{x_i\}$, which, if the energy matrix model is accurate, will largely account for our experimental observations. Here we take the simplest (and crudest) approach, declaring a sequence $\sigma_i$ to be *bound* ($x_i = 1$) if it contains at least one site with energy below some binding threshold $\mu$, and otherwise declare it to be *not bound* ($x_i = 0$). In our analysis we set this threshold to unity, thus fixing the overall scale of the energy matrix elements. While this hard energy cutoff is only a crude approximation to the more complicated physics of binding site occupancy, we find that it can account for real data surprisingly well. It also greatly speeds the necessary computations. As described in the first chapter, we also fix the

[4] Our choice of 20 values per bin is, of course, arbitrary, but as we will show, the results of our analysis depend only weakly on this choice.

smallest element $\epsilon_{bl}$ in each column $l$ of the energy matrix to zero. By also setting $\mu = 1$, we fix all of the degrees of freedom in our model to which EMA likelihood is insensitive. We note that, while fixing the smallest element in each column pins down an unphysical *gauge* degree of freedom, setting $\mu = 1$ artificially fixes a very meaningful energy scale. This energy scale is one example of a *diffeomorphic mode*, the theoretical nature of which is discussed in the next chapter.

### 2.2.2 MMC sampling

In order to use MMC to sample model parameters, one must first define a prior distribution $p(\theta)$ on the space of models $\Omega$. This choice of prior is largely arbitrary, so to speed up computations we choose to define $p(\theta)$ in terms of the projection of a uniform prior on the space of models $\tilde{\Omega}$ hosting parameters $\tilde{\theta}$ consisting of energy matrix elements $\{\tilde{\epsilon}_{bl}\}$ – which can take on values anywhere from 0 to 1 independent of each other – as well as an energy cutoff $\tilde{\mu} \in [0, L]$. Any such $\tilde{\theta} = (\{\tilde{\epsilon}_{bl}\}, \tilde{\mu})$ that makes nontrivial predictions can be projected to a normalized model $\theta = \{\epsilon_{bl}\} \in \Omega$, having an energy cutoff of 1 and matrix elements satisfying $\min_b(\epsilon_{bl}) = 0$, without changing the predictions $\{x_i\}$. This is done by performing the following affine transformation on energy matrix elements:

$$\epsilon_{bl} = \frac{\tilde{\epsilon}_{bl} - \min_c \tilde{\epsilon}_{cl}}{\tilde{\mu} - \sum_{l'} \min_c \tilde{\epsilon}_{cl'}}. \tag{2.41}$$

The advantage of doing MMC on $\tilde{\Omega}$ instead of $\Omega$ is the added ease with which the algorithm can explore parameter space. On the other hand, it is convenient to work in $\Omega$ when analyzing MMC results because $\Omega$ lacks degrees of freedom which EMA likelihood cannot pin down. This forces models which make similar predictions to have similar parameters.

In each MMC step $t$, we applied one of the following random perturbations to parameters $\tilde{\theta}_{t-1} \in \tilde{\Omega}$:

1. Adding a small normally distributed number to the energy cutoff $\tilde{\mu}$.
2. Adding a small normally distributed number to one of the matrix elements $\tilde{\epsilon}_{bl}$.
3. Adding the same small normally distributed number to the energy cutoff $\tilde{\mu}$ and all energy matrix elements $\tilde{\epsilon}_{bl}$ in some specific column $l$.

Such perturbations were rejected if they violated the bounds on $\{\tilde{\epsilon_{bl}}\}$ or $\tilde{\mu}$. Otherwise they were accepted with the probability shown in Eq. 2.38. We note that by restricting matrix perturbations of type 2 and 3 to two specific columns for a large number of consecutive steps, then choosing two different columns to focus on, etc., we were able to implement an indexing trick that greatly sped up the MMC sampling algorithm; this trick does not violate the conditions necessary for proper MMC sampling. In this way we built a large sample $\tilde{\Theta} \subset \tilde{\Omega}$ of properly distributed models. These models were then projected onto $\Omega$, giving a set $\Theta = \{\theta_1, \theta_2, \ldots, \theta_T\}$ of $T$ properly normalized energy matrices.

For Abf1 we posited a $4 \times 20$ energy matrix. We then used multiple runs of the above algorithm to sample over $4 \times 10^7$ energy matrices, of which $4 \times 10^4$ were kept for analysis. We denote this energy matrix ensemble by $\Theta_{PBM}$. Appropriate tests were used to verify MMC convergence; these are detailed in the appendix to this chapter.

### 2.2.3 Results

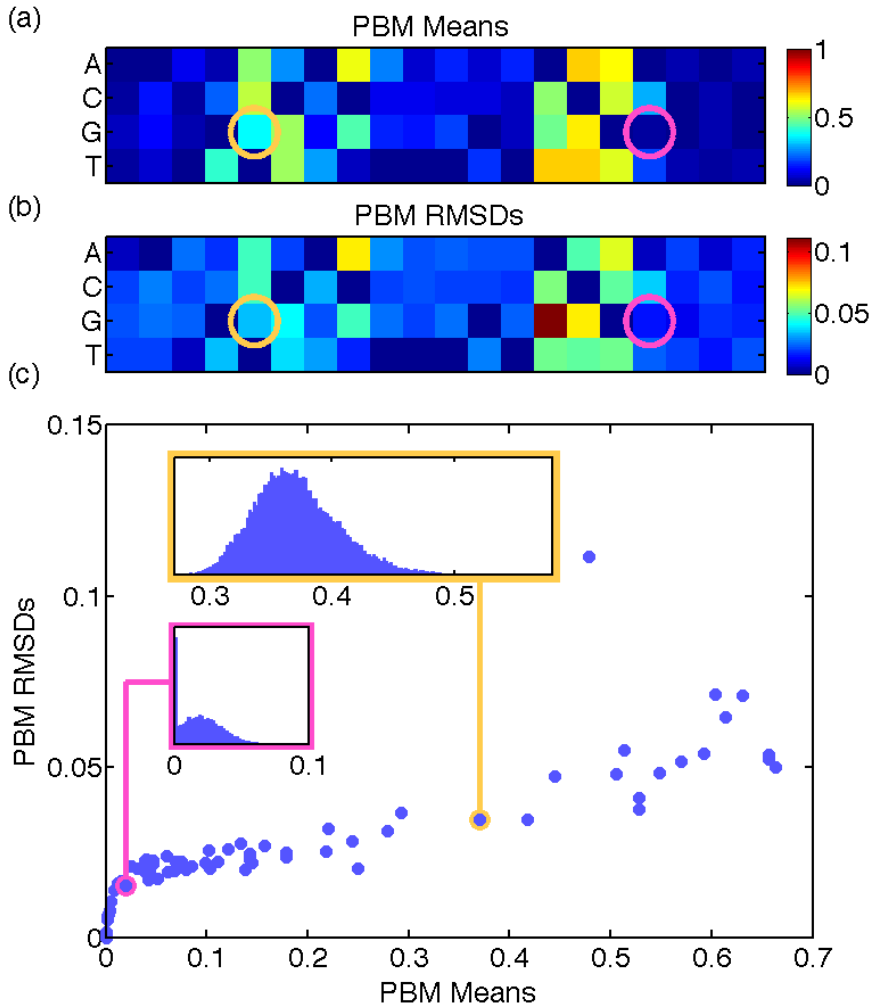


Figure 2.1: PBM data determines Abf1 energy matrix elements with surprisingly high precision. (a,b) Mean and RMSD of energy matrix elements over the $\Theta_{PBM}$ ensemble of Abf1 matrices. All-blue columns contribute little to DNA-binding specificity. Overall, these matrices match the known Abf1 motif RTCRYNNNNNACG well. (c) Scatter plot of matrix element means versus RMSDs: all RMSDs are small compared to the binding cutoff of 1. Insets in (c) display the marginal distributions of two representative matrix elements, circled in corresponding color in panels (a,b).

Our results are generally in line with the qualitative IUPAC sequence RTCRYNN-NNNACG known to describe Abf1 binding sites [45].[5] However, inspection of $\Theta_{PBM}$ shows that the parameters of the Abf1 energy matrix are determined with remarkable precision. The mean and root mean square deviation (RMSD) of each matrix element across all models in $\Theta_{PBM}$ are shown in Figs. 2.1a and 2.1b. A scatter plot of the same data is shown in Fig. 2.1c along with the full $\Theta_{PBM}$ distribution for two representative matrix elements. Matrix elements that differ significantly from zero generally have uncertainties much smaller than their means, and their distributions have a roughly gaussian shape (upper inset, Fig. 2.1c). Other elements are consistently assigned the lowest energy in their respective columns, and are thus set by our normalization scheme to be precisely zero in much of the ensemble (e.g. the lower inset). Even matrix elements in the center of the binding site, in positions which contribute little to overall DNA-binding specificity, are determined with a degree of precision which makes it possible to see meaningful structure.

To test for over-fitting in $\Theta_{PBM}$, we divided Mukherjee et al.'s Abf1 data into two randomly chosen halves: A and B. For each half of the data, we partitioned the probed sequences into equipopulated $z$-bins containing 20 sequences each and ran separate MMC samplings. The mean matrix elements in the two resulting ensembles, $\Theta_{PBM,A}$ and $\Theta_{PBM,B}$ are shown in Figs. 2.2a and 2.2b. Clearly these means are very similar despite being inferred from two completely disjoint sets of data. Even much of the fine structure in the low specificity regions of the binding site is similar. That these ensembles are quantitatively consistent is verified in Fig. 2.2c, which shows the p-values resulting from the $\chi^2$ consistency test discussed in the previous section. These p-values reveal no significant discrepancy between the two parameter distributions for any of the matrix elements, arguing strongly against overfitting.

We also tested whether our results were sensitive to how we binned the data. Figs. 2.3a and 2.3b show the matrix element means deduced from MMC runs performed with 20 and 50 sequences per $z$-bin, respectively. Again, these means are very similar,

[5] IUPAC (International Union of Pure and Applied Chemistry) sequences are nucleotide sequences specified by degenerate IUPAC symbols:

| symbol | nucleotides | description |
|---|---|---|
| A | A | **a**denine |
| C | C | **c**ytosine |
| G | G | **g**uanine |
| T | T | **t**hymine |
| R | A or G | pu**r**ine |
| Y | C or T | p**y**rimidine |
| M | A or C | a**m**ino |
| K | G or T | **k**eto |
| W | A or T | **w**eak |
| S | C or G | **s**trong |
| B | C or G or T | not A |
| D | A or G or T | not C |
| H | A or C or T | not G |
| V | A or C or G | not T |
| N | A or C or G or T | a**n**y |

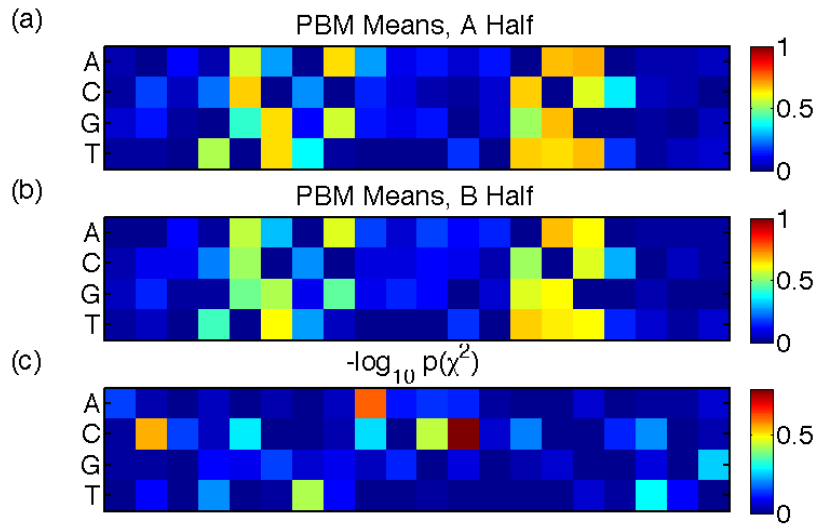


Figure 2.2: EMA Likelihood inference does not over-fit Mukherjee et al.'s Abf1 PBM data. (a) and (b) show the mean matrix elements determined by MMC sampling using two disjoint halves of the data. (c) Corresponding $\chi^2$ consistency p-values. No elements have Bonferroni corrected p-values $< .05$.

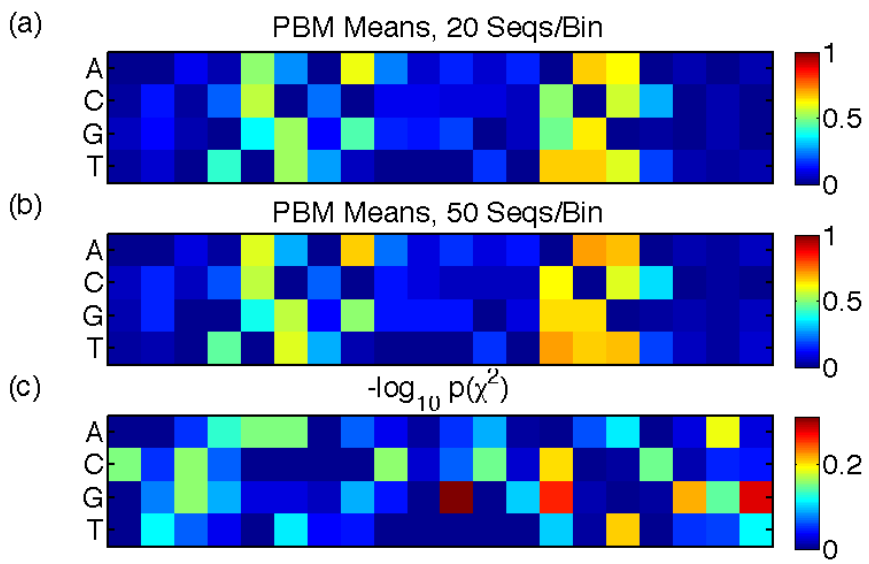


Figure 2.3: Ensemble means for matrix elements inferred from Abf1 PBM data using (a) 20 and (b) 50 sequences per $z$-bin. (c) $\chi^2$ consistency p-values for matrix elements between these two ensembles. No elements have Bonferroni corrected p-values $< .05$.

and the $\chi^2$ consistency p-values shown in Fig. 2.3c confirm the absence of significant discrepancies in any of the matrix elements. However, we note that sampling with MMC using 50 sequences per $z$-bin was much more difficult, and multiple MMC ensembles had to be collated to obtain stable estimates of the posterior distribution.

We further tested the sensitivity of our predictions to the choice of matrix width. MMC runs were performed on Mukherjee et al.'s Abf1 PBM data using matrix widths ranging from 14 to 26; all enough to encompass the primary Abf1 binding site. The MMC ensembles for the shorter matrices tended to have larger matrix elements, but this is expected because the lack of additional positive energy contributions in the flanking bases should result in a lower energy cutoff; when this cutoff is normalized to 1, matrix elements across the entire energy matrix become artificially large. When checking these MMC ensembles for consistency, we therefore allowed an arbitrary rescaling for each. The resulting rescaled energy matrix means from each ensemble are shown in Fig. 2.4. $\chi^2$ p-values for these seven different ensembles are shown at the top (only for matrix elements shared by all MMC ensembles) and reveal no significant discrepancies.

The model ensemble $\Theta_{PBM}$ provides a direct way of predicting putative binding sites. For any 20 bp sequence of DNA, we can determine the fraction of models $\theta$ in $\Theta_{PBM}$ which *hit* that site (i.e. assign it an energy below 1). Fig. 2.5a histograms this *hit fraction* (HF) for every possible 20 bp site in the intergenic DNA of *S. cerevisiae* (as defined in Kellis et al. [89]). This plot shows that the distribution of HFs is strongly bimodal (1469 sites have HF $> 50\%$, while 1182 of these also have HF $> 90\%$). We adopt HF $> 50\%$ as a plausible criterion for predicting a site to be *bound*.

Fig. 2.5b plots the mean fraction of sequences in each $z$-bin declared to be bound by models in $\Theta_{PBM}$ against the mean LIR of those sequences. Error bars show the variation in these predictions across different models in $\Theta_{PBM}$. A histogram of the actual measured LIRs is shown in the background. The most striking feature of this plot is the sigmoidal relationship between model predictions and measured LIRs, showing a rapid transition from mostly *not bound* to mostly *bound* sequences at the beginning of the heavy tail of the LIR distribution. While this general shape is exactly as expected, this outcome is in no way predetermined: Eq. 2.16 places no *a priori* weight on models which make similar predictions for sequences in neighboring $z$-bins or that declare sequences with large LIR to be *bound*. The consistency between the shape of this scatter plot and our physical expectation is an independent confirmation of the validity of EMA likelihood.

The Abf1 models in $\Theta_{PBM}$ appear to account for the data to within Mukherjee et al.'s estimate of the experimental error. The green line in Fig. 2.5b indicates the LIR cut used by Mukherjee et al. to define *bound* regions. Of the 186 regions passing this cut, the experimenters estimated that $7-9\%$ were false positives. We find that 167 (89.8%) of these regions have HF $> 90\%$, 18 (9.7%) have HF $< 10\%$, and only 1 region has an intermediate HF. So while the energy matrix models we employ are simplistic, they appear to account for this PBM data about as well as one could hope for *any* model, regardless of sophistication.

Mukherjee et al. were obliged to adopt a stringent threshold in order to minimize false positives, and in the process they rejected an unknown number of bound se-

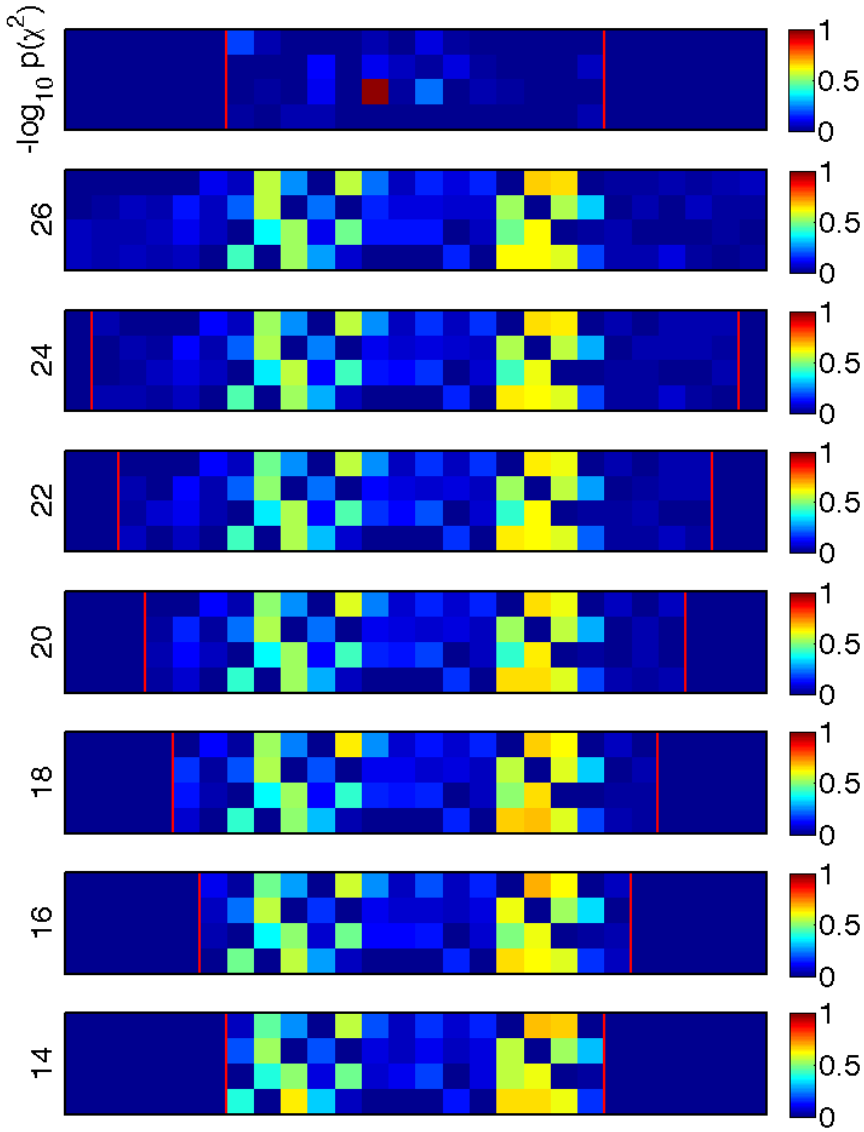


Figure 2.4: Ensemble means for matrix elements inferred from Abf1 PBM data using matrix widths ranging from 14 to 26 bp (indicated on the left side of each plot). Matrices in each MMC ensemble were rescaled to provide maximum consistency between the ensembles. The resulting $\chi^2$ consistency p-values for the central matrix elements common to all ensembles are shown at the top. No elements have Bonferroni corrected p-values $< .05$. Red lines delineate matrix boundaries. The rows of each matrix refer to bases A, C, G, T (top to bottom).

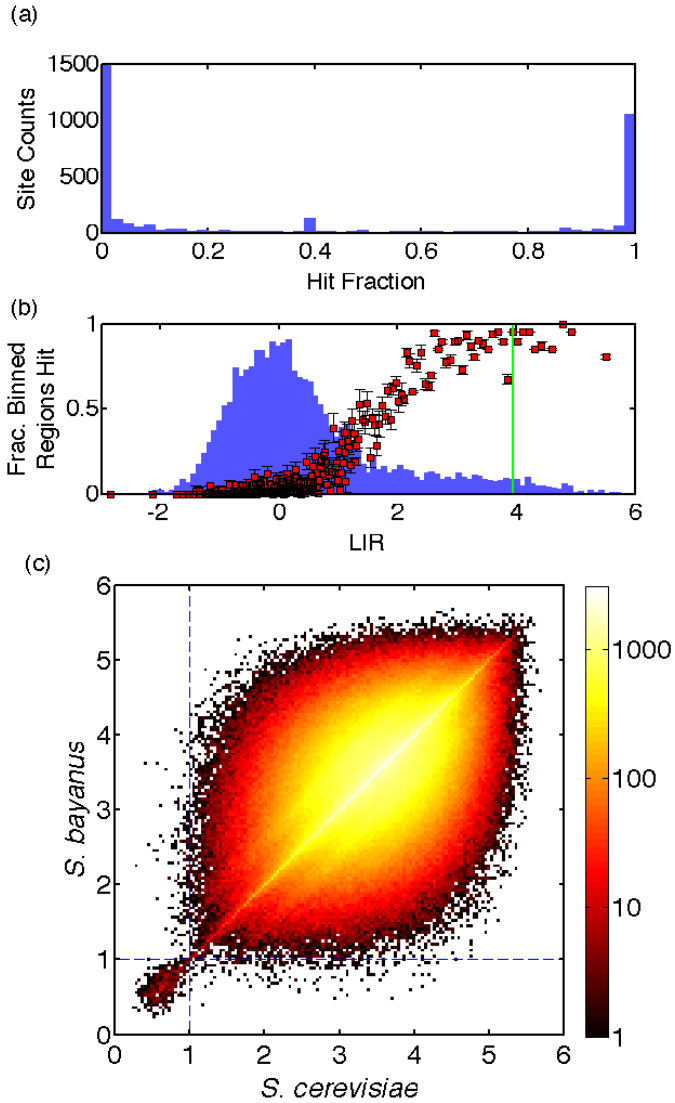


Figure 2.5: Predictions derived from the ensemble of Abf1 energy models. (a) Histogram showing the $\Theta_{PBM}$ hit fraction (HF) of all 20 bp sites in the intergenic DNA of *S. cerevisiae*. This distribution is strongly bimodal, efficiently separating *bound* sites (HF near 1) from *not bound* sites (HF near 0). The left-most bin contains the vast majority of sites and has been truncated for readability. (b) Mean fraction of regions in each $z$-bin declared bound by models in $\Theta_{PBM}$, plotted against the mean LIR of those regions. Error bars show the RMSD variation in this fraction from model to model. The distribution of experimentally-determined LIRs is shown in the background for reference. The green line is the threshold used by Mukherjee et al. to identify bound regions. (c) 2D histogram of mean energies assigned by $\Theta_{PBM}$ to ungapped orthologous intergenic site pairs in *S. cerevisiae* and *S. bayanus*. Sites lying below the binding cutoff (dashed lines) in one species are highly likely to lie below this cutoff in the other species.

quences. Our model-based approach, by contrast, can tease out regions which are likely to be bound, regardless of where they lie in the raw data distribution. In this case, we find 840 regions with HF $> 50\%$ and with LIR *below* the cut chosen by Mukherjee et al. In short, we find *many* more bound regions lying below the experimenters' threshold than lying above it!

This is a strong statement and one for which we would like independent confirmation. The best evidence would come from the direct *in vitro* measurement of large numbers of putative site binding energies. Energy measurements for Abf1 have been carried out for a small number of sites [90], but the results agree neither with our predictions nor with the analysis of Mukherjee et al. or Lee et al. [17]. It is possible that these measurements are in error, and we believe further *in vitro* studies are necessary to resolve this issue. Recently-developed high-throughput techniques for direct measurement of binding affinities [88] should be sufficient to provide data of the quality and scope needed to test these models in quantitative detail.

Although direct binding energy measurements are lacking, phylogenetic analysis provides alternative evidence for the quality of our predictions. Using the intergenic alignments of Kellis et al. [89], we identified all pairs of ungapped orthologous intergenic 20 bp sequences in *S. cerevisiae* and *S. bayanus*. We then computed the mean predicted binding energy (using $\Theta_{PBM}$) of each site in *S. cerevisiae*, as well as that of its ortholog. A scatter plot of the resulting energies (Fig. 2.5c) reveals a large and well-separated population of sites whose putative energies lie below 1 (i.e. are predicted to be *bound*) in both genomes: of the 676 *S. cerevisiae* sites with energy below 1, 501 (74%) have an *S. bayanus* ortholog whose energy also lies below 1. By contrast, sites with energy greater than 1 in either genome tend to have orthologs with highly randomized energy. In short, the large majority of sites with orthologs predicted to be bound by our models have strongly conserved energy. The fact that our models are found directly from *in vitro* binding data provides a compelling case that they also describe the free energy of Abf1-DNA binding *in vivo*, and that this binding energy plays a major role in determining which sites have biological function.

## 2.3 Analysis of ChIP-chip data

Next we performed a similar analysis of Lee et al.'s ChIP-chip data [17] to determine whether they give a description of Abf1 consistent with that obtained from the PBM data. In these ChIP-chip experiments, transcription factors were cross-linked *in vivo* to their binding sites, after which factor-bound fragments of DNA were isolated, amplified, labeled, and hybridized in competition with reference DNA to a microarray containing all yeast intergenic regions. The enrichment observed in each microarray spot was characterized by an *X-statistic* based on the single array error model of Hughes et al. [91]. Assuming a gaussian distribution for these X-statistics in the absence of factor binding, Lee et al. reported an enrichment p-value for each region.

We assigned these probed sequences to $z$-bins (using 50 regions per $z$-bin instead of 20) according to their p-values (equivalently, according to their X-statistics). MMC sampling produced an ensemble $\Theta_{ChIP}$ of $4 \times 10^4$ matrix models. Note that, by

integrating over all possible error models, we avoided having to model the noise contributions from each individual step in the ChIP-chip protocol. We also avoided having to estimate *in vivo* contributions to the experimental noise, such as the fraction of valid binding sites likely to be obscured by chromatin.

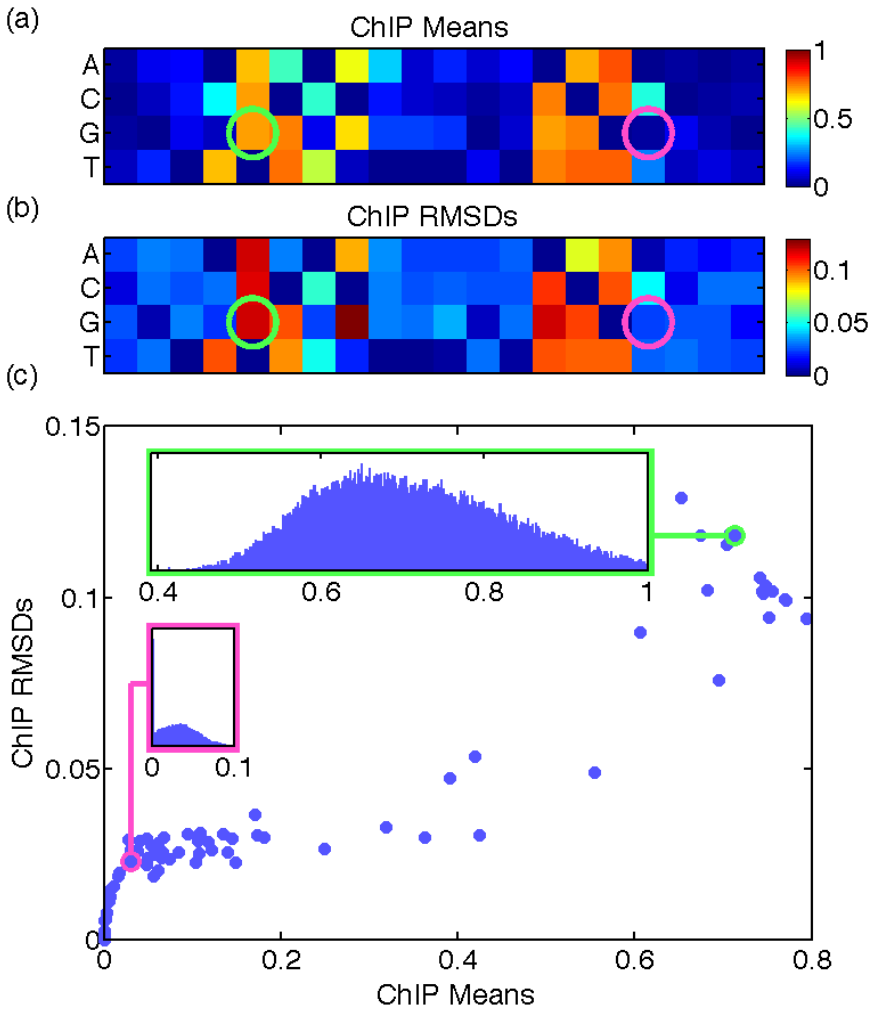


Figure 2.6: Energy matrix parameters inferred from Abf1 ChIP-chip data. (a) Mean values and (b) RMSDs of energy matrix elements across models in the MMC ensemble $\Theta_{ChIP}$. These are plotted against each other in (c). Insets display raw MMC histograms of the values obtained for the matrix elements circled in (a,b) and highlighted in (c). These are the same matrix elements highlighted in Fig. 2.1.

Our analysis of $\Theta_{ChIP}$ gave results similar to those of $\Theta_{PBM}$. Figs. 2.6 and 2.7 mirror Figs. 2.1 and 2.5 in the PBM analysis, and reveal very similar results. Fig. 2.8 tests overfitting $\Theta_{ChIP}$ (finding none), and Figs. 2.9 and 2.10 demonstrate the insensitivity of these results to the level of $z$-bin quantization and the choice of matrix width. We note that, for unknown reasons, MMC sampling was much easier on this ChIP-chip data than on PBM data when coarser $z$-bin quantizations were used. This allowed us to test quantizations ranging from 20 to 200 regions per $z$-bin.

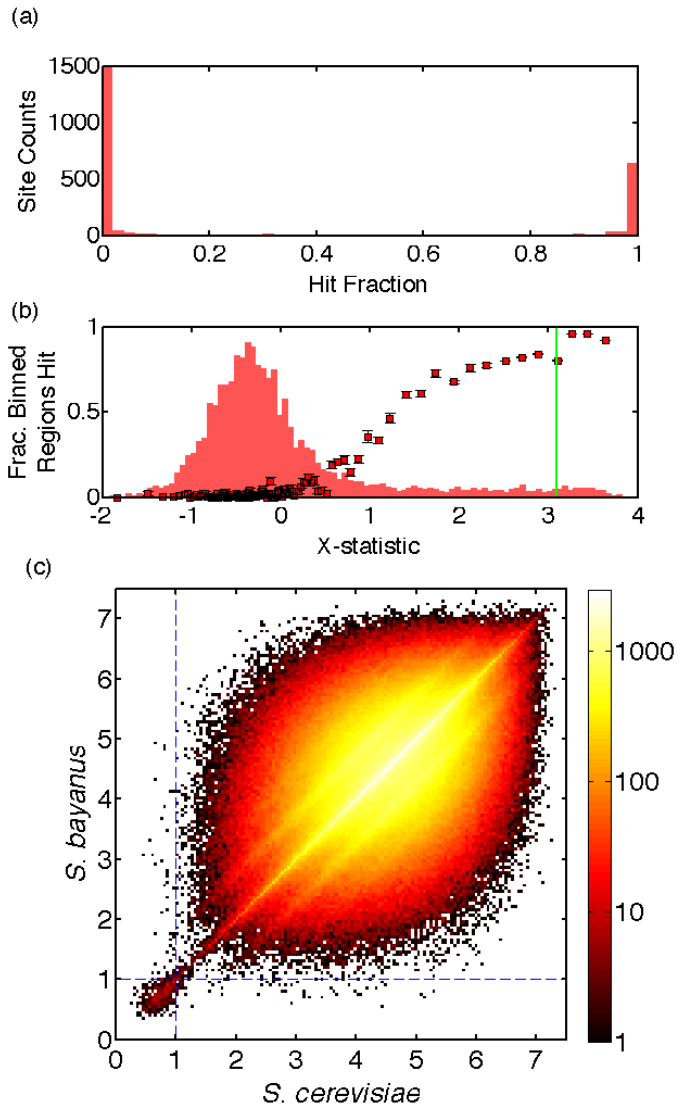


Figure 2.7: Abf1 model predictions using $\Theta_{ChIP}$. (a) Histogram of $\Theta_{ChIP}$ HFs for all 20 bp sites in the intergenic DNA of *S. cerevisiae*. The left-most bar of the histogram contains the vast majority of intergenic sites and has been truncated. (b) Mean X-statistic of regions within each $z$-bin plotted against the mean (red square) and RMSD (error bar) fraction classified as *bound* by models in $\Theta_{ChIP}$. The green line indicates the cutoff used by Lee et al. to delineate putatively bound regions. The X-statistics of all probed sequences are histogrammed in the background for reference. (c) This 2D histogram shows the mean putative energies assigned by $\Theta_{ChIP}$ to ungapped orthologous intergenic site pairs in *S. cerevisiae* and *S. bayanus*. Dashed lines indicate the energy cutoff of 1.

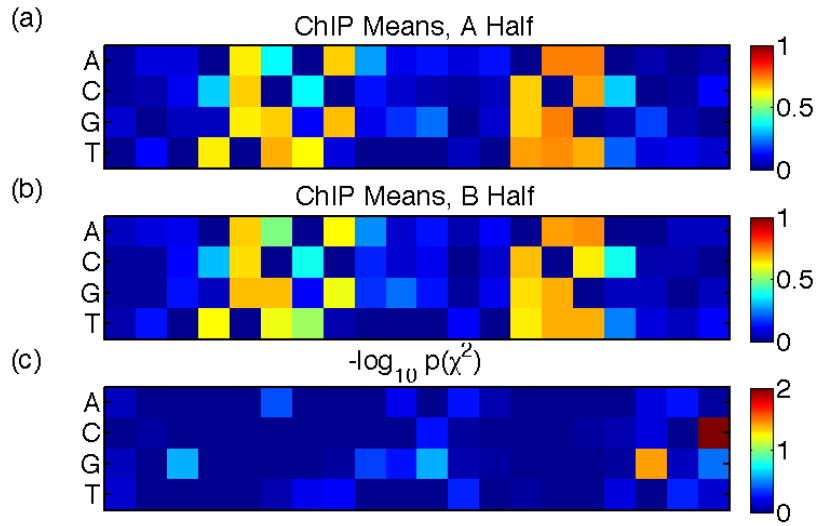


Figure 2.8: EMA Likelihood inference does not over-fit Lee et al.'s Abf1 ChIP-chip data. (a) and (b) show the mean matrix elements determined by MMC sampling using two disjoint halves of the data. (c) Corresponding $\chi^2$ consistency p-values. No elements have Bonferroni corrected p-values $< .05$.

## 2.4 ChIP-chip and PBM comparison

While the ChIP-chip and PBM results are quite similar, the energy matrix elements derived from $\Theta_{ChIP}$ are systematically larger than those of $\Theta_{PBM}$. Our procedure, though, produces energy matrices artificially scaled so that the energy cutoff is equal to 1. But differences between the energy cutoffs measured for different experiments has a natural interpretation: binding site occupancy, which we approximate by a step function at the energy cutoff, should vary with transcription factor concentration and may well differ between experiments; the energy matrix itself, on the other hand, reflects an intrinsic property of the transcription factor which should, in principle, *not* vary between experiments (if other factors, such as ion concentration and pH, are kept at similar levels). Thus, when comparing ensembles, we are free to rescale the matrices in one ensemble so as to bring them into accord with the matrices of the other ensemble. In the case at hand, we found that rescaling the $\Theta_{ChIP}$ energy cutoff to .75, while keeping the $\Theta_{PBM}$ cutoff at 1, brought the $\Theta_{ChIP}$ energy matrix elements into close agreement with those of $\Theta_{PBM}$. Figs. 2.11a and 2.11b illustrate this close agreement between the mean matrix elements in the two ensembles. Fig. 2.11d provides a direct comparison of the $\Theta_{ChIP}$ and $\Theta_{PBM}$ distributions for each matrix element. In most cases, values which could plausibly have been drawn from either the $\Theta_{ChIP}$ or $\Theta_{PBM}$ distribution can be identified. This is illustrated in the upper inset, which shows the raw $\Theta_{ChIP}$ and $\Theta_{PBM}$ histograms for the orange-circled matrix element in Figs. 2.11a and 2.11b. While the two histograms are not identical, they overlap enough that a matrix element consistent with either distribution can be found.

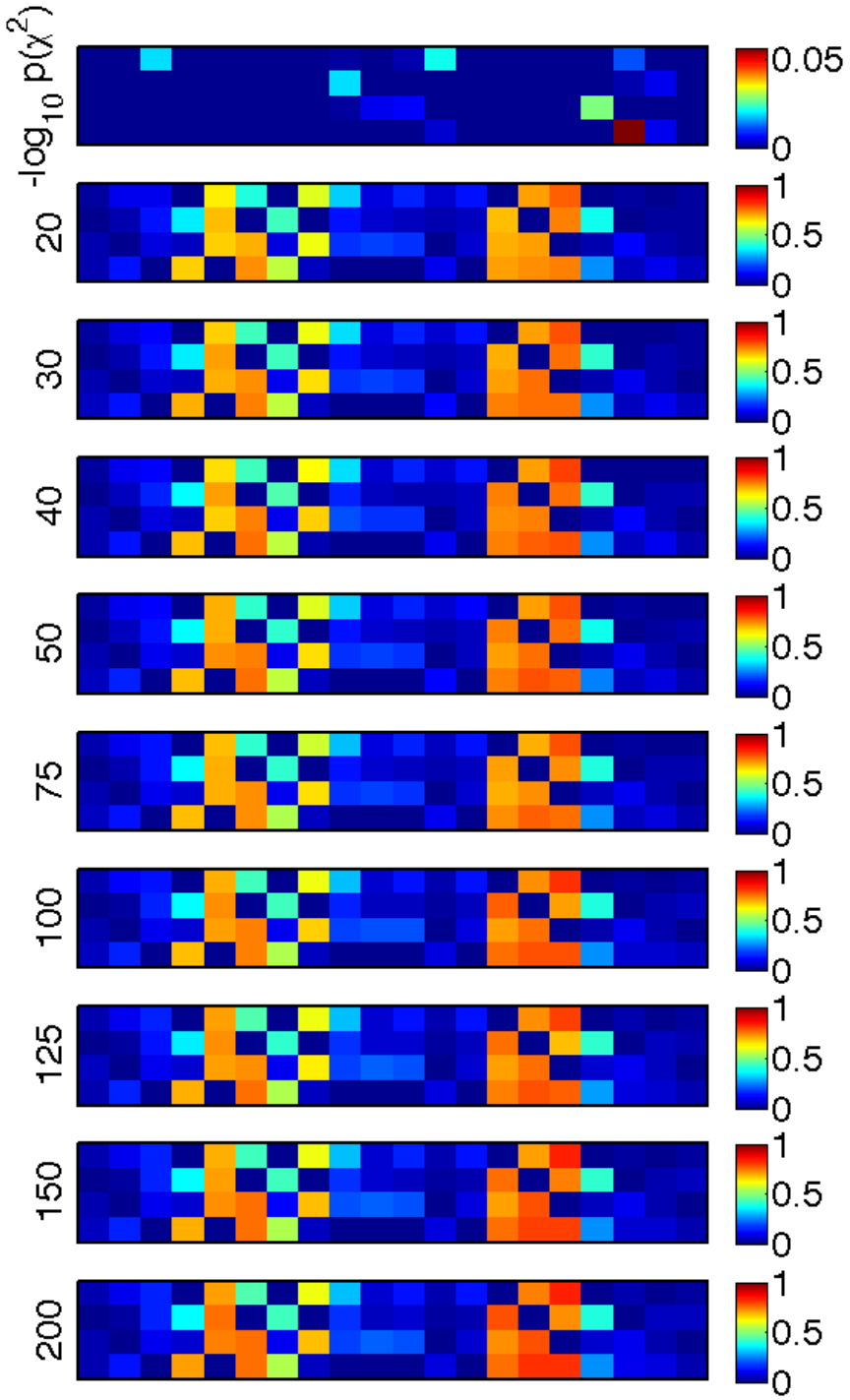


Figure 2.9: Ensemble means for matrix elements inferred from Abf1 ChIP-chip data using quantizations ranging from 20 to 200 sequences per $z$-bin (indicated on the left side of each plot). $\chi^2$ consistency p-values for matrix elements between these ensembles are shown at the top. No elements have Bonferroni corrected p-values $< .05$. The rows of each matrix refer to bases A, C, G, T (top to bottom).

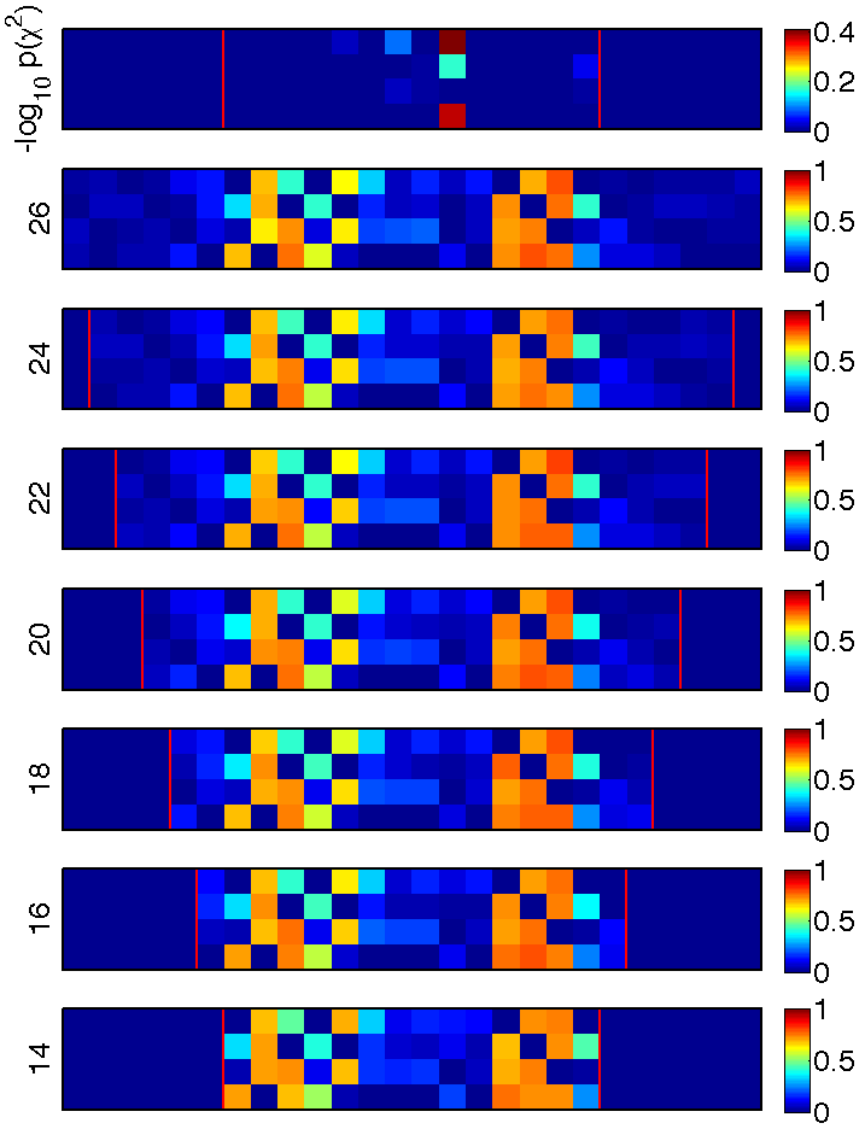


Figure 2.10: Ensemble means for matrix elements inferred from Abf1 ChIP-chip data using matrix widths ranging from 14 to 26 bp (indicated on left side of each plot). Matrices in each MMC ensemble were rescaled to provide maximum consistency between the ensembles. The resulting $\chi^2$ consistency p-values for the central matrix elements common to all ensembles are shown at the top. No elements have Bonferroni corrected p-values $< .05$. Red lines delineate matrix boundaries. The rows of each matrix refer to bases A, C, G, T (top to bottom).

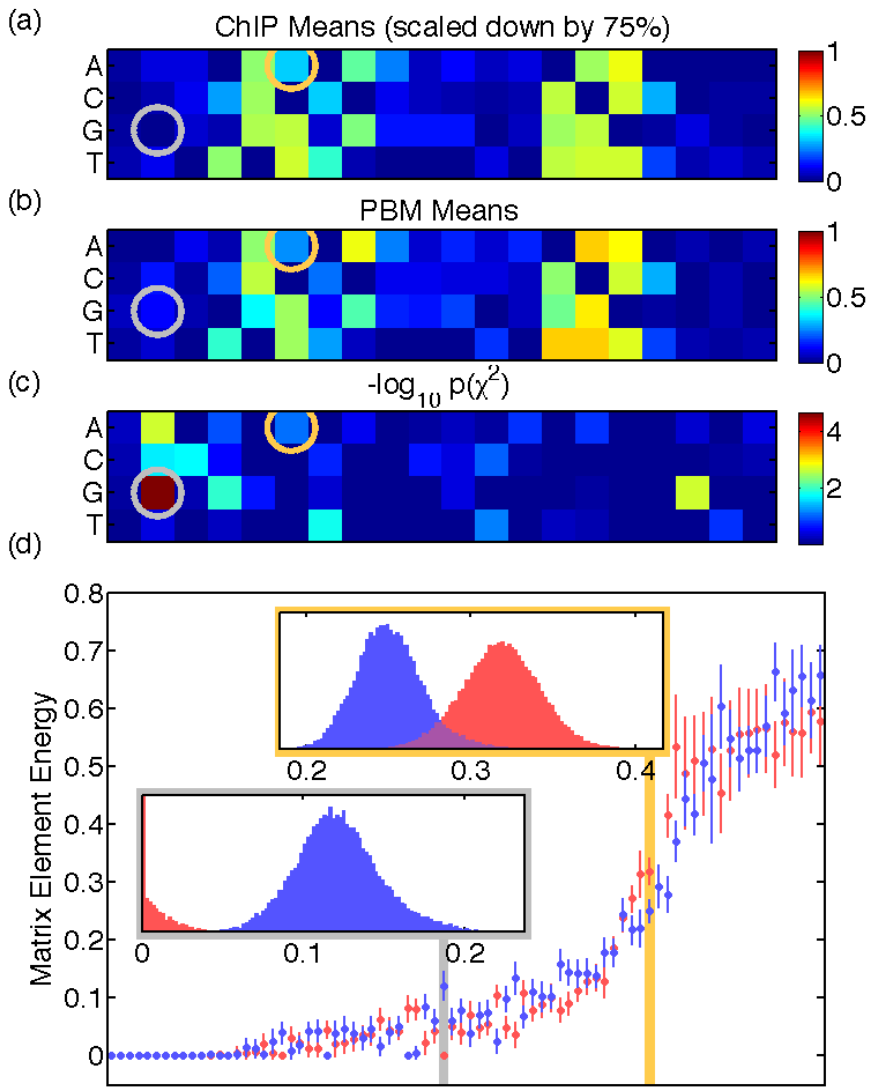


Figure 2.11: Comparison of $\Theta_{PBM}$ and $\Theta_{ChIP}$ parameter distributions. (a) Mean values of rescaled matrix elements in $\Theta_{ChIP}$. (b) Mean matrix elements in $\Theta_{PBM}$ (same as Fig. 2.1a). (c) $\chi^2$ p-values quantifying the element-by-element consistency of the $\Theta_{ChIP}$ and $\Theta_{PBM}$ distributions. (d) Mean and RMSD uncertainty of each matrix element according to the $\Theta_{PBM}$ (blue) and rescaled $\Theta_{ChIP}$ (red) distributions. Elements are arranged from left to right in order of increasing mean. Insets display raw MMC histograms of the values obtained for the matrix elements circled in (a-c) and highlighted below each inset in (d). The $\Theta_{ChIP}$ matrix element distribution in the lower inset has most of its weight at precisely 0; the corresponding histogram bin has been truncated.

The $\chi^2$ consistency p-values for each matrix element are shown in Fig. 2.11c. There are a few matrix elements for which $\Theta_{ChIP}$ and $\Theta_{PBM}$ give inconsistent distributions by this test (the red and yellow squares in Fig. 2.11c). The raw histograms for the most inconsistent matrix element (the red square), plotted in the lower inset of Fig. 2.11d, display the poor overlap between the two distributions. While this inconsistency is significant (even accounting for multiple hypotheses), it occurs outside the core binding site and may not have much practical impact.

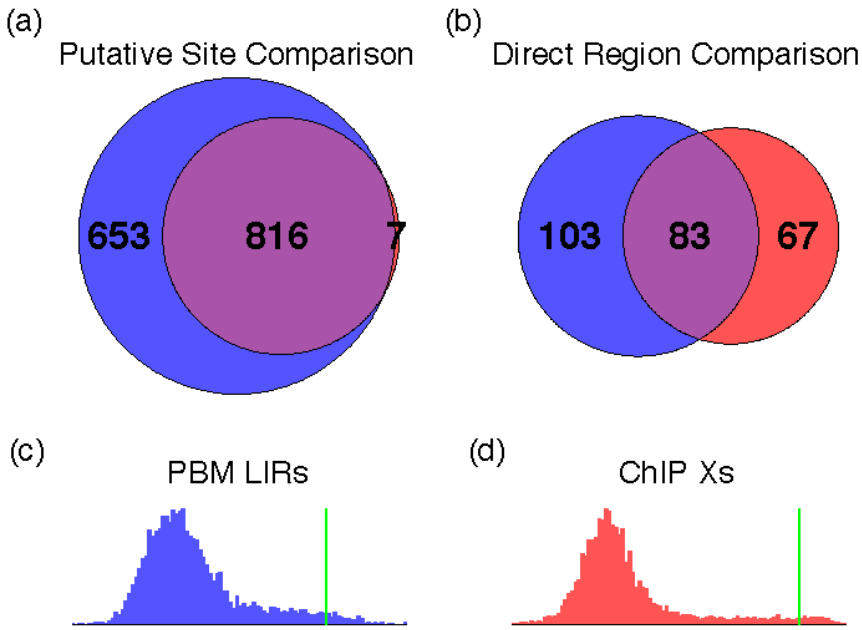


Figure 2.12: EMA likelihood analysis leads to compatible binding site predictions from different experimental data sets, while the more standard method of thresholding the experimental signal leads to substantial disagreement. (a) The 20 bp intergenic *sites* in *S. cerevisiae* having $\Theta_{ChIP}$ hit fraction (HF) $> 50\%$ (red) are a nearly perfect subset of those with $\Theta_{PBM}$ HF $> 50\%$ (blue). (b) By contrast, the intergenic *regions* selected by Mukherjee et al.'s LIR threshold on PBM data (blue) overlap poorly with those selected by Lee et al.'s p-value threshold on ChIP-chip data (red). The thresholds chosen by the experimenters are indicated by the green lines on the experimental LIR histogram of PBM data in (c) and on the X-statistic histogram of ChIP-chip data in (d).

One way to assess the importance of such discrepancies is to compare the binding site predictions of $\Theta_{ChIP}$ and $\Theta_{PBM}$. If the matrices in both ensembles were identical, the sites selected by the smaller (ChIP-chip) cutoff would be a perfect subset of the sites selected by the larger (PBM) cutoff. Fig. 2.12a shows that this is, in fact, nearly the case: of the 823 $\Theta_{ChIP}$-predicted sites, all but 7 are predicted by $\Theta_{PBM}$. We reiterate the fact that these two sets of predictions were made using data from different experimental platforms and using separate analyses involving essentially no free parameters.

Our consistency analysis contrasts with that of Mukherjee et al. [16], who compared their putatively bound regions (identified by a stringent LIR threshold; see Fig. 2.12c) with those of Lee et al. (identified by a stringent p-value threshold; see Fig. 2.12d) and obtained the Venn diagram reproduced in Fig. 2.12b. While the overlap between the two sets of regions is certainly significant, neither is even approximately a subset of the other. This problem cannot be avoided by choosing lower thresholds, as this will result in more false positive predictions. Even though both data sets are of high quality, this discrepancy is an inevitable result of experimental noise, *in vivo* effects on binding, etc. However, by bottlenecking PBM and ChIP-chip data through a simple parametric model, we obtain almost entirely consistent pictures of Abf1 specificity.

## 2.5 Discussion

A quantitative understanding of biological systems will require the ability to deduce quantitative models from data in a transparent and principled way. Not only must optimal model parameters be found, but the confidence one has in the values of these parameters must be characterized, and parameters determined from different data sets must be compared for consistency. For these and other reasons, we believe likelihood inference using MMC, an approach which has had enormous success in physics, should become an important tool in 21st century biology.

Modern biology, however, faces challenges very different from those of physics. While physicists often take great care to understand and minimize experimental noise, high-throughput assays have sources of noise which can vary from system to system or even from lab to lab, and may be impractical to characterize quantitatively. Here we have shown that, given *enough* data, one can precisely characterize interesting biological phenomena without having to explicitly model experimental noise. This is likely to be of particular importance for the study of transcriptional regulation.

Furthermore, the high-throughput platforms currently available for probing transcription factor binding appear, at least for broad-acting factors, to provide sufficient data for such analyses. In our study of Mukherjee et al.'s Abf1 PBM data, we were essentially able to infer the values of 352 independent parameters describing both the transcription factor and the experimental error model (though the error model parameters were integrated out analytically). Nonetheless, most of the 80 energy matrix elements we fit were determined to a precision of $\leq 5\%$ of the functional range. This precision is not due to over-fitting; it simply reflects the large amount of information contained in the $\sigma z$ data we analyzed.

Our analysis is meant only as a proof-of-principle demonstration and could be improved in many ways. Perhaps our most unrealistic assumption is that transcription factor binding is binary – either *on* or *off*. This was done primarily to keep the analysis simple and speed up the MMC algorithm, but a more physically realistic computation of region occupancy, such as that used by the program GOMER [92], could be used instead. The assumption that each nucleotide contributes independently to Abf1's sequence-dependent binding energy (see [87] for a critique) could

also be relaxed by allowing couplings between positions. Indeed, a great advantage of likelihood inference is the way it accommodates such refinements without changing the conceptual basis of the analysis. The only obstacle one faces when implementing such changes is the need for computational power.

Another important aspect of likelihood inference is the possibility of refining quantitative models by using data from multiple experiments. For example, if two $\sigma z$ data sets $\{\sigma_i\}, \{z_i\}$ and $\left\{\sigma'_j\right\}, \left\{z'_j\right\}$ are available for some transcription factor, one can perform a likelihood analysis using the *product* of each data set's likelihood as the combined likelihood of *both* data sets, i.e.

$$p_{ema}(\{z_i\}, \left\{z'_j\right\} \mid \{\sigma_i\}, \left\{\sigma'_j\right\}, \theta) = p_{ema}(\{z_i\} \mid \{\sigma_i\}, \theta) p_{ema}(\left\{z'_j\right\} \mid \left\{\sigma'_j\right\}, \theta). \quad (2.42)$$

Extending this to more than two data sets is straight-forward, and data from different experimental platforms may also be combined in this way. For example, ChIP-chip data might be combined with low-throughput electrophoretic motility shift assay measurements and with high-throughput *in vitro* data using synthetic DNA probes [36, 37, 40, 38]. Also, any knowledge one has about the experimental errors in any of these experiments may be used, either by assuming an explicit error model or by biasing the error model prior. The combination of data from different experiments is a powerful analysis technique, and the use of EMA likelihood should help facilitate its application to large $\sigma z$ data sets.

It is worth while to step back, however, and ask if all this apparatus – in particular the use of EMA-likelihood-based Metropolis Monte Carlo sampling – is really necessary. For most biologists wanting to know which genes are targeted by a specific transcription factor, the answer is probably no. First, ChIP-Seq experiments have greatly improved (with respect to ChIP-chip) the resolution with which one can identify binding sites *in vivo*; such data can often be used with minimal statistical analysis to identify a given factor's target genes.

Also, since the publication of (most of) the material in this chapter [1], an easy-to-use motif finding algorithm called FIRE[6] has been published [93]. Given a set of $\sigma z$ data, this algorithm exhaustively searches IUPAC sequences for ones whose presence in each sequence $\sigma_i$ gives substantial mutual information with the measurement $z_i$. From these IUPAC sequences, FIRE then builds informative motifs. While the results this software returns are largely qualitative (e.g. it does not return error bars on the resulting motif parameters), they appear to be sufficient for the purposes of most biologists.

It is important to note that, while FIRE returns sequence motifs, it does not, fundamentally, seek to produce a full generative model of DNA sequence. In particular, it requires no model of *background* DNA. This fact appears to have allowed FIRE to succeed where other motif finders have previously foundered. In particular, FIRE has found a number of sequence motifs informative about gene expression during the intraerythrocytic life cycle of *Plasmodium falciparum* – a task at which many other algorithms have failed. Since the prediction of relevant sequence motifs in *Plasmod-*

[6]Finding Informative Regulatory Elements

*ium* were made, a number of these motifs have been verified, through PBM assays, to bind computationally predicted DNA-binding domains [94].

Nevertheless, there is still value in understanding what, in principle, we can learn from data. In particular, such an understanding can motivate new experiments whose results would otherwise seem hopelessly uninterpretable. In the next chapter we propose one such experiment.

## 2.6 Appendix: MMC sampling and convergence

Here we give additional details about the MMC sampling used in the preceding text. MMC sampling was performed using Mukherjee et al.'s Abf1 PBM data [16] with 20 intergenic sequences per $z$-bin, and Lee et al.'s Abf1 ChIP-chip data [17] with 50 intergenic sequences per $z$-bin. For each of these data sets, 10 MMC runs were started with a seed model derived from the known Abf1 recognition sequence NNNRT-CAYTNNNNACGWNNN by assigning an energy of 0 to allowed nucleotides and 1 to disallowed nucleotides. We then let each run go for $5 \times 10^6$ steps, and every 1,000th model visited by MMC in each run was recorded. The last 4,000 models recorded for each of the 10 runs were then concatenated to give a representative ensemble $\Theta_{PBM}$ (or $\Theta_{ChIP}$) of $4 \times 10^4$ models.

MMC was seeded with the known Abf1 motif only to reduce the amount of time it took for the algorithm to "burn-in" to the true posterior distribution. Other than the particular alignment of matrix elements within the 20-bp window, this choice of starting point does not affect the distribution of models found after burn-in has occurred. In particular, we have been able to achieve burn-in *de novo* through simulated annealing, starting from seed models with randomly chosen parameters (data not shown).

Evidence for convergence of our MMC routine is presented in Figs. 2.13a and 2.13c where we plot, for PBM and ChIP-chip data respectively, the mean intra-run variance (the mean over the 10 runs of the variance within each MMC ensemble) against the inter-run variance (the variance across models in the 10 MMC ensembles concatenated together) for the 80 energy matrix elements. The scatter plot of these two numbers should lie along the diagonal if the different MMC runs thoroughly sampled the same distribution, and should lie above the diagonal if the MMC runs did not have time to converge. This computation was done for different numbers of models (20, 100, 1,000, 5,000) taken from the beginning of each run and the results for each of these sample sizes (distinguished by color) are shown. The scatter plot collapses convincingly to the diagonal by the time the sample size reaches 5,000 models, and we take this as evidence that MMC indeed provided a thorough sample of the true marginal distribution of each matrix element in both data sets.

Another way to assess MMC convergence is to look at the per-datum log likelihood achieved by models from each MMC run as a function of the order in which these models were sampled. For the MMC runs on PBM and ChIP-chip data, shown in Fig. 2.13b and 2.13d respectively, the per-datum log likelihood saturated by the time the 50th model was recorded. The fact that no large jumps in the per-datum log

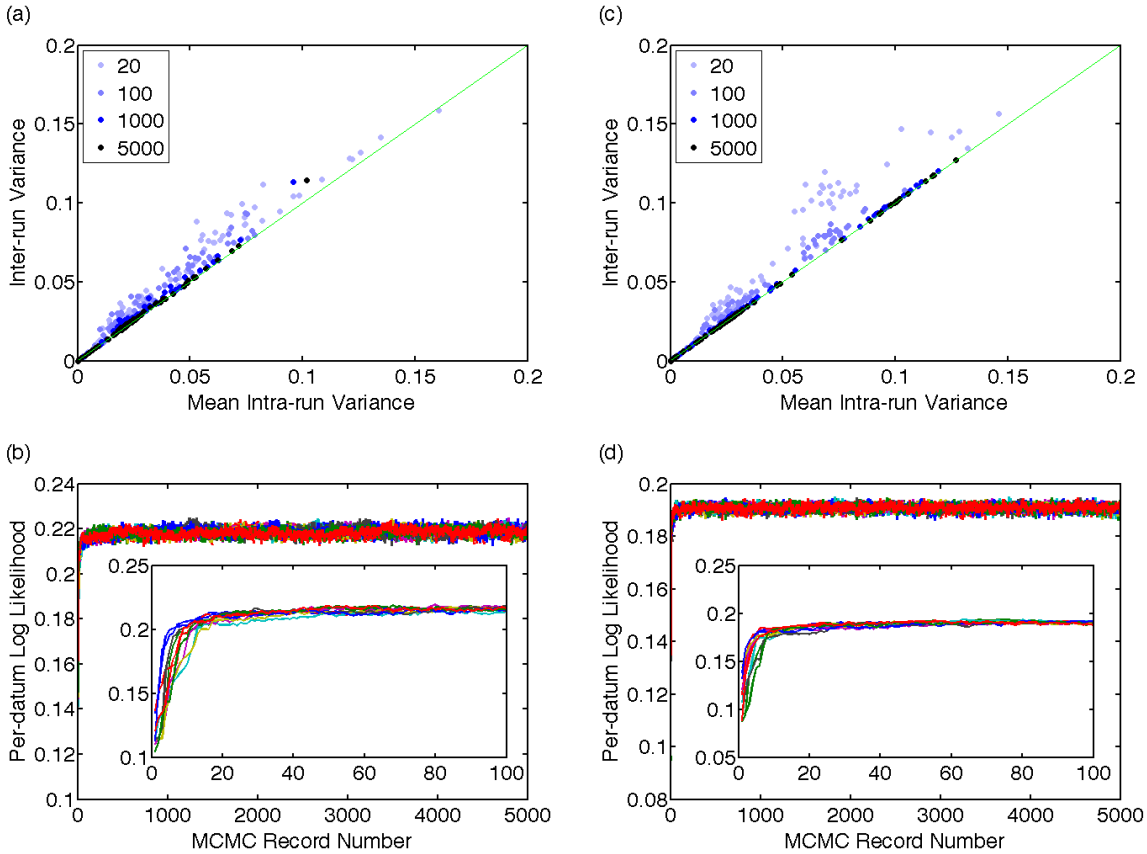


Figure 2.13: MMC convergence. (a) The mean intra-run variance vs. inter-run variance of each matrix element given by MMC runs on Mukherjee et al.'s PBM data. These were computed using the first 20, 100, 1000, and 5000 models recorded in each run. The green line has slope 1. (b) The per-datum log likelihood of models in each of the 10 PBM MMC runs as a function of the order in which those models were sampled. The inset shows a blow-up of results for the first 100 models sampled in each run. (c,d) Similar plots for MMC runs using Lee et al.'s ChIP-chip data. Note: (b) and (d) plot the per-datum log likelihood *relative* to that of a model which makes random predictions; this quantity is approximately equal to $I(z;x)$ (expressed in nats).

likelihood occurred in any of the runs after that point strongly suggests that all MMC runs quickly found the bulk of the posterior distribution and had ample time to sample it before terminating.

# Chapter 3

# Sort-Seq

## 3.1 Overview

We now propose a new experimental technique – called Sort-Seq – for probing the *in vivo* functional architecture of transcriptional regulatory sequences (TRSs). The laboratory techniques required are simple, reliable, and stereotyped. When combined with 454 pyrosequencing, Sort-Seq has the potential to generate massive amounts of $\sigma z$ data to which one can fit quantitative thermodynamic models of transcriptional regulation.

The basis for Sort-Seq is the idea that one can characterize a TRS's functional architecture – the sites within the TRS which affect transcription, the identities of the proteins that bind these sites, and the interaction energies of these regulatory proteins with each other and with the DNA – by assaying the affect that mutant transcriptional regulatory sequences (mTRSs) have on the expression of a reporter gene (Fig. 3.1a). Sort-Seq involves four main steps (Fig. 3.1b):

1. A library of plasmids is created, each plasmid containing a mTRS regulating the expression of a fluorescent reporter gene.

2. Cells are transformed with this library of plasmids and regulated expression of the reporter gene is induced.

3. Using fluorescence-activated cell sorting (FACS), cells are sorted into different batches according to their fluorescence.

4. A large number of mTRSs from each resulting batch of cells is sequenced.

The resulting set of sequences $\{\sigma_i\}$, together with the identity $z_i$ of the FACS batch in which each sequence $\sigma_i$ was found, provides $\sigma z$ data probing the *in vivo* functionality of the TRS.

Creating the library described in step 1 requires only standard cloning techniques; later in this chapter we describe protocols for creating a library containing millions of different mTRSs, each mTRS mutated at a well-defined rate within a well-defined region. The need to transform a large number of plasmids into the cells of interest

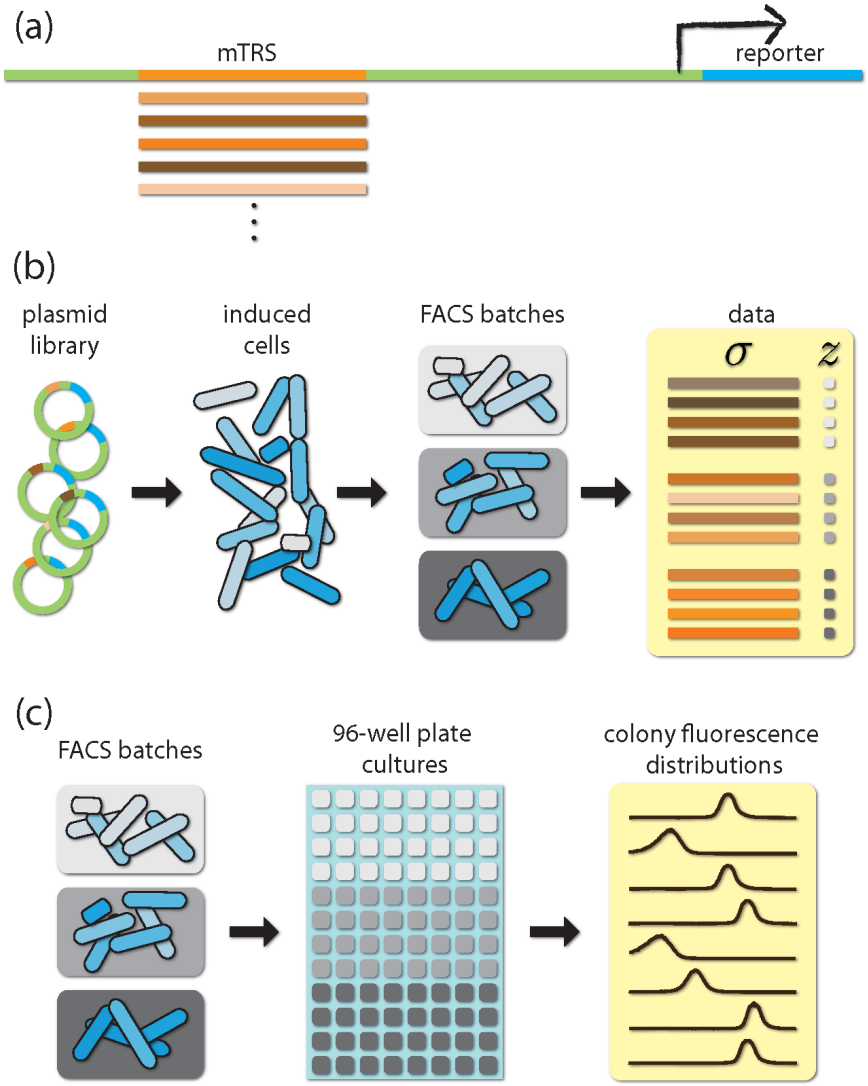


Figure 3.1: Outline of the Sort-Seq assay. (a) Different mTRS, indicated here in shades of orange, are placed upstream of a fluorescent reporter gene. (b) A library of plasmids carrying different mTRSs is created. These plasmids are transformed into cells and expression is induced. Cells are then FACS sorted into a small number of batches and a large number of mTRSs found within each batch is then sequenced. The resulting set of mTRSs, along with each sequence's associated FACS batch, provides $\sigma z$ data for later analysis. (c) Cultures are inoculated with single cells from the sorted FACS batches and grown in 96-well plates. After induction, the fluorescence distribution of cells in each clonal population is measured. These fluorescence distributions allow one to compute the sequence information $I(z;\sigma)$ in the $\sigma z$ data.

may limit the organisms in which Sort-Seq can be applied, but obtaining a large number of independent transformants is not a problem in *E. coli* or *S. cerevisiae*. If temporary transfection suffices, Sort-Seq might be feasible for culturable cell lines as well. And while the need to induce regulation of the reporter gene by the TRS may restrict the TRSs one can assay with this technique, the subsequent sorting of cells and sequencing of mTRSs is straight-forward.

Sort-Seq dovetails particularly well with 454 pyrosequencing, which will soon have ability to produce $\sim 10^6$ sequences, each $\sim 400$ bp long, in one 12 hr run of one machine. So far, ultra-high-throughput sequencers like that of 454 and Illumina have been applied primarily for genome sequencing, resequencing, and various genome-wide *sequence census* assays [15]. Sort-Seq, by contrast, uses the power of ultra-high-throughput sequencing to gain a highly detailed, biophysical picture of how specific genomic sequences function.

The $\sigma z$ data generated by Sort-Seq should allow one to identify the binding sites of all DNA-binding proteins involved in regulation by the TRS of interest, infer perturbative models of each protein's sequence-dependent binding energy, and fit quantitative models describing how the various protein-DNA and protein-protein interactions occurring at the TRS result in gene regulation. More specifically, given the resulting set of mTRSs $\{\sigma_i\}$ and corresponding FACS batches $\{z_i\}$, one can infer models $x(\sigma;\theta)$ of arbitrary parametric form using the techniques detailed in the previous chapter. The large amount of data produced by Sort-Seq, however, should allow one to infer more than just energy matrices; it should be possible to fit complex thermodynamic models of transcriptional regulation, involving multiple protein-DNA and protein-protein interactions.

In this chapter we also detail a method (Fig. 3.1c) for directly measuring the sequence information $I(z;\sigma)$ produced by the Sort-Seq assay. One can then compare this quantity to the mutual information $I(z;x)$ between the FACS batches $\{z_i\}$ and the sequence-dependent predictions $\{x_i = x(\sigma_i;\theta)\}$ of any inferred model. The fraction of sequence information $I(z;\sigma)$ recovered by $I(z;x)$ provides a rigorous assessment of the model's completeness, allowing one to determine whether there are additional sequence features in the TRS which need to be understood.

We demonstrate Sort-Seq's feasibility by sorting cells driving GFP expression from a library of mutant lacZ promoters. By sequencing $\sim 400$ mutant promoters found in these batches of sorted cells, we generate $\sigma z$ data probing the *in vivo* binding of CRP to its site on the *lacZ* promoter. EMA likelihood and Metropolis Monte Carlo sampling allow us to use these data to fit energy matrices describing CRP's sequence-dependent binding energy. By measuring the fluorescence distributions of clonal populations grown from sorted cells, we then determine $I(z;\sigma)$, and find that the inferred energy matrices account for only about half of the sequence-dependent information in the data.

We then discuss various theoretical issues raised by Sort-Seq. In particular, we show that $\sigma z$ data probing the simultaneous binding of multiple proteins per sequence can, because of natural nonlinearities in the formulas of statistical mechanics, be used to infer interaction energies in absolute thermal units – something we were unable to do in our analysis of ChIP-chip and PBM data.

## 3.2 Background

The idea of using libraries of randomized sequences to probe transcriptional regulation is quite old. The replacing of a TRS by a library of mTRSs, followed by an assay of sequence function, was first pursued by Horwitz and Loeb [95]. They replaced the -42 to -21 region of the *tetR* promoter (on a pBR322 plasmid) with a library of random sequences. They then selected promoters which conveyed resistance to tetracycline. Their intention was to characterize the sequence requirements of RNAP's -35 DNA-binding region. They also assayed semi-quantitative values for each clone's resistance to tetracycline (by estimating the antibiotic concentration at which 50% of colonies survived). Horwitz and Loeb's assay differs fundamentally from Sort-Seq, however, since the mutant sequences they tested contained completely randomized nucleotides. Complete sequence randomization might allow one to characterize the sequence features needed to achieve a given function, but does not allow one to characterize a specific sequence of interest.

Around the same time, Oliphant, Hill, and Struhl published a series of papers describing the uses of similar sequence libraries in deciphering the sequence requirements of RNAP [96, 97, 98, 99]. Struhl and colleagues then used a novel cloning method to create a library of mutant *E. coli* promoters driving the expression of *HIS3*, a yeast gene which allows the *E. coli hisB*463 strain to grow in the absence of histidine [97, 98]. While the authors aptly make the case for using such libraries of mTRSs, the DNA sequences they use are again completely randomized. In a separate paper [100], these authors created libraries of mutant promoter sequences which were strongly biased toward the wild type sequence, but no functional assays were done with this library.

The work most resembling Sort-Seq was published by Schneider and Stormo in 1989 [101]. The aim of this work was to determine the sequence specificity encoded at promoters for T7 RNA polymerase (the RNAP encoded by the T7 phage). Alignment of known T7 promoters had been previously revealed $\sim 35$ bits of functional sequence information at each promoter. This amount of sequence information, however, is much greater than the $\sim 18$ bits needed to specify locations within the intergenic DNA of *E. coli*.[1] The authors' aim was to directly measure, *in vivo*, the amount of sequence information needed to specify a T7 promoter.

To do this, Schneider and Stormo cloned *specifically randomized* $\phi 10$ promoters[2] – i.e. $\phi 10$ promoters synthesized with a per-nucleotide substitution rate of 15% – into high-copy-number plasmids. The promoter itself was not used to drive the expression of either a reporter or a selective marker; transcription resulting from the mere presence of a working T7 promoter on a high copy number plasmid will, in the presence of T7 RNAP, use up so many ribonucleotides that it kills the host cell! By sequencing 53 specifically randomized $\phi 10$ promoters which caused cell death in the presence

[1] The *E. coli* genome is $\sim 5 \times 10^6$ nucleotides in length, only about 10% of which is noncoding DNA.

[2] The $\phi 10$ promoter initiates transcription of the T7 bacteriophage gene 10. This gene actually codes for two proteins, 10A and 10B, which make up the virus capsid. 10A is expressed at particularly high levels in infected cells [102].

of T7 RNAP, Schneider and Stormo were able to estimate $18 \pm 2$ bits of functional information at the wild type $\phi 10$ promoter.

This work stands out in the literature as the only one to do a quantitative analysis of the sequences resulting from the selection of specifically randomized TRSs. It is also eerily similar to our proposed assay (which was, by and large, developed independently) in many of the experimental details. However, the quantitative analysis presented in Schneider and Stormo's paper is limited, and the authors do not discuss the full potential of such experiments. Of course such experiments are much easier now than they were in 1989 thanks to advancements in DNA sequencing technology.

The line of investigation opened up by Horwitz, Loeb, Struhl, Oliphant, and Hill has advanced modestly over the last two decades. For instance, the *E. coli* one-hybrid assay [41, 42, 43], reviewed in chapter 1, clearly builds off of the idea of selecting functional sequences from a randomized library. This assay, however, is designed to characterize the specificity of DNA-binding proteins outside of their native environments, and, as is the case with its predecessors, does not probe the functional architecture of specific TRSs. Only the work of Schneider and Stormo [101] comes close to the spirit of Sort-Seq, but it appears that the potential for such experiments has laid dormant since the publication of this paper.

## 3.3 Proof-of-principle experiment

We performed a proof-of-principle Sort-Seq experiment on the *E. coli lacZ* promoter. As discussed in the introduction, transcription at the lacZ promoter is regulated by the transcription factor CRP, which binds just upstream of and interacts favorably with the RNAP holoenzyme (Fig. 1.3). We investigated whether $\sigma z$ data probing the activity of *lacZ* promoters with polymorphic CRP binding sites would allow us to infer an energy matrix describing CRP's *in vivo* sequence-dependent binding energy.

### 3.3.1 Experimental methods

#### Cloning vector

This proof-of-principle experiment was greatly facilitated by the previous work of Zaslaver et al. [103]. These authors created a library of 1820 plasmids, each carrying a different *E. coli* promoter driving the expression of *gfpmut2*.[3] Each of their plasmids also contains a *kanR* gene (enabling selection with the antibiotic kanamycin), as well as a pSC101 origin of replication, which maintains the plasmid at very low copy number ($\sim 5$ copies per cell [105]).

To generate a library of mutant *lacZ* promoters driving *gfpmut2* expression, we started with the *lacZ* promoter plasmid, pUA66lacZ, created by Zaslaver et al.. Using the procedure detailed in Fig. 3.2, we generated a cloning vector pJK10 in which 81 bp of the *lacZ* promoter region (containing the RNAP and CRP binding sites) was

[3] *gfpmut2* is a mutant version of GFP, optimized for fluorescence and expression in *E. coli* [104].

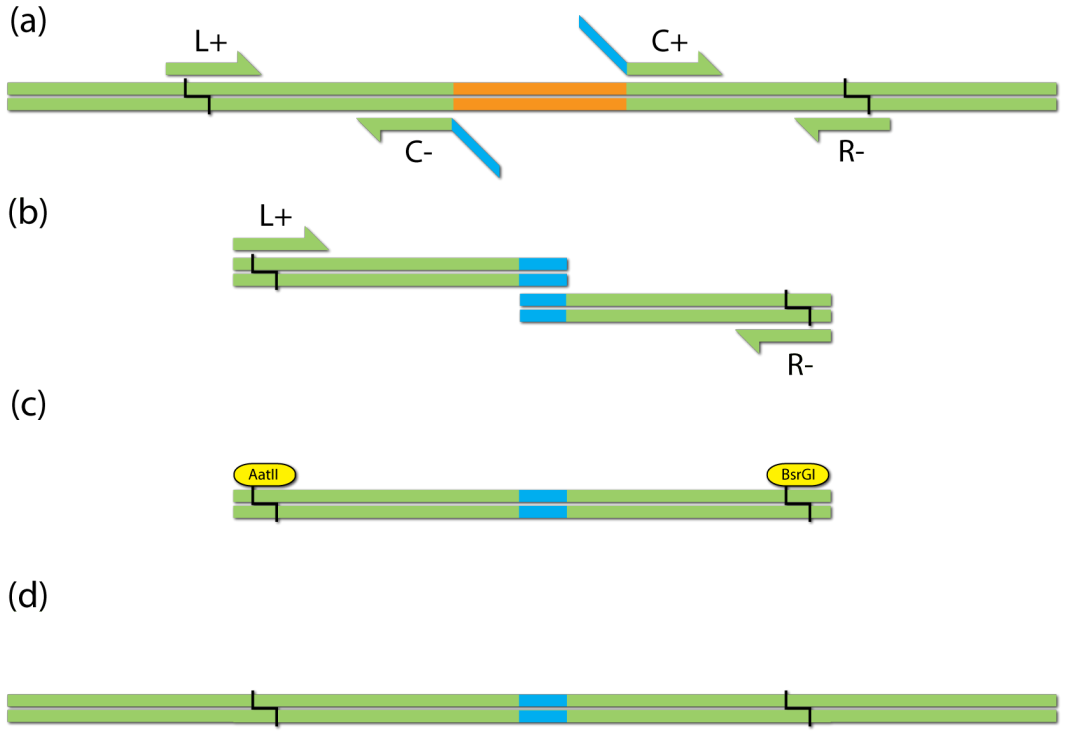


Figure 3.2: Strategy used to create pJK10 from pUA66lacZ. (a) Four PCR primers were used: L+ and R- overlap restriction sites present in the starting plasmid, while C+ and C- bound the region to be replaced (orange). C+ and C- carry complimentary tails containing the polylinker sequence (blue). PCR reactions using primer pairs (L+, C-) and (C+, R-) were performed separately. (b) A small amount of both PCR products were then used as template in a PCR reaction using primer pair (L+, R-). (c) The ends of the resulting PCR product were clipped using the appropriate restriction enzymes (AatII and BsrGI), and cloned back into the original plasmid. (d) The the polylinker sequence exactly replaces the target region in the resulting plasmid.

replaced with a polylinker sequence (see Fig. 3.3c) containing one central restriction site (NheI) and two bounding type IIs restriction sites (BsmBI).

### Insert library

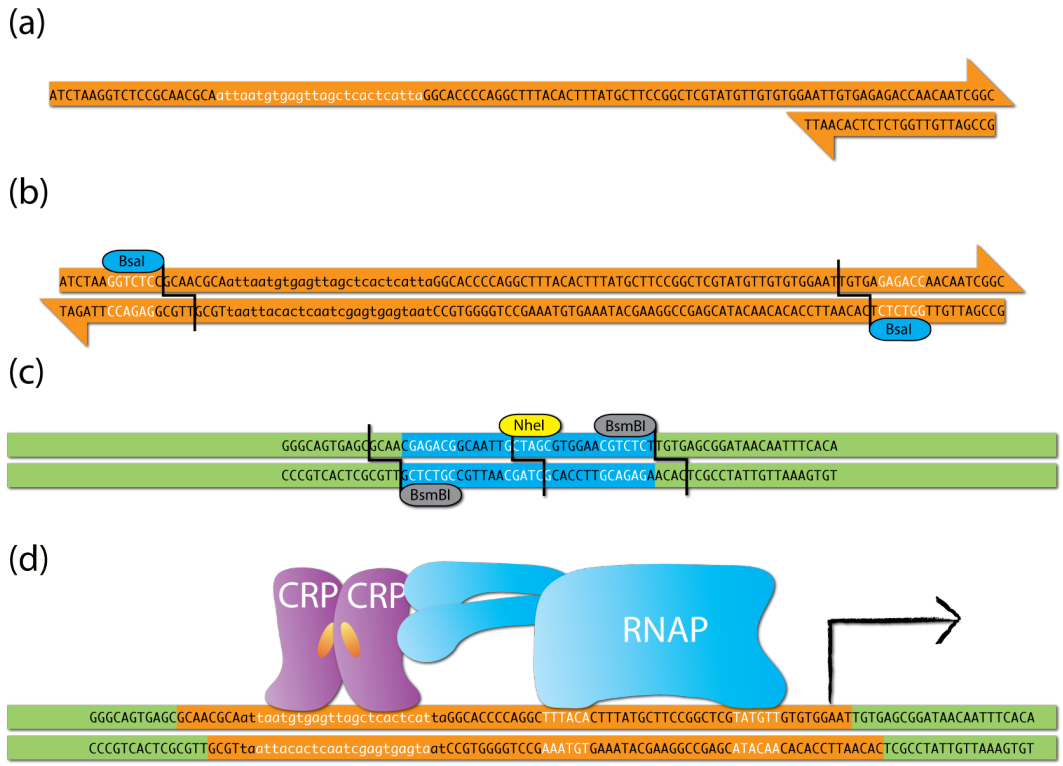


Figure 3.3: Cloning procedure. (a) A library of single-stranded oligos was synthesized with a 26 bp sub-sequence (lower-case bases; shown in white) exhibiting at a 24% per-nucleotide substitution rate. Klenow extension from an annealed primer was used to make this library double-stranded. (b) The double-stranded insert library was digested with BsaI, excising the binding sites of this restriction enzyme and creating non-palindromic sticky ends. (c) The cloning vector pJK10 was digested first with NheI (opening up the plasmid), then with BsmBI (removing the polylinker sequence and creating non-palindromic sticky ends). (d) Digested inserts were ligated into digested vector, effecting exact replacement of the 26 bp target sequence in pUA66lacZ with a library of polymorphic sequences.

We then synthesized a library of single-stranded 119 bp inserts, containing 89 bp of the *lacZ* promoter, as well as (inward-facing) binding sites on either end for the type IIs restriction enzyme BsaI. While most of this insert was synthesized with precisely-specified nucleotides, 26 bp centered on the CRP binding site were synthesized at a 24% per-nucleotide substitution rate.[4] This library was then made double-stranded by Klenow extension from an annealed primer (Fig. 3.3a).

[4]Synthesizing oligonucleotides with a defined substitution rate at specific positions is simple. In a typical DNA synthesis reaction, specified phosphoramidites (nucleotides with special protection

### Cloning

The cloning vector, pJK10, was digested with NheI, gel purified for open plasmid, then digested with BsmBI, thereby completely excising the polylinker and leaving non-palindromic sticky ends (Fig. 3.3c). The insert library was then digested with BsaI, removing the BsaI binding sites and generating nonpalindromic sticky ends compatible with the vector. Insert and vector were ligated using T4 DNA ligase (Invitrogen) according to manufacturer's protocols. The resulting ligation reaction was desalted using agarose gel desalting wells [106], after which 5 $\mu$l was used to transform 20 $\mu$l of Electromax DH5$\alpha$-E *E. coli* (Invitrogen) via electroporation (1.8kV, 200$\Omega$, 25$\mu$F). This protocol consistently yielded $\sim 5 \times 10^6$ transformants. Transformants were cultured in $LB_{kan}$ (50 ng/$\mu$l kanamycin) and maxiprepped (Sigma), yielding $\sim 60$ $\mu$g of supercoiled plasmid library in which $\sim 62\%$ of the plasmids contained cloned inserts in the proper orientation and with no indels (Fig. 3.3d; also see Results).

### Retransformation

5 ng of plasmid library was then used to re-transform Electromax DH5$\alpha$-E *E. coli* via electroporation (1.8kV, 200$\Omega$, 25$\mu$F), generating $\sim 1.3 \times 10^7$ transformants.[5] These retransformed cells, labeled sJK11 LIB, were cultured and stored as a glycerol stock until further use. sJK11 + cells were generated by transforming DH5$\alpha$-E cells with pUA66lacZ, sJK11 - cells were generated by transformation with pJK10, and seven strains, sJK11 1 to sJK11 7, were grown from randomly picked colonies of sJK11 LIB.

### Library induction and sorting

To induce CRP-dependent transcription of *gfpmut2* from the polymorphic *lacZ* promoters in our library, we inoculated 1 ml cultures of $LB_{induce}$ media (50 ng/$\mu$l kanamycin, 1 mM cAMP, 1 mM IPTG) with $\sim 10^7$ sJK11 LIB cells. Similar cultures were prepared for sJK11 +, sJK11 -, and sJK11 1-7 cells. These 10 cultures were incubated for 18 hr 45 min, shaking at 37°C . Each culture was then diluted to $\sim 10^7$ ml$^{-1}$ in 1 ml of PBS before FACS sorting; samples were also stored as glycerol stocks.

We used a FACS Vantage SE to measure the fluorescence of $\sim 10^5$ individual cells from each culture. Cells from sJK11 LIB were sorted into nine batches, LB1 to LB9; each batch received $\sim 1.5 \times 10^4$ cells in 1 ml of $LB_{kan}$. About $\sim 200$ cells from LIB,

groups) are added sequentially to the 5' ends of a population of growing oligonucleotides. Typically, only adenine, cytosine, guanine, or thymine phosphoramidites are present in the reaction solution at any given time. This ensures a well-defined nucleotide (either A, C, G, or T) is added to the end of all growing oligos. Extending with mixtures of phosphoramidites, however, allows one to extend a predictable fraction of the oligos with each of the four possible nucleotides. Since different mixtures of nucleotides can be used at each position of the oligonucleotide, it is a simple matter to synthesize a library of oligonucleotides containing a subregion with a well-defined substitution rate.

[5]This retransformation was not strictly necessary, since we already had DH5$\alpha$-E cells carrying the plasmid library. However, retransformation will generally be necessary when assaying promoter function in *E. coli* with other genotypes, such as *E. coli* MG1655 (a wild type K12 strain), or mutants thereof.

LB1 - LB9 were then plated on $LB_{kan}$ agar and incubated overnight; the rest of each FACS batch was incubated overnight then stored as a glycerol stock. Plated colonies were picked into 96-well culture plates containing 150 $\mu$l $LB_{kan}$ per well; 96 clonal cultures from LIB were grown, along with 32 clonal cultures from each FACS batch LB1 - LB9. These cultures were grown overnight in a 37°C stationary incubator and stored as glycerol stocks.

#### Sequencing

Colony PCR (cPCR) using primers ZAprimer1 (CCATTAACATCACCATCTAA) and ZAprimer2 (CCAGCTGGCAATTCCGACGT) was performed for each of the 384 individual 96-well plate cultures; these are the same primers that Zaslaver et al. [103] used to verify cloned promoters. The resulting cPCR products were run on 3% agarose gels (for quality assessment) and sequenced (from ZAprimer1) by the High-Throughput Genomics Sequencing Unit at the University of Washington. cPCR products from each of the sJK11 1-7 cultures were sequenced (from ZAprimer1) by Genewiz.

Resulting cPCR sequences containing the sequences ACGTCGG and CTCACAA were deemed *recognizable*; sequences in which markers were 155 bp apart were deemed *uncloned vector*; sequences in which they were 204 bp apart were deemed *exactly cloned*. Some exactly cloned sequences had substitutions elsewhere in the sequence; these were rare, though, and are ignored henceforth. All other recognizable sequences were deemed *erroneously cloned*; these sequences consisted mostly of plasmids containing inserts, but with some variety of indel.

#### 96-well plate induction and sorting

Four 2 ml-deep 96-well plates containing 1 ml $LB_{induce}$ per well were inoculated from the 96-well glycerol plates containing LIB and LB1 - LB9 cultures. These deep-well plates were sealed with Breathe-Easy gas-permeable covers (USA Scientific), and incubated overnight at 37°, shaking at 200 rpm. These cultures, which grew to a density of $\sim 5 \times 10^8$ ml$^{-1}$, were then diluted to $\sim 10^6$ ml$^{-1}$ in 200$\mu$l of PBS in 96-well flat-bottom plates. The fluorescence values of $\sim 2 \times 10^4$ cells per well were then measured using an LSR II with a 96-well plate attachment.

### 3.3.2 Computational methods

#### Fluorescence histograms

Raw cellular fluorescence measurements are positive integers representing the number of photons received from each cell. Histograms of these integers are choppy and hard to interpret, particularly when plotted on a log fluorescence scale. The distribution of these quantized values, however, is not of direct interest to us; instead we wish to know the distribution of cellular brightness (a continuous variable) likely to have produced these discrete photon counts.

So when histogramming cellular fluorescence measurements, we assign to each integer $n$ reported by the FACS machine a random variable $\lambda$, drawn according to a posterior distribution conditioned on the measurement $n$. This is done in two steps. First, the mean $\mu$ and variance $\sigma^2$ of all $n$-values reported for a given population of cells is determined. We then posit the gamma distribution with identical mean and variance as the prior distribution on cellular brightnesses:

$$p(\lambda|\alpha,\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)}\lambda^{\alpha-1}e^{-\beta\lambda}, \quad \alpha = \frac{\mu^2}{\sigma^2}, \quad \beta = \frac{\mu}{\sigma^2}. \tag{3.1}$$

The posterior distribution of $\lambda$ given a single fluorescence measurement of $n$ thus has a simple form:

$$p(\lambda|n,\alpha,\beta) = p(\lambda|\alpha+n,\beta+1) = \frac{(\beta+1)^{\alpha+n}}{\Gamma(\alpha+n)}\lambda^{\alpha+n-1}e^{-(\beta+1)\lambda}. \tag{3.2}$$

We note that these $\lambda$ values are used solely for making histograms and not, for example, in the calculation of $I(z;\sigma)$.

### Berg von Hippel CRP energy matrix

To infer a Berg von Hippel energy matrix for CRP, we used the sequences of 49 CRP binding sites (all 26 bp long) which were listed in the DPInteract database and had been verified by DNase I footprinting [30].[6] Adding the reverse-complement of each sequence to this list, we computed energy matrix elements using Berg and von Hippel's formula with a pseudocount of 1:

$$\epsilon_{bl} = -\log\frac{q_{bl}}{q_{b,null}}, \quad q_{bl} = \frac{c_{bl}+1}{N+4}, \tag{3.3}$$

where $c_{bl}$ is the number of sequences in this list with base $b$ in position $l$, and $q_{b,null}$ is the frequency with which base $b$ appears in the noncoding DNA of *E. coli*.[7]

### MMC sampling of CRP energy matrices

We then sampled 26 bp CRP energy matrices using MMC on the $\sigma z$ data produced by this experiment. All MMC runs were initiated from random matrices, and reverse-complement symmetry was enforced throughout each run's duration. Because of the small number of sequences in our data set, we used only 5 $z$-bins, each bin corresponding to the union of two FACS batches: LIB with LB2, LB3 with LB4, LB5 with LB6, LB7 with LB8, and LB9 with LB1.

Unlike our analysis of PBM and ChIP-chip data in chapter 2, where we assumed transcription factors were simply *bound* or *not bound*, here we quantized predicted CRP binding energies into 5 $x$-bins. The boundaries of these $x$-bins were treated

[6] The CRP sites in DPInteract are only 22 bp long; we extended these by 2 bp on either side by retracing their locations in the *E. coli* genome.

[7] The intergenic DNA of *E. coli* has a GC content of 43.4%.

as model parameters themselves and were allowed to vary during MMC sampling. We could have easily introduced finer binning for $x$, but we found that additional bin boundaries tended to cluster together; MMC didn't seem to *want* any more than $\sim 3-5$ $x$-bins.

**Measuring $I(z;\sigma)$**

In order to assess the sequence information $I(z;\sigma)$ in the $\sigma z$ data, it is useful to write

$$I(z;\sigma) = \sum_{\sigma} p(\sigma) D(p(z|\sigma)||p(z)) \tag{3.4}$$

where $D$ is the Kullback Leibler divergence

$$D(p(z|\sigma)||p(z)) = \sum_{z} p(z|\sigma) \log \frac{p(z|\sigma)}{p(z)}. \tag{3.5}$$

One can therefore estimate $I(z;\sigma)$ by measuring $p(z|\sigma)$ for each sequence $\sigma_j$ in *some subset of sequences* $\{\sigma_j\}_{j=1}^{M}$ *representative of the full sequenced set* $\{\sigma_i\}_{i=1}^{N}$:

$$I(z;\sigma) = \langle D_\sigma \rangle_\sigma \pm \sqrt{\frac{\mathrm{var}(D_\sigma)_\sigma}{M}} \tag{3.6}$$

$$D_\sigma = \sum_{z} p(z|\sigma) \log \frac{p(z|\sigma)}{p(z)} \tag{3.7}$$

$$p(z) = \langle p(z|\sigma) \rangle_\sigma . \tag{3.8}$$

Here, $\langle \cdot \rangle_\sigma$ and $\mathrm{var}(\cdot)_\sigma$ respectively denote the mean and variance taken over the $M$ sequences $\{\sigma_j\}$. To characterize $p(z|\sigma)$ for each sequence $\sigma \in \{\sigma_j\}$, it suffices to measure the fluorescence distribution $p(f|\sigma)$ observed for a clonal population of cells (all carrying sequence $\sigma$) induced under the same conditions as the sorted library.[8]

To be precise, the distribution $p(z|\sigma)$ is the probability that a sequence $\sigma$ randomly drawn from our $\sigma z$ data set came from FACS batch $z$. To infer $p(z|\sigma)$ from $p(f|\sigma)$, we first define the quantity

$$w_{z|\sigma} = \int df \; p(f|\sigma) \Theta_z(f), \tag{3.9}$$

where $\Theta_z(f)$ is unity if $f$ is within the fluorescence range of $z$, and is zero otherwise. $w_{z|\sigma}$ therefore gives the fraction of cells carrying sequence $\sigma$ which fall within the fluorescence range of FACS batch $z$. The distribution $p(z|\sigma)$ is then given in terms of these $w_{z|\sigma}$ by

$$p(z|\sigma) = \frac{\alpha_z w_{z|\sigma}}{\sum_{z'} \alpha_{z'} w_{z'|\sigma}} \tag{3.10}$$

for some as-of-yet unknown constants $\alpha_z$. These constants account for the fact that whether a cell with fluorescence $f$ in the range of FACS batch $z$ actually ends up

[8] We emphasize that $M$ can be much less than $N$; indeed, while 454 sequencing should enable $N \sim 10^6$, one can practically measure the fluorescence distributions of only $M \sim 10^2$ clonal populations.

being sorted into batch $z$ and sequenced can depend quite strongly on which batch $z$ one considers. For instance, in our proof-of-principle experiment we sorted $1.5 \times 10^4$ cells into each of 9 FACS batches; all the rest of the cells in our library (i.e. the vast majority of them) were thrown away. As shown in Fig. 3.5, batch LB2 spanned a much more densely populated region of the library fluorescence distribution than, say, LB9. Since an equal number of cells were sorted into each batch, cells with fluorescence in the range of LB9 were much more likely to be sorted (and thus sequenced) than cells in the range of LB2. We therefore expect $\alpha_z \sim w_z^{-1}$ where $w_z = \left\langle w_{z|\sigma} \right\rangle_\sigma$. This relation is only approximate, though, as there are other factors which influence $\alpha_z$ which are difficult to quantify directly. Fortunately, this is not necessary.

Since the subset of sequences $\{\sigma_j\} \subset \{\sigma_i\}$ for which we have fluorescence distributions is representative of the entire $\sigma z$ data set, it makes sense that $p(z)$, given by averaging $p(z|\sigma_j)$ over $\{\sigma_j\}$ (see Eq. 3.8), should match the fraction $f(z)$ of sequences in the full $\sigma z$ data set that are assigned to batch $z$. More explicitly, we should find

$$\left\langle \frac{\alpha_z w_{z|\sigma}}{\sum_{z'} \alpha_{z'} w_{z'|\sigma}} \right\rangle_\sigma = \frac{c_z}{N}. \tag{3.11}$$

We can therefore use this system of equations to solve for the coefficients $\alpha_z$. These equations are nonlinear and generally cannot be solved analytically, but the author has found that this system can (apparently) be solved through the iterative algorithm

$$\alpha_z^0 = 1, \quad \alpha_z^{n+1} = c_z \left\langle \frac{w_{z|\sigma}}{\sum_{z'} \alpha_{z'}^n w_{z'|\sigma}} \right\rangle_\sigma^{-1}. \tag{3.12}$$

**Estimating $I(z;x)$**

Applying some model $x(\sigma;\theta)$ to our sequences $\{\sigma_i\}$ produces a set of continuous $x$-values, each assigned a corresponding $z$-value. To compute $I(z;x)$, we first discretize $x$ so that each of $n$ possible $x$-values occur with identical frequency in our data set. Using these discretized $x$-values, we then fill out a counts table, producing a matrix of values $\{c_{zx}\}$. Estimating the mutual information $I(z;x)$ from a table of counts $\{c_{zx}\}$ is a nontrivial problem, though. While we can express $I(z;x)$ as a difference of entropies

$$I(z;x) = H(z) + H(x) - H(z,x), \tag{3.13}$$

no unbiased estimators for entropy are known analytically. For instance, while $H(z)$ has a simple form

$$H(z) = -\sum_z p(z) \log p(z), \tag{3.14}$$

simply plugging the observed frequencies $f(z) = c_z/N$ into this equation,

$$H_{naive}(\{c_z\}) = -\sum_z f(z) \log f(z), \tag{3.15}$$

will give a systematically under-estimated entropy [107]. This bias will typically cause one to overestimate $I(z;x)$.

Progress can be made if we treat each distribution as a set of variables to be learned from the data. For instance, let $q = \{q_z\}$ be the $m$-dimensional vector of probabilities specifying $p(z)$, and let $c = \{c_z\}$ be the vector of counts. We can analytically compute $p(q|c)$, the probability of a distribution $q$ given counts $c$ using Bayes's rule,

$$p(q|c) = \frac{p(c|q)p(q)}{\int dq\ p(c|q)p(q)}, \tag{3.16}$$

if we assume the prior $p(q) = p(q|\beta)$ is a Dirichlet distribution,

$$p(q|\beta) = \frac{\Gamma(\beta m)}{\Gamma(\beta)^m}\delta\left(1 - \sum_z q_z\right)\prod_z q_z^{\beta-1}. \tag{3.17}$$

The probability of counts $c$ produced from a distribution $q$ drawn from $p(q|\beta)$ is

$$p(c|q) = \left(\frac{N!}{\prod_z c_z!}\right)\prod_z q_z^{c_z} \tag{3.18}$$

Using this and Eq. 3.17 in Eq. 3.16, one gets the posterior distribution for $q$,

$$p(q|c,\beta) = \frac{\Gamma(N+\beta m)}{\prod_z \Gamma(c_z+\beta)}\delta\left(1 - \sum_z q_z\right)\prod_z q_z^{c_z+\beta-1}. \tag{3.19}$$

As shown by Wolpert and Wolf [108], the mean and variance of the entropy $H(q) = -\sum_z q_z \log q_z$ of distributions drawn from this posterior can be computed analytically. Letting $A_z = c_z + \beta + 1$ and $B = N + \beta m + 1$, these quantities can be written in terms of polygamma functions[9] (see Appendix for derivation),

$$\begin{aligned}
\langle H\rangle_{q|c,\beta} &= \int dq\ H(q)p(q|c,\beta) && (3.21)\\
&= \psi_0(B) - \sum_z \frac{A_z-1}{B-1}\psi_0(A_z), && (3.22)\\
\mathrm{var}\,(H)_{q|c,\beta} &= \left\langle H^2\right\rangle_{q|c,\beta} - \langle H\rangle^2_{q|c,\beta} && (3.23)\\
&= \sum_z \frac{A_z(A_z-1)}{B(B-1)}\psi_1(A_z) - \psi_1(B) && (3.24)\\
&\quad + \sum_z \frac{A_z-1}{B(B-1)}\psi_0^2(A_z) - \frac{1}{B}\left[\sum_z \frac{A_z-1}{B-1}\psi_0(A_z)\right]^2.
\end{aligned}$$

[9] Here we use the digamma function $\psi_0$ and the trigamma function $\psi_1$, where

$$\psi_n(x) = \frac{d^{n+1}}{dx^{n+1}}\log\Gamma(x). \tag{3.20}$$

In particular, the entropy mean and entropy variance for a Dirichlet prior is

$$\langle H\rangle_{q|\beta} = \psi_0(\beta m+1)-\psi_0(\beta+1) \tag{3.25}$$

$$\mathrm{var}\,(H)_{q|\beta} = \frac{\beta+1}{\beta m+1}\psi_1(\beta+1)-\psi_1(\beta m+1). \tag{3.26}$$

Nemenman, Shafee, and Bialek [81] (henceforth NBS) noted, though, that Wolpert and Wolf's entropy estimate,

$$H_{WW}(c) = \langle H\rangle_{q|c,\beta} \pm \sqrt{\mathrm{var}\,(H)_{q|c,\beta}}, \tag{3.27}$$

is typically quite bad unless one has distribution-saturating data. This is because Dirichlet priors $p(q|\beta)$ typically generate distributions with entropies $H(q)$ that fall only within a narrow range.[10] So while Dirichlet priors (especially the $\beta = 1$ prior) might seem like "flat" distributions of distributions, they are typically very strongly peaked in entropy – the variable we wish to estimate.

NSB argued that, when estimating entropy, it is vital that one use a prior on the space of distributions that supports a broad range of entropies. To this end, NSB proposed using a mixture of Dirichlet priors

$$p(q) = \int_0^\infty d\beta\; p(\beta)p(q|\beta), \quad p(\beta) = \frac{1}{\log m}\frac{d\,\langle H\rangle_{q|\beta}}{d\beta} \tag{3.28}$$

This prior effectively integrates $p(q|\beta)$ uniformly over $\langle H\rangle_{q|\beta}$, and thanks to the fact that $p(H|\beta)$ is nearly a delta-function centered around $\langle H\rangle_{q|\beta}$, this mixture model prior indeed produces a very uniform distribution of entropies $p(H) = \int d\beta\, p(\beta)p(H|\beta)$.

Computing the mean entropy given a set of counts $c$ and using this NSB prior $p(q)$,

$$\langle H\rangle_{q|c} = \int dq\; H(q)p(q|c) \tag{3.29}$$

$$= \int d\beta\; p(\beta|c)\left(\int dq\; H(q)p(q|c,\beta)\right) \tag{3.30}$$

$$= \left\langle \langle H\rangle_{q|c,\beta}\right\rangle_{\beta|c} \tag{3.31}$$

where

$$p(\beta|c) = \frac{p(c|\beta)p(\beta)}{p(c)} = \frac{p(\beta)}{Z_c}\frac{\Gamma(\beta m)}{\Gamma(\beta)^m}\frac{\prod_z \Gamma(c_z+\beta)}{\Gamma(N+\beta m)}. \tag{3.32}$$

for appropriate normalization constant $Z_c$. The variance is given by

$$\mathrm{var}\,(H)_{q|c} = \left\langle H^2\right\rangle_{q|c} - \langle H\rangle^2_{q|c} \tag{3.33}$$

$$= \left\langle \left\langle H^2\right\rangle_{q|\beta,c} - \langle H\rangle^2_{q|\beta,c}\right\rangle_{\beta|c} + \left\langle \langle H\rangle^2_{q|\beta,c}\right\rangle_{\beta|c} - \left\langle \langle H\rangle_{q|c}\right\rangle^2_{\beta|c} \tag{3.34}$$

$$= \left\langle \mathrm{var}\,(H)_{q|c,\beta}\right\rangle_{\beta|c} + \mathrm{var}\left(\langle H\rangle_{q|c,\beta}\right)_{\beta|c}. \tag{3.35}$$

[10] As $m$ grows with $\beta$ fixed, $\mathrm{var}\,(H)_{q|\beta} \sim 1/m$ while $\langle H\rangle_{q|\beta} \sim \log m$.

While the $\beta$ integrals in these formulas (including that needed to compute $Z_c$) cannot be done analytically, they can be integrated numerically without much fuss.

We note that, instead of the NSB prior $p(q)$, one would ideally like to use a prior of the form

$$p(q) = \delta\left(1 - \sum_z q_z\right) e^{f(H(q))} \tag{3.36}$$

where the function $f(H)$ is chosen so that the induced entropy distribution is flat on the interval $[0, \log m]$. Such a prior would be the maximum entropy distribution on the space of distributions that is consistent with a flat distribution in entropy. The NSB prior does not have this form, and one might worry that this discrepancy could be a problem. In practice, though, the NSB entropy estimate

$$H_{NSB}(c) = \langle H \rangle_{q|c} \pm \sqrt{\mathrm{var}\,(H)_{q|c}} \tag{3.37}$$

appears to work very well, and the author is unaware of any limits in which it produces problematic results.

To estimate the mutual information between a set of measurements $\{z_i\}$ and a set of sequence-dependent predictions $\{x_i\}$ we will henceforth use

$$I(z;x) = H_{naive}(\{c_z\}) + H_{naive}(\{c_x\}) - H_{NSB}(\{c_{zx}\}). \tag{3.38}$$

Here, the naive entropy is used to estimate the entropy of the marginal distributions, which are fixed by how we choose to sample/quantize the data. The NSB entropy estimate, on the other hand, is used to estimate the term which varies with the choice of model parameters. Note: the numerical values given by Eq. 3.38 for mutual information are in *nats* – i.e. natural information units. These values can be converted to bits by multiplying by $\log_2 e$.

### 3.3.3 Results

Using the procedures described in Sec. 3.3.1, we created a library of plasmids, each plasmid containing a *lacZ* promoter driving the expression of *gfpmut2*. These promoters were engineered to be polymorphic within a defined 26 bp region centered on the CRP binding site; on each plasmid, each position within this polymorphic region had a 24% probability of having a nucleotide which differed from that of the wild type sequence. A library of $\sim 5 \times 10^6$ independently cloned promoters was transformed into *E. coli* (q.v. Sec. 3.3.1) to produce the strain library sJK11 LIB.

96 mutant promoters drawn randomly from sJK11 LIB were sequenced; 91 of the resulting sequences were recognizable.[11] Of these, 23 were uncloned vector, 8 were erroneously cloned, and 60 were exactly cloned, giving a cloning success rate of $62\% \pm 5\%$. These 60 sites are aligned in Table 3.1.

[11] See the Sequencing subsection of Sec. 3.3.1 for definitions of the terms *recognizable, uncloned vector, erroneously cloned*, and *exactly cloned*.

| Exactly cloned sequences from sJK11LIB |
|---|
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTAGCTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTTATGTGCGCATGGGAACGCATTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAAGGTGAGGAAGCTATGTCAGTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATAATTGTGTGTTAACTCACTCGTTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTTCGTTATCACACTCACTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAAATAATGTCAGCTAGGTCATTCATCAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTGATATTATTGACCTCTCTCATAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTAGCAAGCTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTTATGATAGTTAGCAGGCTCATTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTGGTGTGAGTTACCTCAATCGTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTCAATGTGAATTAGCTCACTCTCGAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTCCATTGAGTTAGATTGCTCATAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATGAATCTAAGTTAGCTCACTTATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGCTTGCTCACACACAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATAAATGTGAGTCAGATCACTGACTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATATGGCTTAGTTCATTTATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAATGCAATTAATGTGCGGTATCTCAATCGTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGGGAGTTAGCTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCACTTCGTGTCATTTCGCTAACTTGTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTAACTCACGCACTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGTAATTAATCTTAGTTAGTTCTCCCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATACATGGGCGTTAGCTAACTCCCTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAAGCATTTTGATTTAAACGACTGATGAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAGTGTTAGTTAACCCACTCATCGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTGAAGGTTAATTGGCCGAAGCTTTTGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTATTGTGTTTTAGGGGACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAAATAATCATAGCTAGCTCATTCATAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGGGTTAGCAGACACATTTGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATTTTAGTTAGCTCATTCCTTTGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTTATAGGAACGAGCTGAGAAATTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAACGGCTTTGAGTTACCTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTATTACTAGTTAACTCCCTAACTTGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAACTAATGTGTGTTAGCTCACTCAATAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTAATGTGAGGAATCTCGCTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCACTCAATGTGAGTTTGTTCAGTCATTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTGATGTCGGTTCAATCACACTTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAGGGTTTGTGCCTTAGTTCACTGAATAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAGTGTGTGTTAGCTCACTCATTCGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGCGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTGAACATGCGTTAGCTCTCTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAAGCAATTTGAGGCAGCTGACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTAGCTGGCTCATTGGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTAATGTGCGGTAGGTCACTCAGTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATAGAATGTGAGTTTGGTCAGTGATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTCAATTAGCTCACAAAGTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTAGCACACTCTTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATCTGAGTTAGCTCTTTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAAAGGTCAGTTAGCTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAAACAATATGCGTTATCTAACTCAGCCGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTATTCTGTTTTAACTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTAATAAGGGCTAGCTCACTCATTCGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATAAATGAGAGTTAACTCACTCATAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAGTTAATATGAGAAAACTCATTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTATCATGAGGTAGCTCTATGGATAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGGAGACTTACGGATAAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATTTGTTTTAGGGTTCTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCACTTAGTGTGGGTTAACCCAATTTCCAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCATTTAATGTTAGTTGGTGCATTGATGAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTATTAAGAGTTCCATCAGTGATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGGGAGCTACCTCAATCGTTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTAATGTGAGTTACCTCACTCATTAGGCACCCCAGGCTTTACACTTTATGCTTCCGGCTCGTATGTTGTGTGGAA |
| TCGAGCACGACAGGTTTCCCGACTGGAAAGCGGGCAGTGAGCGCAACGCAATTACTGTCAGTTAGCTCACTCATTAGGCACCCTAGGCTTTACACTTTATGTTTCCGGCTCGTATGTTGTGTGGAA |

Table 3.1: An alignment of the 60 exactly cloned sJK11 LIB promoters; the 26 bp variable region is colored blue. For reference, the wild type sequence is shown at the top.

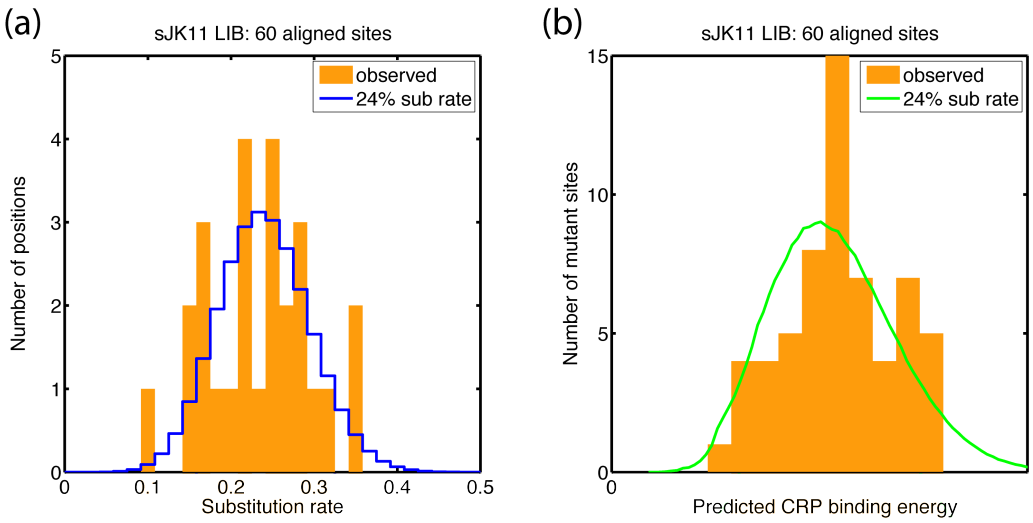


Figure 3.4: Polymorphic site statistics. (a) Distribution of substitution rates observed at the 26 variable positions, together with the distribution expected from a binomial substitution rate of 24%. (b) Distribution of binding site energies predicted by the Berg von Hippel CRP energy matrix, together with the distribution expected from a 24% substitution rate.

We examined the 26 bp mutant region of these 60 exactly cloned plasmids to make sure the sequence variation was as expected. Indeed, the 26 polymorphic positions exhibited a mean and RMSD substitution rate of $0.23 \pm 0.06$ among these 60 sites (see Fig. 3.4a); both of these statistics are consistent with a 24% binomial substitution model. Using a Berg von Hippel (BvH) energy matrix inferred from the sequences of 49 verified CRP binding sites (see Sec. 3.3.2), we also computed the distribution of BvH-predicted binding energies for these 60 sites. The mean and RMSD of this distribution is also consistent with that expected from the 24% binomial substitution model; see Fig. 3.4b. So in terms of both the position-specific substitution rate and a plausible distribution of binding energies, these sequences appear consistent with the way they were designed.

We then induced *gfpmut2* expression in the library sJK11 LIB. We also induced expression in a positive control population sJK11 + (carrying the wild type *lacZ* promoter), a negative control population sJK11 - (carrying a *lacZ* promoter with both the CRP and RNAP binding sites deleted), and seven clonal populations, sJK11 1 to sJK11 7, picked at random from sJK11 LIB (see Sec. 3.3.1). The fluorescence distribution from each of these populations is shown in Fig. 3.5.

The fluorescence distributions for sJK11 + and sJK11 - are well separated, demonstrating FACS fluorescence signals carry substantial information about *lacZ* promoter activity. The seven clonal populations sJK11 1-7 exhibit a nice variety of fluorescence distributions, all of which have widths substantially less than the total range of fluorescence observed across all samples. This shows that the fluorescence of a single cell can convey a substantial amount of information about the sequence of its plasmid's *lacZ* promoter region.

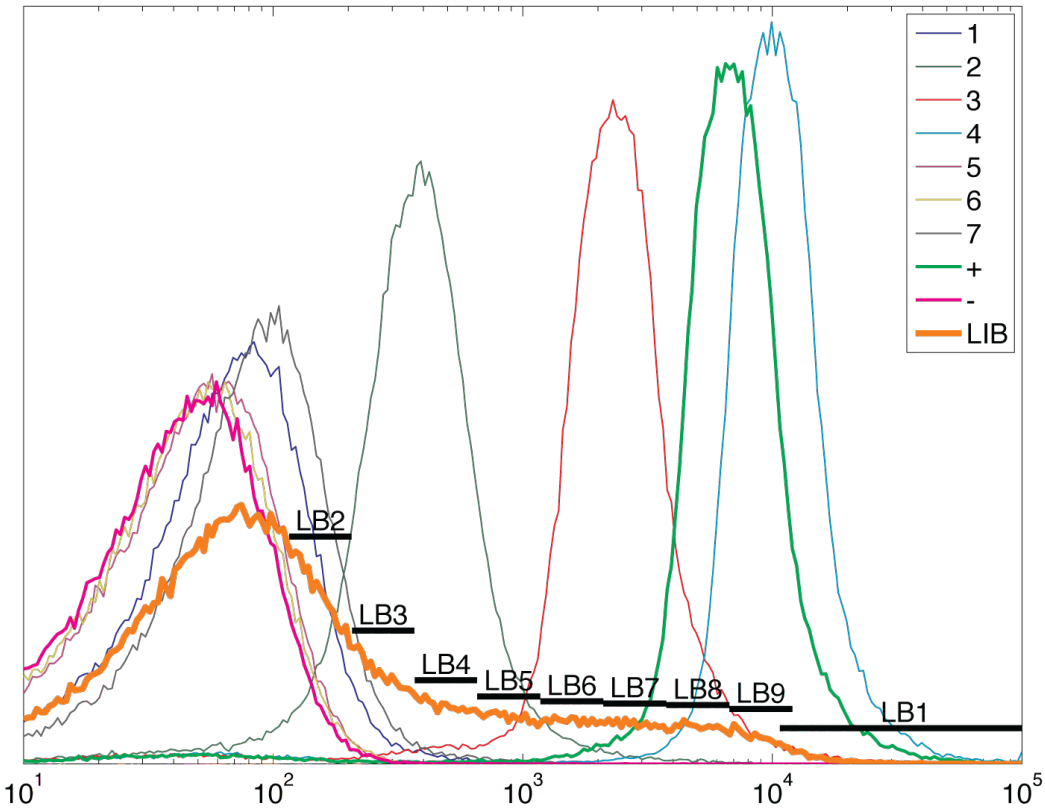


Figure 3.5: Fluorescence distributions for the library sJK11 LIB, the positive control sJK11 +, the negative control sJK11 -, and seven clonal populations sJK11 1 to sJK11 7. Black bars indicate the fluorescence range corresponding to each FACS batch, LB1 to LB9; $\sim 1.5 \times 10^4$ cells were sorted into each of these batches.

Upon further inspection we notice a few interesting features. The fluorescence distributions of clones sJK11 5 and sJK11 6 are nearly coincident with that of sJK11 -; sequencing of cPCR products confirmed that sJK11 5 and sJK11 6 indeed contain uncloned vector, and that all other clonal populations contain exactly cloned plasmids.

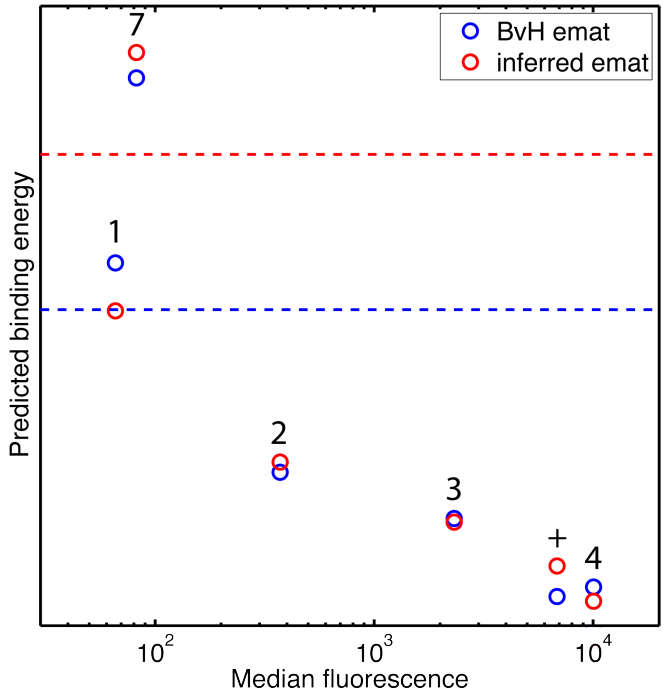


Figure 3.6: Each point shows the predicted energy of a CRP binding site from one of the clonal populations in Fig. 3.5 vs. the median fluorescence of cells from that population. BvH-predicted binding energies are indicated in blue, while binding energies predicted by an inferred energy matrix (to be described later) are shown in red. The clonal population is noted above each corresponding pair of data points. Dashed lines indicate the largest binding energy (BvH: blue; inferred: red) among the 49 verified CRP binding sites in DPInteract [30]. Note: we do not provide tick marks on the vertical axis because of the affine uncertainty of the energy matrix; indeed, binding energies were all rescaled and shifted so as to fit on the same plot.

We then asked how the BvH-predicted binding energy of each plasmid's CRP biding site corresponded to that plasmid's fluorescence distribution. Fig. 3.6 plots each binding site's predicted energy against the median resulting fluorescence (in blue). Overall, we find a strongly monotonic relationship between fluorescence and predicted binding energy; typically, the lower the predicted energy, the higher the median fluorescence. One exception to this monotonicity is between colonies 1 and 7, both of which have exactly cloned inserts but are barely more fluorescent than the negative control, colony $-$. Colony 7 is marginally more fluorescent than colony 1, while the latter's *lacZ* promoter is predicted to bind CRP more tightly. This is likely a result of both plasmids having CRP binding sites that have essentially been mutated away; both plasmids have sites with predicted binding energy larger than

all of the 49 verified CRP sites in DPInteract [30]. That one clone is marginally brighter than the other may have resulted from more subtle effects, such as other *E. coli* factors weakly binding the variable region and affecting transcription. Another possibility is that these differing sequences allow RNAP to bind alternate -10 and -35 sites (overlapping the 26 bp polymorphic region) at different levels (e.g. see [18]). Of course, this slight difference in colony fluorescence might also have resulted from minor differences in growth conditions.

Another exception is that the BvH energy matrix predicts stronger binding to the wild type (+) binding site than to colony 4's binding site, while colony 4 has brighter median fluorescence. We think this is due to inaccuracies in the BvH energy matrix. First of all, when the BvH matrix is shrunk to 22 bp, the binding site of colony 4 is predicted to have lower energy than that of colony +. Also, the binding energy predicted by the energy matrix we will soon infer from our own sequence data predicts (see red data points) that colony 4's site should have tighter binding energy than the wild type sequence. So while it is not perfect, Fig. 3.6 does indicate a strong relationship between predicted CRP binding energy and cellular fluorescence.

The fluorescence distribution of cells from sJK11 LIB (Fig. 3.5) is much wider than any of the clonal distributions; while the bulk of the distribution peaks near the medians of the colony 1 and colony 7 distributions (suggesting most CRP binding sites were rendered inoperative), $\sim 20\%$ of cells reside in a nearly flat tail extending up to roughly the median fluorescence of the wild type colony. We proceeded to sort cells from sJK11 LIB into nine different batches, labeled LB1 through LB9; each batch was given $\sim 1.5 \times 10^4$ cells. 32 mutant promoters were drawn from each FACS batch and sequenced (from cPCR products, see Sec. 3.3.1). Of these 288 sequences, 274 were recognizable, and 250 of these were exactly cloned, giving a success rate of $87\% \pm 2\%$. Only 4 plasmids were uncloned. It appears that, since only cells of substantial fluorescence were sorted, sorted plasmids were greatly enriched for properly cloned products. The resulting $\sigma z$ data set consisted of 310 sequences: 250 exactly cloned sequences from FACS batches LB1 through LB9, and 60 exactly cloned sequences from LIB.[12]

We checked whether BvH-predicted binding energy was informative about the FACS batch from which each sequence came. To do this, we used the BvH energy matrix to assign an energy to each sequence $\sigma_i$ in our data set. We then partitioned these energies into 5 equipopulated $x$-bins. Fig. 3.7a shows the joint counts $c_{zx}$ for all 310 sequences. Using these counts we estimate $I(z;x) = 0.54 \pm 0.07$ bits (see Sec. 3.3.2).

We then performed MMC sampling of energy matrices ($L = 26$ bp) using EMA likelihood as described in chapter 2 (see Sec. 3.3.2 for details). Fig. 3.7b shows the $\{c_{zx}\}$ counts for one energy matrix randomly drawn from this MMC ensemble; we estimate that, for this energy matrix, $I(z;x) = 0.67 \pm 0.07$ bits. This is typical

[12]For purposes of analysis, we will refer to cells from the initial library sJK11 LIB as being in FACS batch LIB. While these cells were not technically sorted, it is nevertheless useful (and kosher) to use their mutant promoter sequences in our analysis.

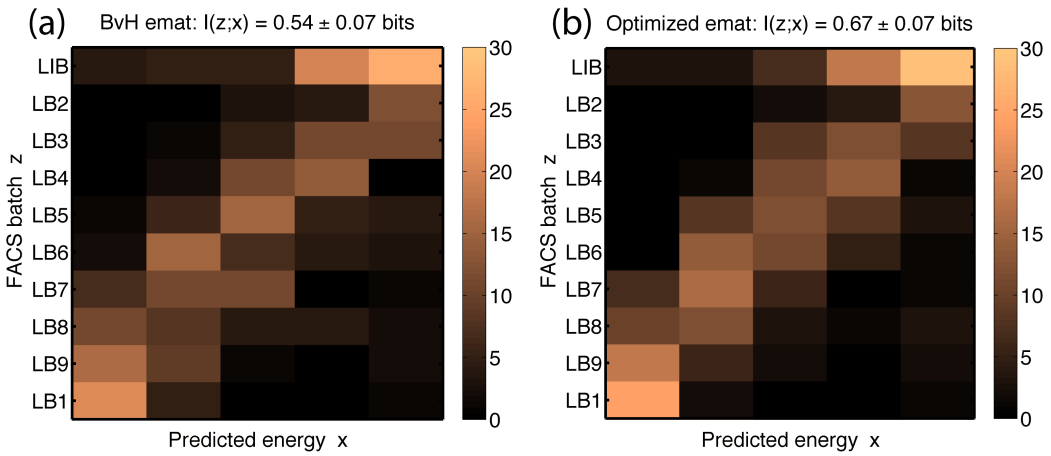


Figure 3.7: Predicted binding energy vs. FACS batch for (a) the BvH energy matrix and (b) a representative MMC-sampled energy matrix. Heat plots show raw $c_{zx}$ counts. FACS bins are ordered by median fluorescence (for visualization only; this ordering does not affect $I(z;x)$). Energies run from low (left) to high (right) on the horizontal axis of each panel; a numerical scale is not provided due to the affine uncertainty in predicted binding energy.

of matrices in the MMC ensemble. While this appears somewhat larger than the information $I(z;x)$ given by the BvH matrix, this difference is not significant.

We also tested our energy matrix predictions against previously published measurements of the CRP binding energies of a small number of sites (from [76]). Fig. 3.8 shows the relationship between these measured affinities and the binding energies predicted by (a) the BvH energy matrix and (b) our inferred energy matrix. Correlation coefficients and corresponding p-values are given in the figure caption. It appears that our inferred energy matrix gives energy predictions that are about as good as those of the BvH matrix.

We then sought to determine if our inferred energy matrix fully accounted for all of the the sequence information in the $\sigma z$ data; i.e. whether $I(z;x) \approx I(z;\sigma)$. To do this, we cultured the already-sequenced bacterial clones in 96-well plates. We measured the fluorescence distributions resulting from each of the 250 exactly cloned sequences from LB1 - LB9, as well as the fluorescence distributions of 43 of the 60 exactly cloned sequences from LIB (see Sec. 3.3.1). Bulk cells from the sJK11 LIB library were also cultured in this manner. Fig. 3.9 compares the fluorescence distribution from the plate-cultured library (Fig. 3.9b) to that from the sorted library (which was induced in a 15 ml tube, not in a 96-well plate; Fig. 3.9a). Fluorescence distributions from 96-well plate cultures were measured using a different machine (an LSR II) with different gain settings than the FACS Vantage SE used to sort the cells. Comparing the library distributions measured on these two machines allows us to calibrate one instrument relative to the other. In particular, it allows us to map the boundaries of the FACS bins onto the fluorescence scale produced by the LSR II; these boundaries were mapped by matching the quantiles spanned by each FACS batch in each of the

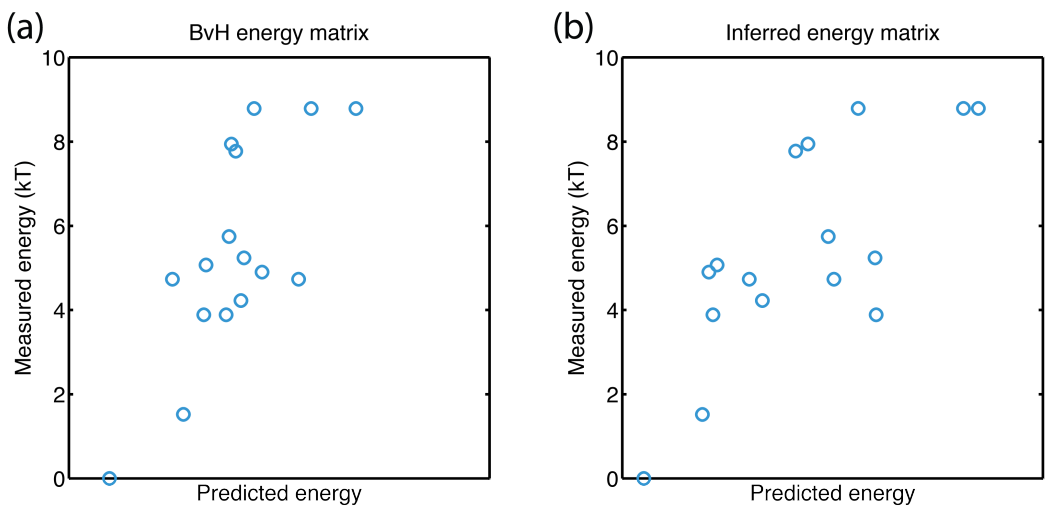


Figure 3.8: The predicted binding energy of CRP sites vs. their measured affinities (from [76]) in units of $kT$. (a) Energy predictions made using the BvH energy matrix. The resulting correlation coefficient is $\rho = 0.74$, with a corresponding p-value of $P = 9.3 \times 10^{-4}$, (b) Energy predictions made using the MMC-sampled energy matrix; $\rho = 0.75$, $P = 8.8 \times 10^{-4}$.

two library fluorescence distributions. It also allows us to verify that cultures grown in 96-well plates were induced under conditions similar to those used prior to cell sorting.

All of the fluorescence distributions measured for exactly cloned sequences are shown in Fig. 3.10, and are grouped according to the FACS batch from which each cloned sequence was drawn. While there are some outlier distributions, most overlap well with the fluorescence range of their native FACS batch. For each of these sequences $\sigma$, we then computed the fraction $w_{z|\sigma}$ of $\sigma$'s fluorescence distribution which overlaps the fluorescence range spanned by *each one* of the $z$-bins. We then used this set of weights $\left\{w_{z|\sigma}\right\}$ to compute $p(z|\sigma)$ for each of these sequences (see Sec. 3.3.2). The conditional distributions $p(z|\sigma)$ are shown in Fig. 3.11. As explained in Computational methods, these conditional distributions allowed us to estimate $I(z;\sigma) = 1.25 \pm 0.02$ bits. This sequence information is substantially larger than $I(z;x)$ produced by either the BvH energy matrix ($0.54 \pm 0.07$ bits), the MMC sampled energy matrix used in Fig. 3.7 ($0.63 \pm 0.07$ bits), or any of the MMC sampled energy matrices. Does this discrepancy between $I(z;x)$ (for the MMC-sampled energy matrix) and $I(z;\sigma)$ really indicate the need for a more complex model of how the 26 bp region in question affects transcription?

Before addressing this point directly, we note that there are clear differences between the two library distributions in Fig. 3.9; the sorted library has a fatter and flatter tail containing about 4/3 as many cells as the plate-cultured library. This is likely due to poor oxygenation of the cultures grown in the 96-well plates, evidenced by the fact that the plate-cultured library grew to only 1/4 the density of the sorted library. This indicates substantial differences in the growth conditions of sorted cells

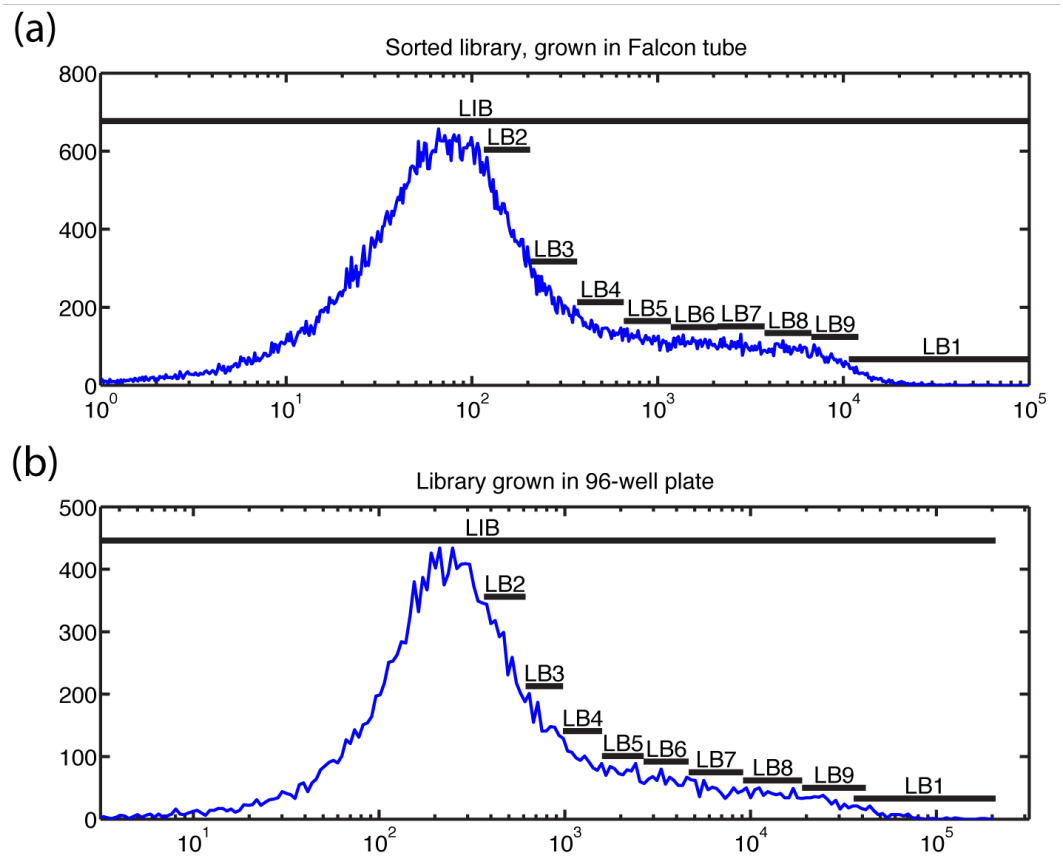


Figure 3.9: Fluorescence distributions for (a) $10^5$ cells from the FACS-sorted library sJK11 LIB, measured on the FACS Vantage SE, and (b) $2 \times 10^4$ cells from sJK11 LIB cultured in a 96-well plate and measured on the LSR II. Note that the fluorescence scale (horizontal axis) is substantially different for these two distributions; this is a result of using two different machines with different gain settings. Fluorescence ranges for each FACS batch were mapped from (a) to (b) by matching up the quantiles these ranges spanned in each of the two distributions.

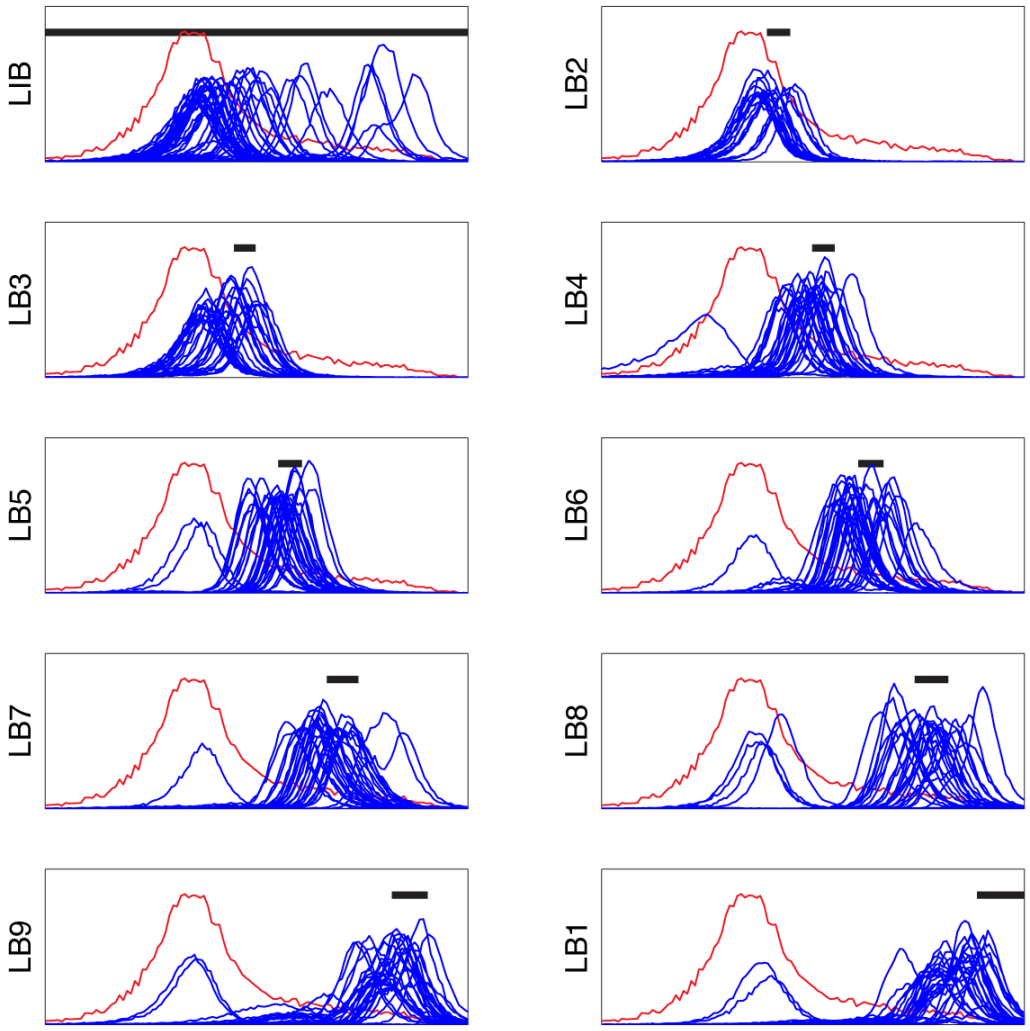


Figure 3.10: Fluorescence distributions (blue) measured for 293 exactly cloned promoters. These distributions are grouped according to the FACS batch from which the corresponding promoters were drawn. The library distribution (shown in red and enlarged for visibility) is shown in the background. The black bar in each plot indicates the fluorescence span of the corresponding FACS batch.

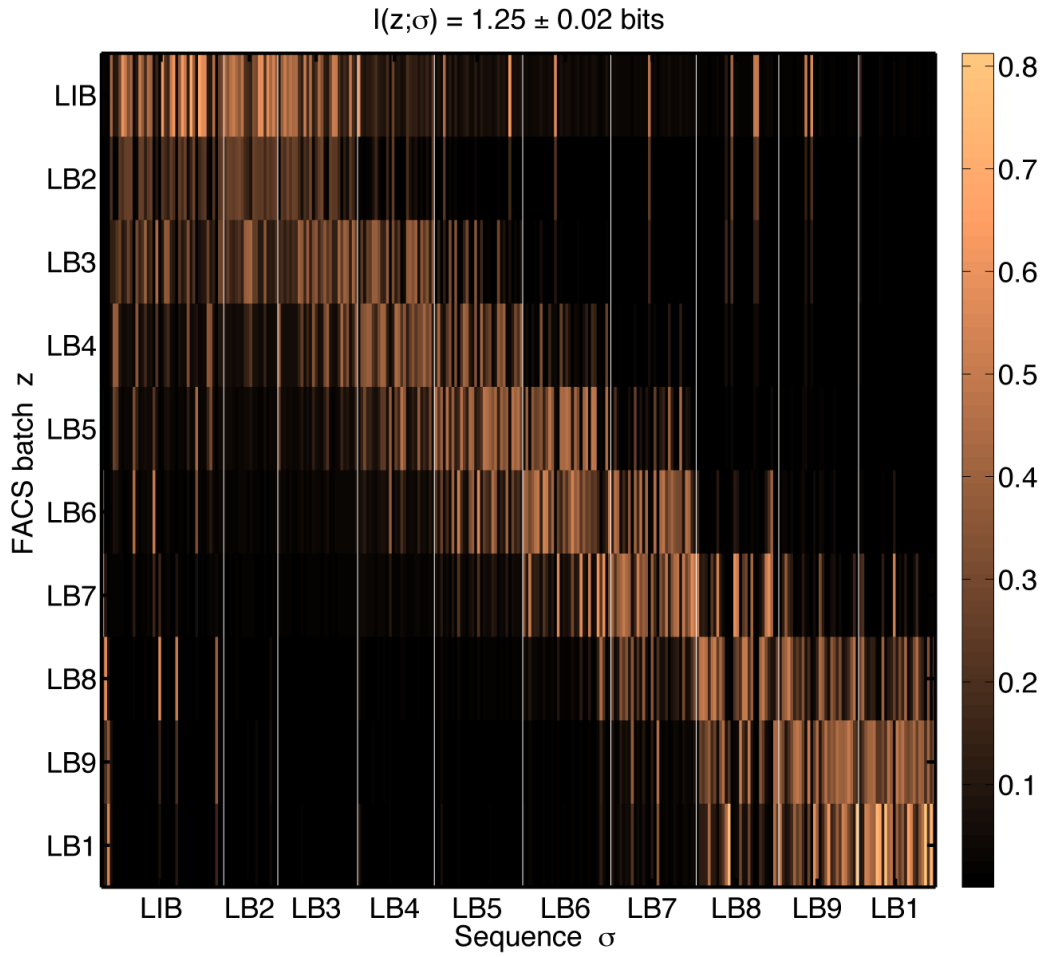


Figure 3.11: This heat plot shows the conditional distribution $p(z|\sigma)$ for each of the 293 sequences for which we measured fluorescence distributions. FACS batches $z$ are listed on the vertical axis, while sequences $\sigma$ are arrayed along the horizontal axis, grouped according to the FACS batch from which they were drawn; sequences from different batches are separated by vertical white lines.

and cells grown in 96-well plates. This discrepancy is also revealed in the fluorescence distributions of highly-fluorescent populations in Fig. 3.10; many of these distributions exhibit either a plateau or two peaks, indicating a heterogeneous population of cells.

This discrepancy between growth conditions used for sorting and those used for culturing in 96-well plates might be cause for concern, and we plan to rectify this problem in the future. However, we suspect that this would most likely cause us to *underestimate* $I(z;\sigma)$. So the insufficiency of the BvH and MMC-sampled CRP energy matrices is likely *worse* than our results suggest. Another possibility is that SNPs elsewhere in the assayed promoters affect promoter function. Some exactly cloned plasmids, as we've defined them, do exhibit SNPs elsewhere in the promoter. The presence of such SNPs appear to be the exception rather than the rule, though, and it is unlikely the small fraction of sequences containing such SNPs could reduce $I(z;x)$ by a factor of two.

In the end, it seems most likely that this disparity between the predictive information of our model and the sequence information in the data truly reflects the inadequacy of our model of promoter function. One possible explanation for this discrepancy is that energy matrices are inadequate (in this context) for describing CRP's sequence-dependent binding energy. Another is the effect that mutations within the 26 bp polymorphic region have on RNAP binding at alternative transcription start sites; at least two such alternative start sites are known, and one of these involves RNAP binding which overlaps this 26 bp region [18]. In any case, we clearly see the importance of having a standard against which to judge the completeness of inferred models.

### 3.3.4 Discussion

This experiment was done to demonstrate the feasibility of Sort-Seq, and to show that biologically relevant information can be inferred from the resulting $\sigma z$ data. We were able to create a library containing millions of different promoters, each promoter mutated at a specific rate within a specified region. Upon induction of GFP expression, cellular fluorescence was shown to provide substantial information about the underlying sequence perturbations. The resulting $\sigma z$ data, characterizing the function of only 310 sequences, was enough to infer an energy matrix for CRP (containing 35 independent parameters) which is in close agreement with the Berg von Hippel CRP energy matrix derived from verified binding sites. We also demonstrated how to estimate $I(z;\sigma)$ by measuring the fluorescence distributions of clonal populations carrying a representative set of promoter sequences. Our results strongly suggest that a simple CRP energy matrix model is not enough to account for the functional influence of the 26 bp region we probed, though this claim needs to be checked with additional data.

The most glaring difference between Sort-Seq as we proposed it and as demonstrated here is the paltry number of promoter sequences (310) we obtained in our data set. Sort-Seq was designed specifically with 454 pyrosequencing in mind; one run of this ultra-high-throughput sequencer could easily produce a $\sigma z$ data set con-

taining hundreds of thousands of promoter sequences. The availability of so much data would allow one to gainfully probe a much larger sequence footprint, and infer/test a wide range of biophysical models. In the next chapter we briefly sketch how such data might be used to characterize the architecture of a transcriptional regulatory sequence *de novo*.

We now return to theoretical issues regarding what one can and cannot infer by maximizing mutual information. In the process we uncover a subtle aspect of the data that Sort-Seq might produce when probing the simultaneous binding of multiple regulator proteins per sequence: natural nonlinearities in corresponding thermodynamic models of transcriptional regulation might allow one to determine protein-DNA and protein-protein interaction energies in absolute thermal units.

## 3.4 Diffeomorphic modes

In chapter 2 we showed that if a deterministic function $x(\sigma;\theta^*)$ defined by parameters $\theta^*$ underlies the sequence-dependent measurements $\{z_i\}$ of a $\sigma z$ data set, i.e.

$$\sigma \xrightarrow{x(\sigma;\theta^*)} x^* \xrightarrow{p(z|x^*)} z,$$

then in the plentiful data limit the mutual information $I(z;x)$ between $\{z_i\}$ and the sequence-dependent predictions $\{x_i = x(\sigma_i;\theta)\}$ will be maximized by choosing $\theta = \theta^*$. Here we address the converse issue: will maximizing $I(z;x)$ lead one to the correct set of model parameters, i.e. $\theta = \theta^*$?

We first show that the invariance of mutual information to monotonic deformations of its input variables can lead to *diffeomorphic modes* in parameter space, along which mutual information is invariant. Values of model parameters along these directions in parameter space cannot, as a result, be pinned down by maximizing $I(z;x)$. In the previous chapter, this problem was essentially the reason we lacked sensitivity to the units of the energy matrices we inferred; ideally we would like to characterize all energy matrix elements in units of $kT$. By contrast, maximum likelihood inference with an assumed error model does not permit diffeomorphic models. Indeed, this is one advantage that maximum likelihood has over mutual information maximization.

Fortunately, this insensitivity of mutual-information-based sequence analysis to binding energies in absolute units of $kT$ is not fundamental, but rather a byproduct of the simplicity of the protein-DNA interaction models we have considered so far. Thermodynamic models describing more than one protein molecule binding to each DNA sequence naturally contain nonlinearities, and these nonlinearities very often break diffeomorphic modes. In many cases, data probing the interactions of multiple proteins per sequence should be able to pin down *all* relevant binding energies in natural thermal units.

### 3.4.1 Origin of diffeomorphic modes

Consider, as before, a model $x(\sigma;\theta)$ which assigns to each sequence $\sigma_i$ a corresponding prediction $x_i$. These model-dependent predictions, together with the set of sequence-

dependent measurements $\{z_i\}$, determine a joint distribution $f(z,x)$, which in turn determines $I(z;x)$,[13]

$$I(z;x) = \sum_z \int dx\ f(z,x) \log \frac{f(z,x)}{f(z)f(x)}. \tag{3.39}$$

Now assume there is a change of model parameters $\theta \to \theta'$ which changes the values of the model's predictions $x_i \to x'_i = x(\sigma_i;\theta')$, but preserves their rank-order, i.e. $x_i > x_j \Rightarrow x'_i > x'_j$. We can then express $x'$ as a monotonic function of $x$ and vice versa. The jacobian of this transformation cancels out in the formula for mutual information, which is therefore left unchanged. Using $f'(z,x')$ to denote the joint distribution of measurements and predictions after this transformation,

$$\begin{aligned} I(z;x') &= \sum_z \int dx' f'(z,x') \log \frac{f'(z,x')}{f'(z)f'(x')} && (3.40)\\ &= \sum_z \int dx f(z,x) \log \frac{|\frac{dx}{dx'}| f(z,x)}{|\frac{dx}{dx'}| f(z)f(x)} && (3.41)\\ &= I(z;x). && (3.42) \end{aligned}$$

If such a transformation exists for all $\theta$, then maximizing mutual information will not allow us to pin down values for all of our model parameters; at best we can determine $\theta$ modulo such transformations.

There are actually two distinct classes of such transformations. One class is dependent on the particular sequences $\{\sigma_i\}$ in our data set. The set of model predictions $\{x_i\}$ will always be a discrete set of real numbers, so one can almost always find transformations $\theta \to \theta'$ so minute that none of the $\{x_i\}$ change rank order. Such microtransformations are of little concern when the number of sequences probed is large.

The more problematic type are macroscopic, continuous transformations $\theta \to \theta'$ for which the corresponding change in model predictions $x \to x'$ depends only on $x$ with no residual dependence on $\sigma$. We call these parameter transformations *diffeomorphic*, since they effect a diffeomorphism on model predictions independent of the underlying sequences. Mutual information maximization is unable to pin down model parameters along diffeomorphic modes regardless of the data set size.

Diffeomorphic modes correspond to nontrivial solutions, $h(\cdot)$ and $\vec{g}(\cdot)$, to the generator equation

$$h(x)\partial_x = \vec{g}(\theta) \cdot \nabla_\theta. \tag{3.43}$$

Here, $h(x)$ is some real function of $x$ and $\vec{g}(\theta)$ is a vector field in $\theta$-space. The right-hand side of this equation is a generator of transformations in parameter space. The left-hand side is a generator of diffeomorphisms of $x$. Equality means that such transformations of model parameters induce a diffeomorphism in $x$. Applying both

[13] For now we will assume that $x$ is a continuous variable while $z$ is discrete.

sides of this equation to $x$, we see that diffeomorphic modes correspond to vector fields $\vec{g}(\theta)$ for which

$$\vec{g}(\theta) \cdot \nabla_\theta x(\sigma;\theta) = h(x) \tag{3.44}$$

can be expressed as a function of $x(\sigma;\theta)$, without any additional dependence on $\sigma$. Indeed, the fact that Eq. 3.44 results in diffeomorphic modes can be seen from the formula for $\nabla_\theta I(z;x)$ (see Appendix for derivation):

$$\nabla_\theta I(z;x) = \left\langle [\nabla_\theta x(\sigma;\theta)] \frac{d}{dx} \log \frac{f(x|z)}{f(x)} \right\rangle_\sigma \tag{3.45}$$

$$= \int dx\ f(x) \sum_z \langle \nabla_\theta x \rangle_{\sigma|z,x} \frac{d}{dx} \frac{f(x,z)}{f(x)}, \tag{3.46}$$

If Eq. 3.44 holds, then $\vec{g}(\theta)\cdot\langle\nabla_\theta x\rangle_{\sigma|z,x} = h(x)$ will have no dependence on $z$. $\langle\nabla_\theta x\rangle_{\sigma|z,x}$ can therefore be moved out of the sum in Eq. 3.46, rendering

$$\vec{g}(\theta) \cdot \nabla_\theta I(z;x) = \int dx\ f(x) h(x) \frac{d}{dx} \frac{\sum_z f(x,z)}{f(x)} = 0. \tag{3.47}$$

On the other hand, if $\vec{g}(\theta)\cdot\nabla_\theta x$ depends on $\sigma$ independently of $x$, then $\vec{g}(\theta)\cdot\langle\nabla_\theta x\rangle_{\sigma|z,x}$ will depend on $z$ (through the set of sequences one averages over) and $\langle\nabla_\theta x\rangle_{\sigma|z,x}$ cannot be factored out of the sum.

We note that diffeomorphic degrees of freedom are qualitatively different from *gauge* degrees of freedom, which correspond to vector fields $\vec{g}(\theta)$ such that

$$\vec{g}(\theta) \cdot \nabla_\theta x(\sigma;\theta) = 0, \tag{3.48}$$

i.e. leave $x$ invariant. No inference method can pin down these degrees of freedom because they are either physically meaningless or they are not probed by the experiment. While gauge modes are technically diffeomorphic modes, they are trivial ones and are not of interest to us. Throughout this section we will concern ourselves only with *proper* diffeomorphic modes, i.e. modes for which $h(x)$ in Eq. 3.43 is not identically zero.

Before moving on to examples, we note that likelihood, unlike mutual information, is generically immune to diffeomorphic modes; this is one important advantage that maximizing likelihood has over maximizing mutual information. As discussed in the last chapter, likelihood is related to mutual information through

$$p(\{z_i\} \,|\, \{x_i\}) \propto \exp N[I(z;x) - D(f||p)] \tag{3.49}$$

where the constant of proportionality is independent of $\theta$, and

$$D(f||p) = \sum_z \int dx f(z,x) \log \frac{f(z|x)}{p(z|x)}. \tag{3.50}$$

A diffeomorphic transformation $\theta \to \theta'$ causes $f'(z, x'(x)) = |\frac{\partial x}{\partial x'}| f(z, x)$ and $f'(z|x'(x)) = f(z|x)$. In other words, $f(z, x)$ transforms as a distribution. However, the error model $p(z|x)$, which is posited *a priori*, remains invariant. Thus,

$$D(f'||p) = \sum_z \int dx' f'(z, x') \log \frac{f'(z|x')}{p(z|x')} \tag{3.51}$$

$$= \sum_z \int dx f(z, x) \log \frac{f(z|x)}{p(z|x'(x))} \tag{3.52}$$

$$= D(f||p) + \sum_z \int dx f(z, x) \log \frac{p(z|x)}{p(z|x'(x))}. \tag{3.53}$$

The right-most term in Eq. 3.53 will not typically be zero. We therefore see that $D(f||p)$ is not generally invariant under diffeomorphic models.

This does not mean, of course, that likelihood doesn't have nullclines in $\theta$-space; typically it has to. The important question is whether or not likelihood exhibits nullclines at $\theta = \theta^*$. The answer is no. When model parameters have their correct values, $f(z|x) = p(z|x)$ for all $z$ and $x$, and so $D(f||p) = 0$. For a change of parameters $\theta \to \theta'$ to leave $D(f'||p) = 0$ as well, the error model must satisfy

$$p(z|x'(x)) = f'(z|x'(x)) = f(z|x) = p(z|x) \tag{3.54}$$

for all $x$ at which $f(z, x)$ has support. The only way this can happen is if the experiment itself is insensitive to the deformation of $x$-values. This would indicate a problem with the experiment, not the method of inference. So as long as the experiment does indeed probe the variable one is modeling, maximizing likelihood will allow one to infer model parameters up to gauge transformations.

### 3.4.2 Single-site binding models

We now discuss the diffeomorphic modes of single-site binding models, that is, models describing the binding of at most one protein molecule per sequence $\sigma$. Consider again the proof-of-principle experiment in which we modeled only the binding of only one CRP dimer to each 26 bp sequence $\sigma_i$; in doing so we used an energy matrix, arguably the simplest quantitative model of protein-DNA binding. In this model, the SDBE of CRP to a sequence $\sigma$ is given by

$$x(\sigma, \theta) = \sum_{bl} \epsilon_{bl} \sigma_{bl} \tag{3.55}$$

where $\theta = \{\epsilon_{bl}\}$ are the elements of the $4 \times L$ energy matrix.

Energy matrix models, such as this, permit two diffeomorphic modes:

$$G_{add} = \partial_x = \frac{1}{L} \sum_{bl} \frac{\partial}{\partial \epsilon_{bl}} \tag{3.56}$$

generates additive shifts in $x$, while

$$G_{mult} = \partial_{\log x} = x \partial_x = \sum_{bl} \epsilon_{bl} \frac{\partial}{\partial \epsilon_{bl}} \tag{3.57}$$

generates a multiplicative rescaling of $x$. Gauge freedoms allow multiple ways of representing these generators in parameter space; two such ways are shown above.

Whether $G_{add}$ is of concern to us is debatable, since only energy differences are physically relevant here. The absolute value of these energies becomes an issue only when they are presumed to be relative to some chemical potential, and whether this is the case depends on one's data and one's intentions in fitting models to them. On the other hand, the multiplicative generator $G_{mult}$ is problematic. The absolute scale of binding energy has very real physical consequences, and the invariance of $I(z;x)$ to transformations generated by $G_{mult}$ means that a maximally informative analysis of the $\sigma z$ data cannot reveal the values of energy matrix elements in natural thermal units – not without additional information or assumptions about $p(z|x)$.

In fact, $G_{add}$ and $G_{mult}$ are the only diffeomorphic generators that energy matrices of length $L \geq 2$ can have. To show this, take partial derivatives of Eq. 3.44 with respect to the parameters $\epsilon_{bl}$ and $\epsilon_{b'l'}$, $l \neq l'$:

$$\frac{\partial^2 h}{\partial \epsilon_{bl} \partial \epsilon_{b'l'}} = h''(x)\sigma_{bl}\sigma_{b'l'} = \sum_{b''l''} \frac{\partial^2 g_{b''l''}(\theta)}{\partial \epsilon_{bl} \partial \epsilon_{b'l'}} \sigma_{b''l''}. \tag{3.58}$$

Here we used the fact that $\partial x / \partial \epsilon_{bl} = \sigma_{bl}$. Now consider four different sequences $\sigma^1, \sigma^2, \sigma^3, \sigma^4$, all identical except at positions $l$ and $l'$, where they have the following bases ($b \neq c$ and $b' \neq c'$),

| | $\sigma^1$ | $\sigma^2$ | $\sigma^3$ | $\sigma^4$ |
|---|---|---|---|---|
| $l$ | $b$ | $b$ | $c$ | $c$ |
| $l'$ | $b'$ | $c'$ | $b'$ | $c'$ |

Using Eq. 3.58, we find

$$\frac{\partial^2}{\partial \epsilon_{bl} \partial \epsilon_{b'l'}} \Big[ (h(x^1) - h(x^2) - h(x^3) + h(x^4) \Big] = h''(x^1) = 0. \tag{3.59}$$

$h(x)$ is thus constrained to be a linear function of $x$, proving that $\partial_x$ and $x\partial_x$ (and linear combinations thereof) are the only diffeomorphic modes that such energy matrices can have.

The presence of these diffeomorphic modes is not an artifact of assuming an energy matrix, i.e. assuming each nucleotide within a binding site contributes independently to the protein's sequence-dependent binding energy. For instance, allowing couplings between all positions,

$$x(\sigma;\theta) = \sum_{bb'll'} \epsilon_{bb'll'} \sigma_{bl} \sigma_{b'l'}, \tag{3.60}$$

still permits both additive and multiplicative diffeomorphic modes:

$$G_{add} = \frac{1}{L^2} \sum_{bb'll'} \frac{\partial}{\partial \epsilon_{bb'll'}}, \quad G_{mult} = \sum_{bb'll'} \epsilon_{bb'll'} \frac{\partial}{\partial \epsilon_{bb'll'}}. \tag{3.61}$$

Indeed, additive and multiplicative diffeomorphic modes result from the model being linear in $\theta$; linearity in $\sigma$ is irrelevant.

One can of course employ a model $x(\sigma;\theta)$ that is nonlinear in $\theta$ simply by choosing to represent protein-DNA occupancy instead of binding energy, e.g.

$$x(\sigma;\theta) = \left(1 + \exp\left[\sum_{bl} \epsilon_{bl}\sigma_{bl}\right]\right)^{-1} \tag{3.62}$$

But this does nothing to dispel diffeomorphic modes; every monotonic function of $x(\cdot;\theta)$ retains the same diffeomorphic modes as $x(\cdot;\theta)$.

In order to eliminate diffeomorphic modes, the models that one considers need to be *naturally nonlinear* in their parameters. The author is unable to think of any naturally nonlinear model for one factor binding to a single site. In fact, it seems reasonable that, in some fundamental sense, the energy matrix really is the simplest model of protein-DNA interaction, and that complicating this model will not rid it of any diffeomorphic modes. We conclude that $\sigma z$ data probing the binding of a factor to only one binding site per sequence is fundamentally incapable of determining, through a maximally informative analysis, the absolute scale of transcription-factor-DNA binding energy; for such systems, energy scales can be inferred from $\sigma z$ data only after making assumptions about the experimental error model $p(z|x)$.

### 3.4.3 Exploiting natural nonlinearities

On the other hand, thermodynamic models of transcriptional regulation involving the simultaneous binding of multiple proteins per sequence typically *are* naturally nonlinear in their parameters (e.g. [11, 9, 10, 13]). This introduces the possibility that Sort-Seq data probing such systems might be used to infer protein-DNA and protein-protein interaction energies in natural thermal units. We illustrate this explicitly with two theoretical examples and one simulation.

#### Recruitment of RNAP by a single transcription factor

Consider the model of RNAP recruitment by a transcription factor TF, depicted in Fig. 3.12a. The equilibrium occupancy of RNAP at its binding site is given by

$$\mathcal{O}(\sigma;\theta) = \frac{1}{1 + x(\sigma;\theta)^{-1}}, \;\; x(\sigma;\theta) = e^{-\alpha(\sigma)}\frac{1 + e^{-\beta(\sigma)-\gamma}}{1 + e^{-\beta(\sigma)}} \tag{3.63}$$

where $x(\sigma;\theta)$ is the *regulation factor* (after Bintu et al. [9, 10]; see chapter 1). The parameters $\theta$ include $\gamma$, as well as all parameters defining the functions $\alpha(\cdot)$ and $\beta(\cdot)$.

Imagine a Sort-Seq experiment in which the sequence $\sigma$, spanning both the RNAP and TF binding sites (Fig. 3.12a), is specifically randomized in the mTRS library. We now show that, on the resulting $\sigma z$ data, the regulation factor $x(\sigma;\theta)$ will have only one diffeomorphic mode, corresponding to changes in the concentration of RNAP.

First notice that any diffeomorphic generator $G$ of $x$ must also be a diffeomorphic generator of *both* $\alpha$ and $\beta$. For instance, $\{\sigma_i\}$ might contain sequences for which $\beta$ is fixed and $\alpha$ varies. Since $x(\sigma;\theta)$ is a monotonic function of $\alpha(\sigma)$ for fixed $\beta$, any diffeomorphic generator $G$ of $x$ must be such that $G\alpha(\sigma)$ is sequence-independent

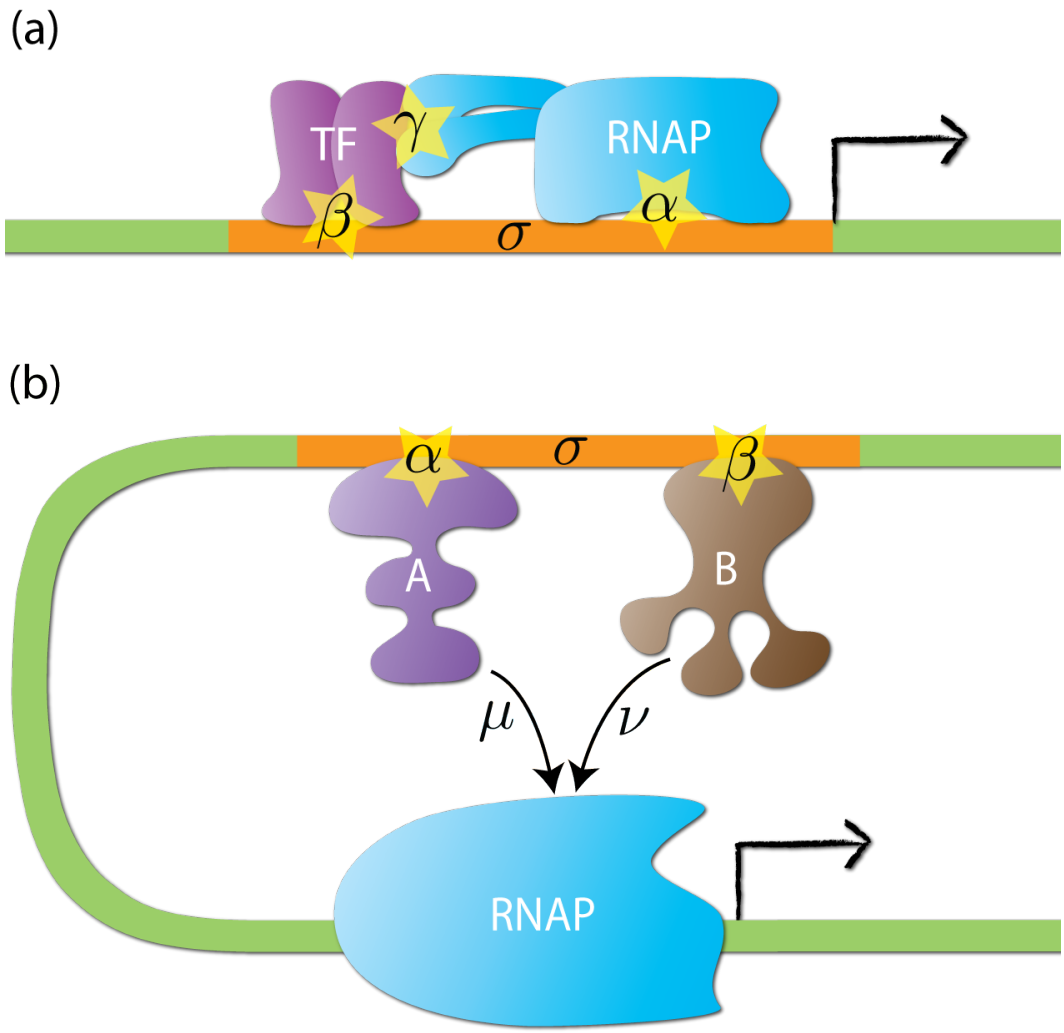


Figure 3.12: Two simple models of transcriptional regulation. (a) A model of RNAP recruitment by a transcription factor TF. $\sigma$ (shown in orange) is the sequence of the region spanning both the TF and RNAP binding sites, $\alpha(\sigma)$ is the sequence-dependent binding energy of RNAP (with the chemical potential subtracted out), $\beta(\sigma)$ is the binding energy of TF (ditto), and $\gamma$ is the interaction energy between RNAP and TF. All of these energies are assumed to be in units of $kT$. (b) A model of a billboard enhancer of sequence $\sigma$. $\alpha(\sigma)$ is the binding energy of factor $A$, and $\beta(\sigma)$ is the binding energy of factor $B$. We assume these factors independently activate transcription in proportion to their respective occupancies; $\mu$ and $\nu$ are the corresponding proportionality constants.

for given $\alpha$. Similarly, $G\beta(\sigma)$ must be sequence-independent for given $\beta$. Such a generator must therefore be expressible in terms of the diffeomorphic generators of $\alpha$, $\beta$, and $\gamma$. If $\alpha$ and $\beta$ are energy matrices, $G$ must therefore be of the form

$$G = (A + B\alpha)\partial_\alpha + (C + D\beta)\partial_\beta + E\partial_\gamma \tag{3.64}$$

where the coefficients $A, B, C, D$, and $E$ can depend arbitrarily on any of the parameters $\theta$, but cannot be sequence-dependent. Fixing the value of $x$, we can solve for $\alpha$ in terms of $\beta$ and $\gamma$, giving

$$Gx = -\left[A - B\log\left\{x\frac{1+e^{-\beta}}{1+e^{-\beta-\gamma}}\right\} + \right. \tag{3.65}$$

$$\left. (C + D\beta)\frac{e^{-\beta}\left(1-e^{-\gamma}\right)}{\left(1+e^{-\beta-\gamma}\right)\left(1+e^{-\beta}\right)} + E\frac{e^{-\beta-\gamma}}{1+e^{-\beta-\gamma}}\right] x \tag{3.66}$$

Since $\beta$ is sequence-dependent, $Gx$ is, for fixed $x$, sequence-independent only if $B = C = D = E = 0$.[14] This means the only diffeomorphic mode inherent in this model is $G = \partial_\alpha$, which generates additive shifts in the RNAP chemical potential. That this is, in fact, a diffeomorphic mode can be seen by simply looking at the formula for $x(\sigma;\theta)$, which scales multiplicatively when $\alpha$ is shifted by a constant.

Importantly, $x(\sigma;\theta)$ has no other diffeomorphic modes. This means that fitting maximally informative models of the form Eq. 3.63 to $\sigma z$ from an appropriate Sort-Seq experiment might determine both protein-DNA and protein-protein interaction energies in natural thermal units. Furthermore, the fact that $\partial_\beta$ is not a diffeomorphic mode of $x(\sigma;\theta)$ means that one might also gain access to TF-DNA binding energy relative to the TF chemical potential. Appropriate $\sigma z$ data might therefore allow us to probe intracellular transcription factor concentrations.

### A simple billboard enhancer

Transcriptional regulation in eukaryotes – especially multicellular ones – is largely effected by sequences called *enhancers* [4]. These sequences are often positioned quite far from transcription start sites (e.g. $10^3$ - $10^6$ bp), and the precise way in which the transcription factors they bind interact with RNA polymerase is still poorly understood. At some enhancers, such as that of the human interferon-$\beta$ gene (see [4]), bound proteins interact with each other in a very specific manner, and any substantial alteration to any factor's binding site will destroy enhancer function. Such enhancers have been termed *enhanceosomes* [109], which emphasizes the importance of the protein-DNA complexes they form. Other enhancers, such as the *eve* stripe 2 enhancer in *Drosophila melanogaster* ([21, 22, 23]; see also [4]), appear to be very

[14] That is, unless $\gamma = 0$ (i.e. there is no coupling between RNAP and CRP), in which case

$$G = (A + B\alpha)\partial_\alpha + (C + D\beta)\partial_\beta$$

is diffeomorphic for all $A$, $B$, $C$, and $D$. Of course in this case $\partial_\beta$ and $\beta\partial_\beta$ would be gauge degrees of freedom, as $x$ would be completely independent of $\beta$.

modular and flexible. It is possible to manipulate the sequences of such enhancers, e.g. to eliminate binding sites, while only partially affecting enhancer function. These types of enhancers have been termed *billboard enhancers* [109], since they appear to act as simple information displays as opposed to scaffolds for rigid protein-DNA complexes.

Natural nonlinearities seem to arise in nearly all thermodynamic models describing the interactions of multiple proteins per DNA sequence; in particular, these proteins need not interact with one another to generate nonlinearities. A simple model of a billboard enhancer illustrates this point. Consider the billboard enhancer model shown in Fig. 3.12b. This model enhancer contains two well-separated binding sites for two different transcription factors, and factor molecules bound to either site activate transcription in proportion to their occupancy (with the different factors having different coefficients of recruitment). We further assume that two simultaneously bound transcription factor molecules do not otherwise interact. This leads to transcriptional regulation governed by the quantity

$$x(\sigma;\theta) = \frac{\mu}{1+e^{\alpha(\sigma)}} + \frac{\nu}{1+e^{\beta(\sigma)}}, \tag{3.67}$$

where $\alpha(\sigma)$ and $\beta(\sigma)$ are the respective sequence-dependent binding energies of factors A and B (in units of $kT$ with the chemical potential subtracted out), and $\mu$ and $\nu$ are the corresponding activation constants.

Any diffeomorphic mode $G$ of $x$ must be of the form

$$G = (A + B\alpha)\partial_\alpha + (C + D\beta)\partial_\beta + E\partial_\mu + F\partial_\nu. \tag{3.68}$$

Solving for $\alpha$ in terms of $x$ gives

$$\begin{aligned} Gx &= -\Big(A + B\log[\tau^{-1} - 1]\Big)\,\mu\tau(1-\tau) && (3.69)\\ &\quad -(C + D\beta)\frac{\nu e^{\beta}}{(1+e^{\beta})^2} + \frac{Ex}{\mu} + \frac{F - E\nu/\mu}{1+e^{\beta}} && (3.70)\end{aligned}$$

where we have defined

$$\tau = \frac{1}{\mu}\left[x - \frac{\nu}{1+e^{\beta}}\right]. \tag{3.71}$$

Since $\tau$ and $\beta$ both depend on $\sigma$ independently of $x$, diffeomorphic modes of $x(\sigma;\theta)$ can occur only when

$$A = B = C = D = F - E\eta/\mu = 0. \tag{3.72}$$

and must therefore be proportional to

$$G = \mu\partial_\mu + \nu\partial_\nu. \tag{3.73}$$

We could have seen this diffeomorphic generator directly from Eq. 3.67; it multiplicatively scales $\mu$ an $\nu$ by the same factor.

The important point is that there are no other diffeomorphic modes. So given the proper $\sigma z$ data set, maximizing $I(z;x)$ should allow one to determine the sequence-dependent binding energies $\alpha(\sigma)$ and $\beta(\sigma)$ – in units of $kT$ and relative to each factor's chemical potential – as well as the ratio $\mu/\nu$ of activation constants.

### A simulated example

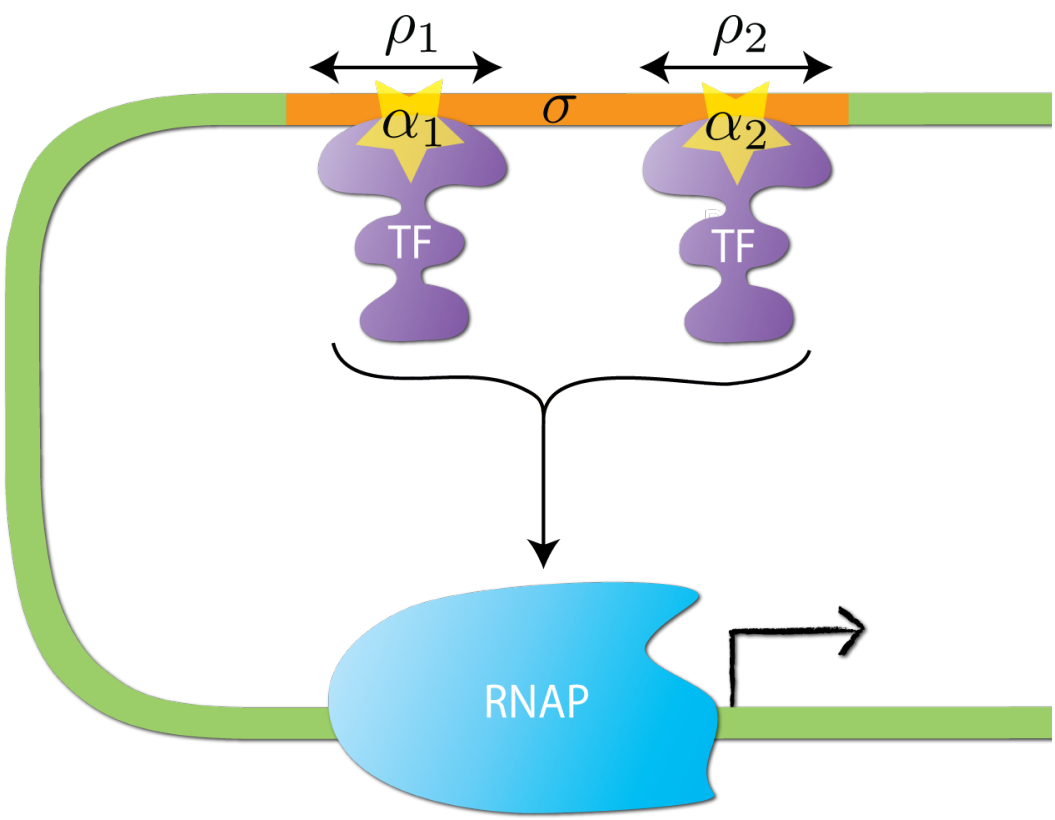


Figure 3.13: A billboard enhancer which binds only one factor, TF. We assume this enhancer activates transcription in proportion to the total occupancy of TF molecules at the two enhancer binding sites $\rho_1$ and $\rho_2$. This occupancy is given in Eq. 3.74.

We now demonstrate, through simulation, the feasibility of breaking diffeomorphic modes. For this we consider the billboard enhancer pictured in Fig. 3.13, which contains two binding sites for a single transcription factor TF. We further assume that there is no interaction between bound proteins and that transcription is activated in proportion to *total factor occupancy*. Let $\alpha(\rho)$ be the sequence-dependent binding energy of TF (expressed in units of $kT$ with the chemical potential subtracted out). Factor occupancy is then given by

$$x^*(\sigma) = \frac{1}{1 + e^{\alpha_1(\sigma)}} + \frac{1}{1 + e^{\alpha_2(\sigma)}} \tag{3.74}$$

where $\alpha_1(\sigma) = \alpha(\rho_1)$ is the sequence-dependent binding energy of TF to subsequence $\rho_1$ of $\sigma$ (shown in Fig. 3.13), and $\alpha_2(\sigma) = \alpha(\rho_2)$ is TF's binding energy to subsequence $\rho_2$.

A Sort-Seq experiment in which only the $\rho_1$ sequence of $\sigma$ is varied would allow one to infer the sequence-dependent binding energy $\alpha(\cdot)$ up to an affine transformation. But by varying both $\rho_1$ and $\rho_2$ simultaneously, the natural nonlinearities in $x^*(\sigma)$

| sequence $\sigma$ | $\alpha_1(\sigma)$ | $\alpha_2(\sigma)$ | shorthand for $x(\sigma;\theta)$ |
|---|---|---|---|
| $\sigma_1$ | 0 | 0 | $x_1(A,B)$ |
| $\sigma_2$ | -5 | 5 | $x_2(A,B)$ |
| $\sigma_3$ | $\log(2/3)$ | $\log 14$ | $x_3(A,B)$ |
| $\sigma_4$ | $\log 2$ | $\log 2$ | $x_4(A,B)$ |

Table 3.2: Hypothetical sequences considered in Fig. 3.14 with corresponding values for $\alpha_1$ and $\alpha_2$. Just to be clear, we note explicitly that $x_i$ is used to denote $x(\sigma_i,\theta)$ for various values of $A$ and $B$.

should allow one to pin these modes down, and to infer $\alpha(\cdot)$ in its quantitative entirety. Affine transformations of $\alpha$ are produced by generators of the form

$$G = (A + B\alpha)\partial_\alpha \tag{3.75}$$

where $A$ and $B$ are constants. As was done in the previous section, it is easy to show that such diffeomorphic modes can be eliminated – at least in principle – by the natural nonlinearities of $x^*$.

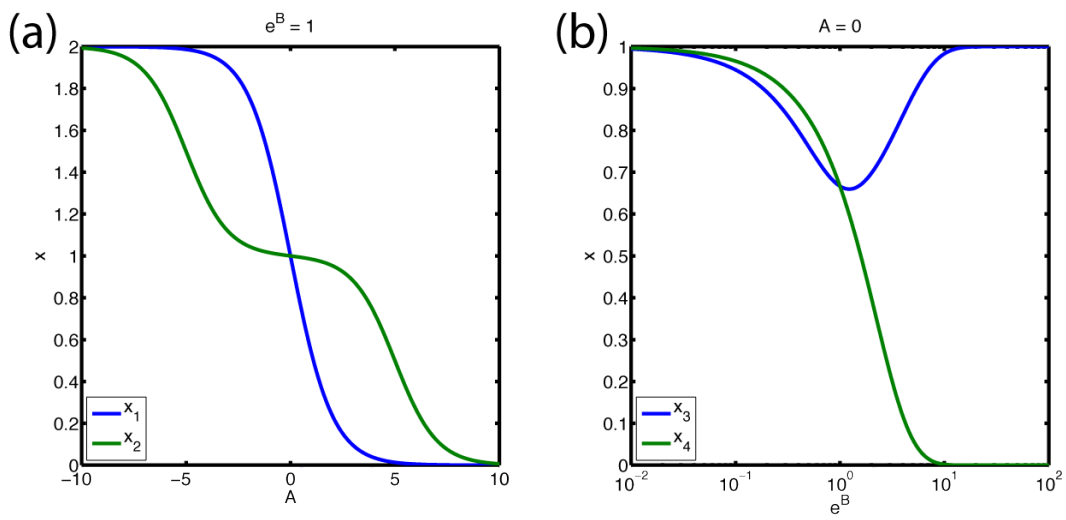


Figure 3.14: Explicit examples of how the rank-order of $x$-values depends on $A$ and $B$ in Eq. 3.76. The binding energies $\alpha_1$ and $\alpha_2$ of the sequences whose $x$-values are plotted here are listed in Table 3.2. (a) $x_1(A,B)$ and $x_2(A,B)$ for $B = 0$ (i.e. $e^B = 1$) and variable $A$. (b) $x_3(A,B)$ and $x_4(A,B)$ for $A = 0$ and variable $B$.

To see this explicitly, though, consider the model $x(\sigma;\theta)$, defined by parameters $\theta = \{A, B\}$, resulting from application of the exponentiated generator $e^G$ to $x^*(\sigma)$:

$$x(\sigma;\theta) = e^G x^*(\sigma) = \frac{1}{1+\exp\left[e^B\alpha_1(\sigma)+A\right]} + \frac{1}{1+\exp\left[e^B\alpha_2(\sigma)+A\right]}. \tag{3.76}$$

Now consider a hypothetical sequence $\sigma_1$, for which $\alpha_1 = 0$ and $\alpha_2 = 0$ (see Table 3.2). Let $x_1$ denote $x(\sigma_1,\theta)$. Fig. 3.14a plots $x_1$ for different values of $A$ (with $B = 0$),

showing a single transition from $x_1 \sim 2$ when $A < -1$ to $x_1 \sim 0$ when $A > 1$. Also shown is the $A$-dependence of $x_2$, the $x$-value of a hypothetical sequence $\sigma_2$ for which $\alpha_1 = -5$ and $\alpha_2 = +5$. For this sequence, $x_2 \sim 2$ for $A < -6$, $x \sim 1$ for $-4 < A < +4$, and $x \sim 0$ for $A > +6$. Clearly, $x_1 > x_2$ for $A < 0$, while $x_1 < x_2$ for $A > 0$. This demonstrates explicitly how variations in the value of $A$ can change the rank-order of $x$-values and thereby alter $I(z; x)$. Similarly, Fig. 3.14b shows how altering $B$ (and thus the scale $e^B$ of the energy matrix $\alpha$) can also change the rank order of $x$-values for the hypothetical sequence $\sigma_3$ and $\sigma_4$ in Table 3.2.

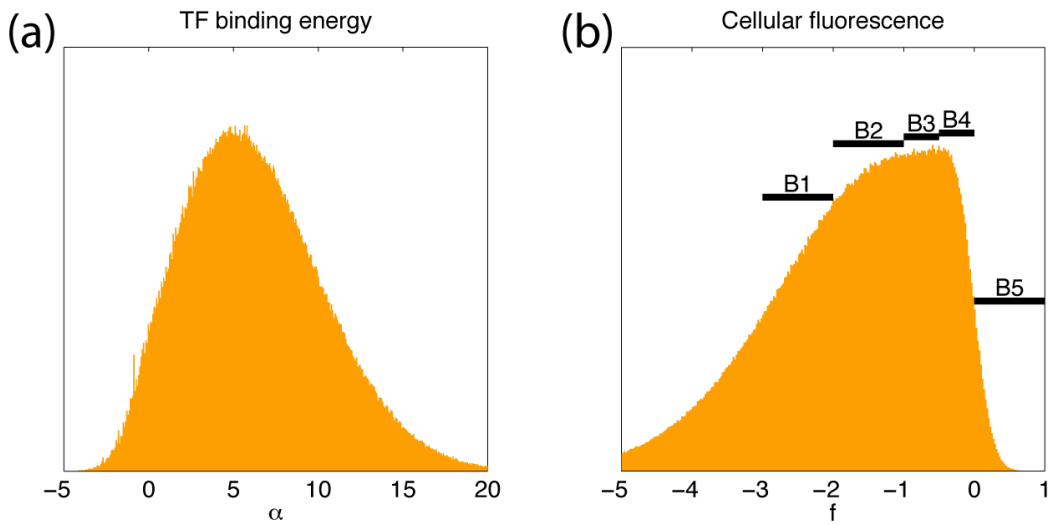


Figure 3.15: Sort-Seq simulation. (a) Histogram of binding energies $\alpha(\rho)$ for polymorphic binding sites $\rho$ in the unsorted enhancer library. (b) Histogram of cellular fluorescence for unsorted cells. The fluorescence range of each FACS batch (B1, B2, B3, B4, and B5) is indicated by a black bar above the corresponding part of the distribution.

Appropriate Sort-Seq data may, in fact, be able to exploit this breaking of diffeomorphic modes and thus pin down the values of $A$ and $B$. To illustrate this through simulation, we use the CRP energy matrix inferred from our proof-of-principle experiment as the the sequence-dependent binding energy $\alpha$ of the factor TF.[15] We hypothesized a wild type enhancer $\sigma_{wt}$ containing two TF binding sites.[16] From $\sigma_{wt}$ we generated a library of enhancer sequences in which the TF binding sites were 24% polymorphic at each position. The resulting distribution of energies $\alpha$ assigned to these sites is shown in Fig. 3.15a. To every sequence $\sigma$ in this polymorphic library we then assigned a log fluorescence

$$f(\sigma) = \log_{10} x^*(\sigma) + 0.2 * \delta \tag{3.77}$$

where $\delta$ is a normally distributed random number accounting for experimental noise. The resulting distribution of fluorescence is shown in Fig. 3.15b. We then sorted

[15] This energy matrix was affinely transformed to give a seemingly reasonable range of binding energies.

[16] For these sites we chose the sequence of the *lacZ* promoter's CRP binding site.

$2 \times 10^4$ sequences into each of five FACS batches (B1, ..., B5); the fluorescence range of each batch is indicated in the figure. This resulted in simulated $\sigma z$ data consisting of $N = 10^5$ sequences $\{\sigma_i\}$ with each sequence $\sigma_i$ assigned a FACS batch $z_i$. For these data we measured $I(z;\sigma) = I(z;x^*) = 1.53$ bits.

To show how the EMA likelihood of these data depended on our choice of $A$ and $B$, we computed the log EMA likelihood ratio of $x^*(\cdot) = x(\cdot,\theta^*)$ vs. $x(\cdot,\theta)$ for different values of $A$ and $B$, i.e.

$$\Delta = \log \frac{p_{ema}(\{z_i\} \,|\, \{\sigma_i\}, \theta)}{p_{ema}(\{z_i\} \,|\, \{\sigma_i\}, \theta^*)} \approx N[I(z;x) - I(z;x^*)] \tag{3.78}$$

where $\theta = \{A, B\}$ and $\theta^* = \{0, 0\}$. For the EMA likelihood computation, $x$-values were quantized into 50 equipopulated bins.[17] The resulting dependence of $\Delta$ on $A$ and $B$ is plotted in Figs. 3.16a-c.[18]

For $A = B = 0$, $\Delta = 0$ (indicated by a white cross in each panel). If all other values of $A$ and $B$ are less consistent with the data, then they should cause $\Delta < 0$. Figs. 3.16a-c show that the rate at which $\Delta$ decreases as one moves away from the origin, though, depends dramatically on the direction in $AB$ that space one moves. Fig. 3.16a shows that, roughly speaking, the strongest constraint on $A$ and $B$ is a lower-bound on $B$ (and thus the energy scale, $e^B$). However, Figs. 3.16b and 3.16c, in which the color scaled is respectively magnified by one and two orders of magnitude, show that the data actually specify this energy scale to $\sim 5\%$ on both sides. EMA likelihood appears much less sensitive to changes in $A$ than in $B$, in Figs. 3.16a and 3.16b, but Fig. 3.16c seems to indicate the ability to determine $A = 0 \pm .15$, thus fixing the binding energy of TF relative to its chemical potential.

Fig. 3.16c, however, reveals a clear problem with EMA likelihood. The maximum value of $\Delta$ in this plot is about 16. While it is not surprising that $\Delta > 0$ for some choices of $A$ and $B$, the fact that $\Delta \gg 1$ occurs for *any* choice of $A$ and $B$ is worrying. $\Delta = 16$ implies that some nonzero value for $A$ and/or $B$ is $e^{16} \sim 10^7$ times more likely than the correct model. Even accounting for multiple hypothesis testing this is easily enough to falsify the claim that $A = B = 0$. This inconsistency is likely due to our quantization of $x$ – the most obvious step in our analysis which violates the assumptions of EMA likelihood. Indeed, the author has found that coarser quantizations of $x$ often lead to $\Delta \gg 1$ at values $A$ and $B$ further from the origin.[19] However, using very fine quantization of $x$ does not seem to eliminate these $\Delta \gg 1$ inconsistencies. This fact might be due to our choice of error-model prior, Eq. 2.15. As discussed earlier in this chapter, Dirichlet priors exhibit a large bias towards models with specific entropy values. This bias will worsen as the number of possible values for $x$ increases. Computing EMA likelihood using a mixture of Dirichlet priors,

[17] Not 2 bins, as was done for the analysis of ChIP-chip and PBM data in chapter 2.

[18] Alternatively, one could directly ask how the mutual information $I(z;x)$ varied with choice of $A$ and $B$. In these simulations we found little difference between $\Delta$ and $N[I(z;x^*) - I(z;x)]$, where $I(z;x)$ is computed using the naive formula for mutual information. We show $\Delta$ because it has a clear Bayesian interpretation.

[19] In retrospect, it is surprising that our analysis of ChIP-chip and PBM data, where we quantized $x$ into only 2 bins, worked at all.

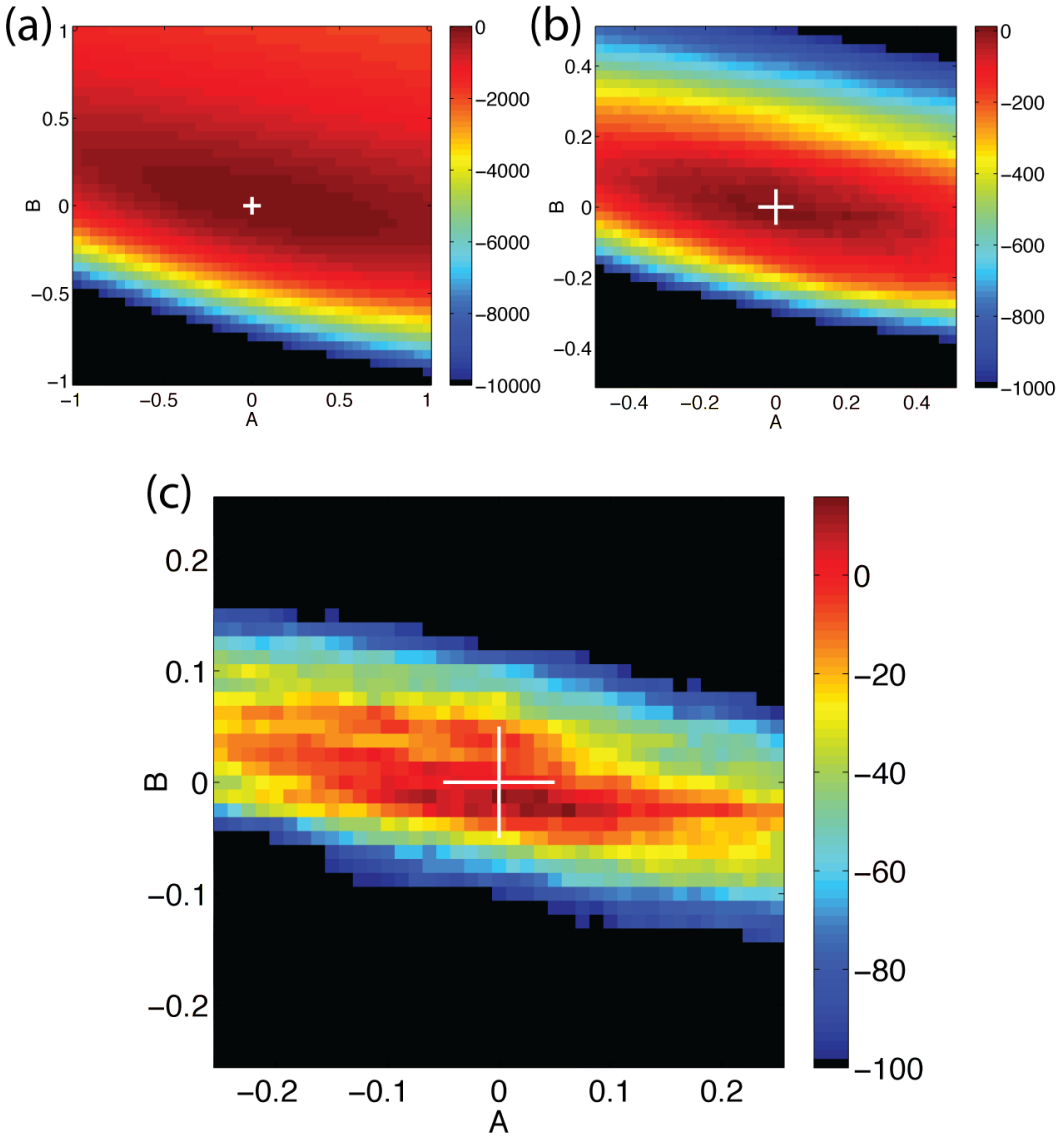


Figure 3.16: Log EMA likelihood ratio $\Delta$ (defined in Eq. 3.78) plotted for different values of $A$ and $B$. These panels show $\Delta$ for different ranges of $A$ and $B$, with the color scale ranging from a maximum of 16 to a minimum of (a) $-10^4$, (b) $-10^3$, and (c) $-10^2$. Values of $\Delta$ which fall below these lower bounds are shown in black. A white cross in each plot indicates $A = B = 0$.

such as that proposed by Nemenman, Shafee, and Bialek [83], might alleviate this problem. Another approach might be to treat continuous $x$ values naturally through a field-theoretic computation of EMA likelihood (as was proposed in chapter 1). Doing so, though, would require additional analytic results.

Despite these problems, EMA likelihood appears to be of great utility. The author finds it impressive that, while $\Delta$ takes on negative values spanning four or five decades, no values of $\Delta$ are $\gg 10$. And while straight MMC sampling using EMA likelihood would produce results inconsistent with the known answer, the EMA likelihood landscape clearly does specify both $|A|$ and $|B|$ to $\lesssim 0.1$. We conclude that the breaking of diffeomorphic modes by natural nonlinearities may well allow one to infer factor binding energies in real thermal units, as well as probe the concentration of regulatory proteins in the cell. We demonstrated this capability on a simulated $\sigma z$ data set of roughly the size that might be generated by 454 pyrosequencing. There are many questions that still need to be addressed: can this breaking of diffeomorphic modes be exploited when one has only an approximate model of factor-DNA binding energy? How should one do inference given EMA likelihood's inconsistencies? And how should one choose the fluorescence range of one's $z$-bins to maximize the statistical power of the data one produces? Despite these lingering problems, the tools presented in this thesis should provide a starting point from which such questions can be addressed.

### 3.4.4 Discussion

In this section we have shown that Sort-Seq has the potential to measure – in real thermal units – the protein-DNA and protein-protein interaction energies underlying transcriptional regulation. How important this capability is depends on one's goals. If one simply wants a description of sequence-dependent binding energy good enough to identify likely genomic binding sites, then being able to assign proper units to an inferred energy matrix won't help. On the other hand, knowing actual interaction energies in units of $kT$ is essential for biophysical modeling – especially when trying to understand the cooperative binding of transcription factors to DNA. Understanding cooperatively is particularly important in eukaryotes, since many eukaryotic transcription factors do not have enough sequence specificity to pinpoint functional binding sites without cooperative interactions (e.g. see [27]).

## 3.5 Appendix

### 3.5.1 Entropy moments of Dirichlet distributions

Here we derive the formulas Eq. 3.22 and Eq. 3.24 for $\langle H \rangle_{q|c,\beta}$ and $\mathrm{var}\,(H)_{q|c,\beta}$.

$$\langle H(q)|c,\beta\rangle = \int dq \left(-\sum_z q_z \log q_z\right) p(q|c,\beta) \tag{3.79}$$

$$= -\frac{d}{d\lambda} \int dq \left(\sum_z q_z^\lambda\right) p(q|c,\beta)\Big|_{\lambda=1} \tag{3.80}$$

$$= -\frac{d}{d\lambda}\sum_z \frac{\Gamma(N+\beta m)}{\Gamma(c_z+\beta)}\frac{\Gamma(c_z+\beta+\lambda)}{\Gamma(N+\beta m+\lambda)}\bigg|_{\lambda=1} \tag{3.81}$$

$$= -\sum_z \frac{\Gamma(B-1)}{\Gamma(A_z-1)}\frac{\Gamma(A_z)}{\Gamma(B)}\left[\psi_0(A_z)-\psi_0(B)\right] \tag{3.82}$$

$$= \psi_0(B) - \sum_z \frac{A_z-1}{B-1}\psi_0(A_z) \tag{3.83}$$

where we have used $A_z = c_z + \beta + 1$, $B = N + \beta m + 1$ to ease notation. This gives Eq. 3.22.

$$\left\langle H^2\right\rangle_{q|c,\beta} = \int dq \left(-\sum_z q_z \log q_z\right)^2 p(q|c,\beta) \tag{3.84}$$

$$= \frac{d}{d\lambda}\frac{d}{d\eta}\sum_z \frac{\Gamma(N+\beta m)}{\Gamma(c_z+\beta)}\frac{\Gamma(c_z+\beta+\lambda+\eta)}{\Gamma(N+\beta m+\lambda+\eta)}\bigg|_{\lambda=\eta=1} + \dots \tag{3.85}$$

$$\frac{d}{d\lambda}\frac{d}{d\eta}\sum_{z\neq y} \frac{\Gamma(N+\beta m)}{\Gamma(c_z+\beta)\Gamma(c_y+\beta)}\frac{\Gamma(c_z+\beta+\lambda)\Gamma(c_y+\beta+\eta)}{\Gamma(N+\beta m+\lambda+\eta)}\bigg|_{\lambda=\eta=1}$$

$$= \sum_z \frac{A_z(A_z-1)}{B(B-1)}[\psi_1(A_z+1)-\psi_1(B+1)+\cdots \tag{3.86}$$

$$(\psi_0(A_z+1)-\psi_0(B+1))^2] + \dots$$

$$\sum_{z\neq y}\frac{(A_z-1)(A_y-1)}{B(B-1)}\left[-\psi_1(B+1)+\cdots\right.$$

$$\left.(\psi_0(A_z)-\psi_0(B+1))(\psi_0(A_y)-\psi_0(B+1))\right].$$

Using the fact that $\sum_z(A_z-1) = B-1$, as well as $\psi_0(x+1) = \psi_0(x) + 1/x$ and $\psi_1(x+1) = \psi_1(x) - 1/x^2$, we find the simplified answer (after some algebra),

$$\mathrm{var}(H)_{q|c,\beta} = \left\langle H^2\right\rangle_{q|c,\beta} - \langle H\rangle^2_{q|c,\beta} \tag{3.87}$$

$$= \sum_z \frac{A_z(A_z-1)}{B(B-1)}\psi_1(A_z) - \psi_1(B) + \dots \tag{3.88}$$

$$\sum_z \frac{A_z-1}{B(B-1)}\psi_0^2(A_z) - \frac{1}{B}\left[\sum_z \frac{A_z-1}{B-1}\psi_0(A_z)\right]^2.$$

### 3.5.2 Derivation of $\nabla_\theta I(z;x)$

This derivation of $\nabla_\theta I(z;x)$ traces, in part, the derivation in Appendix C of Sharpee, Rust, and Bialek [110]; however, we do not require that $x(\sigma;\theta)$ be linear in either $\sigma$ or $\theta$. To minimize ambiguity in our notation, we will henceforth write $x = X(\sigma;\theta)$, thus distinguishing between the real (continuous) variable $x$ and the function $X$. Also, for any function $g(\sigma)$ we will write $\langle g(\sigma)\rangle_\sigma \equiv (1/N)\sum_i g(\sigma_i)$. Similarly, $\langle g(\sigma)\rangle_{\sigma|z}$ will denote the average of $g(\sigma_i)$ over all $\sigma_i$ for which $z_i = z$, and $\langle g(\sigma)\rangle_{\sigma|z,x}$ will denote the average over all $\sigma_i$ for which $z_i = z$ and $X(\sigma_i;\theta) = x$.

We start by expressing $I(z;x) = H(x) - H(x|z)$ where,

$$H(x) = -\int dx\ f(x) \log f(x) \tag{3.89}$$

$$H(x|z) = -\sum_z f(z) \int dx\ f(x|z) \log f(x|z). \tag{3.90}$$

Writing the observed probability density $f(x)$ as an average of delta functions,

$$f(x) = \langle \delta(x - X(\sigma;\theta)) \rangle_\sigma , \tag{3.91}$$

allows us to compute the gradient of $f(x)$ with respect to model parameters $\theta$:

$$\nabla_\theta f(x) = \langle \nabla_\theta \delta(x - X(\sigma;\theta)) \rangle_\sigma \tag{3.92}$$

$$= -\left\langle \frac{\partial \delta(x - X(\sigma;\theta))}{\partial x} \nabla_\theta X(\sigma;\theta) \right\rangle_\sigma \tag{3.93}$$

$$= -\frac{\partial}{\partial x} \langle \delta(x - X(\sigma;\theta)) \nabla_\theta X(\sigma;\theta) \rangle_\sigma \tag{3.94}$$

$$= -\frac{\partial}{\partial x} \left[ f(x) \langle \nabla_\theta X \rangle_{\sigma|x} \right]. \tag{3.95}$$

This, in turn, allows us to compute the gradient of $H(x)$,

$$\nabla_\theta H(x) = -\nabla_\theta \int dx\ f(x) \log f(x) \tag{3.96}$$

$$= -\int dx\ [\nabla_\theta f(x)] \log f(x) \tag{3.97}$$

$$= -\int dx\ f(x) \langle \nabla_\theta X \rangle_{\sigma|x} \frac{d}{dx} \log f(x) \tag{3.98}$$

$$= -\left\langle [\nabla_\theta X] \frac{d}{dx} \log f(x) \right\rangle_\sigma . \tag{3.99}$$

From $f(x|z) = \langle \delta(x - X(\sigma;\theta)) \rangle_{\sigma|z}$ we can similarly derive,

$$\nabla_\theta f(x|z) = -\frac{\partial}{\partial x} \left[ f(x|z) \langle \nabla_\theta X \rangle_{\sigma|x,z} \right], \tag{3.100}$$

$$\nabla_\theta H(x|z) = -\left\langle [\nabla_\theta X] \frac{d}{dx} \log f(x|z) \right\rangle_\sigma , \tag{3.101}$$

from which we get Eq. 3.45 in the main text:

$$\nabla_\theta I(z;x) = \left\langle [\nabla_\theta X] \frac{d}{dx} \log \frac{f(x|z)}{f(x)} \right\rangle_\sigma . \tag{3.102}$$

# Chapter 4

# Outlook

The underlying theme of this dissertation has been the potential for sequence data to provide high-resolution information about specific biophysical systems. In chapter 2 we proposed some basic principles for DNA sequence analysis, and applied these principles to gain a detailed picture of Abf1's sequence-dependent binding energy using previously published data. In chapter 3 we proposed the Sort-Seq assay which, when combined with a principled approach to sequence analysis, has the potential to provide detailed biophysical pictures of the mechanisms underlying the function of transcriptional regulatory sequences. We also argued that Sort-Seq data probing the simultaneous binding of multiple regulatory proteins might allow one to infer interaction energies in natural thermal units, as well as measure relative transcription factor concentrations.

We note that Sort-Seq may be able to reveal the functional architecture of transcriptional regulatory sequences *de novo*. Suppose one performs Sort-Seq using an initial sequence library in which nucleotides at different positions appear with independent probabilities (e.g. as was the case for the 26 bp region probed in our proof-of-principle experiment). By computing $I(z;\sigma_{bl})$[1] – the mutual information between FACS batch $z$ and the base $b$ occurring at position $l$ – one could immediately tell which positions of the mutagenized region substantially influence transcription: $I(z;\sigma_{bl})$ should be noticeably larger than background at all positions $l$ for which the identity of the nucleotide has a detectable effect on gene expression.

Alternatively, we can think of energy matrices as vectors in sequence space, just as the receptive fields of retinal ganglia cells can be thought of as dimensions in stimulus space. We can therefore use the technique of maximally informative dimensions [110, 111] to infer energy matrices for all regulatory proteins which bind to the mutagenized region of the transcriptional regulatory sequence. It is possible, in fact, to infer multiple informative dimensions simultaneously, which in our case would correspond to multiple proteins binding the same mutagenized region. And unlike trying to infer biological function by looking at $I(z;\sigma_{bl})$ position-by-position, maximally informative dimensions require no *a priori* assumptions about the statistics of the initial sequence library.

[1] The notation $I(z;\sigma_{bl})$ is regrettably ambiguous; we mean the mutual information between $z$ and the *column vector* of the matrix $\{\sigma_{bl}\}$ in which $l$ is fixed and $b$ varies.

Energy matrices inferred from Sort-Seq data could then be used as "fingerprints": they could be cross-referenced against PBM or *E. coli* one-hybrid data to identify the corresponding proteins. In this way, PBM or *E. coli* one-hybrid data for all transcription factors in a given organism might allow one to use Sort-Seq to identify the precise location *and identity* of all DNA-binding proteins involved in transcriptional regulation at a specific locus. Higher-order thermodynamic models which incorporate these energy matrices could then be used to discern the precise influence that each of these DNA-binding proteins has on transcription.

By proposing the Sort-Seq assay, we seek to move the difficulty of studying transcriptional regulation out of the laboratory and onto the computer. We believe that, quite generally, there is substantial merit to transforming experimental problems into problems of data analysis. Fully understanding the data produced by Sort-Seq is likely to require nontrivial computational effort, as well as substantial trial and error. But we feel that the relative ease with which Sort-Seq data can be produced, as well as the potential for this data to probe transcriptional regulation with unprecedented resolution, more than compensates for these difficulties.

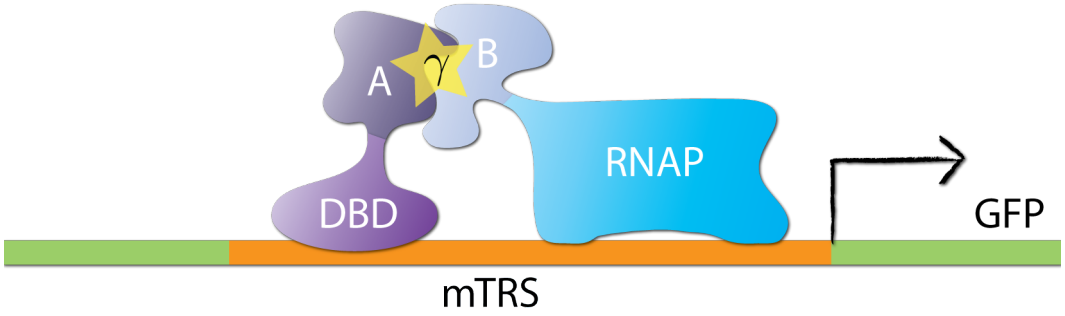


Figure 4.1: A possible two-hybrid experiment for measuring protein-protein interaction energies. Protein A is fused to a DNA-binding domain (DBD), while protein B is fused to a subunit of the RNA polymerase (RNAP). Performing Sort-Seq with mTRSs that cover the DNA binding sites of both the DBD and RNAP might allow one to measure the interaction energy $\gamma$ between A and B.

The ability to measure energies using sequence data raises a number of other possibilities. For instance, the ability to infer the interaction energy $\gamma$ between the factor and RNA polymerase in Fig. 3.12 suggests that sequence data might be used to measure interaction energies between arbitrary proteins. This might be done through a modified two-hybrid assay (sketched in Fig. 4.1), in which one of the proteins of interest is fused to a DNA binding domain while the other protein is fused to RNA polymerase. Performing Sort-Seq using mutant sequences which bind both the RNA polymerase and the DNA binding domain might allow one to infer the interaction energy $\gamma$ between these two proteins. Sort-Seq's ability to measure protein-DNA binding energies relative to a protein's chemical potential also suggests the possibility of using sequence data to monitor the intracellular concentrations of activated transcription factors. Of course, models inferred from Sort-Seq data are inherently

fit to the population as a whole; Sort-Seq cannot be used to determine, say, protein concentrations in individual cells.

Experimental techniques such as Sort-Seq are also likely to find application in areas of biology other than just transcriptional regulation. Sequence perturbations of regulatory RNAs might help reveal the detailed requirements that various modes of regulation place on RNA secondary and tertiary structure. Recent work has also indicated the importance of pair-wise interactions (between amino acids) for protein folding and function [112, 113]; it might be possible to map these interactions in great detail using a Sort-Seq-like assay with polymorphic coding sequences.

It is widely believed that ultra-high-throughput sequencing technologies will soon obviate the need for microarrays in many genome-wide assays. But it is not hard to envisage the day when massively parallel sequencing is also used to routinely make precise quantitative measurements of specific biological systems. We feel that thinking about such applications now is eminently worth while.

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